

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:43:30

Source: data/json/noise_experiments/v2/noise8i_nocoplayer_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_nocoplayer/noise8i_nocoplayer_memprimary_game_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_nocoplayer
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 4
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
dyadic_pairing_schedule: {
  "num_turns": 10,
  "role_targets": {
    "Agent_1": 5,
    "Agent_2": 5,
    "Agent_3": 5,
    "Agent_4": 5,
    "Agent_5": 5,
    "Agent_6": 5,
    "Agent_7": 5,
    "Agent_8": 5
  },
  "strategy": "balanced_random_roles_v1"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths,

follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

=====

ROUND 1

=====

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Sender, Agent_6=Sender, Agent_7=Sender, Agent_8=Sender

Pairings:

dyad_1: Agent_8=Sender, Agent_1=Receiver
dyad_2: Agent_5=Sender, Agent_4=Receiver
dyad_3: Agent_6=Sender, Agent_2=Receiver
dyad_4: Agent_7=Sender, Agent_3=Receiver

Parsed game decisions:

dyad_1: Agent_8 sent \$3; received \$9; Agent_1 returned \$4; payoffs Agent_8=\$6, Agent_1=\$5;
noise/visible: sent visible=\$3.26, received visible=\$9.78, returned visible=\$3.03
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_5 sent \$3; received \$9; Agent_4 returned \$3.5; payoffs Agent_5=\$5.5, Agent_4=\$5.5;
noise/visible: sent visible=\$2.49, received visible=\$7.47, returned visible=\$2.93
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_6=\$7, Agent_2=\$4;
noise/visible: sent visible=\$3.34, received visible=\$10.02, returned visible=\$5.74
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_3 returned \$5.28; payoffs Agent_7=\$7.28, Agent_3=\$3.72; noise/visible: sent visible=\$3.89, received visible=\$11.67, returned visible=\$5.09
Action validation: {"returned": {"amount": 5.28, "clamped": false, "max": 9.0, "min": 0, "raw": 5.28}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,
"Agent_2": 4.0,
"Agent_3": 3.7199999999999998,
"Agent_4": 5.5,
"Agent_5": 5.5,
"Agent_6": 7.0,
"Agent_7": 7.28,
"Agent_8": 6.0

}

Visible balances after round: {

"Agent_1": 5.7799999999999999,
"Agent_2": 5.02,
"Agent_3": 6.39,
"Agent_4": 3.9700000000000006,
"Agent_5": 4.93,
"Agent_6": 7.74,
"Agent_7": 7.09,
"Agent_8": 5.0299999999999999

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:28:59
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_1"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_1": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver (3 × \$3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:29:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_1"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_1": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
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- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep: $\$9.78 - \$4.00 = \$5.78$
- Sender ends with: $\$1.74 + \$4.00 = \$5.74$

This seems fair given their generous initial trust.

{'return': 4.00}

Reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is $\$3.26 \times 3$)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($\$5 - \3.26)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps: $\$5 - \$3.26 = \$1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):

- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):

- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:

- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

Usage: {

"input_tokens": 529,
"output_tokens": 798,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-08-12T09:29:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_4"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_4": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game setup.
2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: $5 - X + (\text{amount returned})$
 - Receiver's final payoff: $3X - (\text{amount returned})$
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

```
...
Usage: {
  "input_tokens": 500,
  "output_tokens": 652,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:29:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_4"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_4": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: $\$2.51 + \$3.50 = \$6.01$ (profit of ~\$1)
- I keep: $\$7.47 - \$3.50 = \$3.97$ (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

{"return": 3.50}

Reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received $\$2.49 \times 3 = \7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Going higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98 ...

Usage: {

"input_tokens": 537,

```
"output_tokens": 1072,
"reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T09:30:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_2"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_2": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

****My reasoning:****

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send ****\$3****. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

{"send": 3}

Reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

****If I send nothing (\$0):****

- I keep: \$5 - \$0 + \$0 = \$5
- Receiver gets: \$0 × 3 = \$0
- Total payoff: \$5

****If I send everything (\$5):****

- Receiver gets: \$5 × 3 = \$15
- If they return half (\$7.50), I get: \$5 - \$5 + \$7.50 = \$7.50
- If they return nothing, I get: \$5 - \$5 + \$0 = \$0
- Maximum total payoff possible: \$20 if I send \$5

****Optimal social outcome:****

If I send \$5, the total pie becomes \$20 (\$5 I keep initially - \$5 sent + \$15 receiver gets). If the receiver returns \$7.50, we each get \$7.50, which is better than the \$5 I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

Usage: {

"input_tokens": 500,
"output_tokens": 761,
"reasoning_tokens": 0

}

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-08-12T09:30:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_2"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_2": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with: $\$1.66 + \$5.00 = \$6.66$ (a profit of \$1.66)
- I end with: $\$10.02 - \$5.00 = \$5.02$ (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

{"return": 5.00}

Reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 ($\$5 - \3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.34 + \$0 = \1.66
- I get: $\$10.02 - \$0 = \$10.02$
- Total: \$11.68

If I return \$5:

- Sender gets: $\$5 - \$3.34 + \$5 = \6.66
- I get: $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return: $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets: $\$1.66 + \$4.18 = \$5.84$
- I get: $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding

g their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1155,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T09:30:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_3"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

Reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent/returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain a good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is \$5 + 2X, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-08-12T09:31:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_3"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_3": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

Reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received $\$3.89 \times 3 = \11.67
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with: $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with: $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total = $\$5 - \$3.89 + \$11.67 = \12.78
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need \$1.11 (what they kept) + what I return = \$6.39
- So I should return: $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received: $\$11.67 \times 0.778 = \9.08
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable

- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation: if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Sender, Agent_4=Sender, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Receiver, Agent_8=Receiver

Pairings:

dyad_1: Agent_4=Sender, Agent_7=Receiver
dyad_2: Agent_1=Sender, Agent_5=Receiver
dyad_3: Agent_2=Sender, Agent_8=Receiver
dyad_4: Agent_3=Sender, Agent_6=Receiver

Parsed game decisions:

dyad_1: Agent_4 sent \$2.5; received \$7.5; Agent_7 returned \$3.5; payoffs Agent_4=\$6, Agent_7=\$4;
noise/visible: sent visible=\$2.08, received visible=\$6.24, returned visible=\$3.93
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_1 sent \$3.25; received \$9.75; Agent_5 returned \$4; payoffs Agent_1=\$5.75, Agent_5=\$5.75; noise/visible: sent visible=\$2.82, received visible=\$8.46, returned visible=\$4.82
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.75, "min": 0, "raw": 4.0}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}
dyad_3: Agent_2 sent \$3.5; received \$10.5; Agent_8 returned \$6.5; payoffs Agent_2=\$8, Agent_8=\$4;
noise/visible: sent visible=\$4.33, received visible=\$12.99, returned visible=\$6.73
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_4: Agent_3 sent \$4; received \$12; Agent_6 returned \$6; payoffs Agent_3=\$7, Agent_6=\$6;
noise/visible: sent visible=\$3.25, received visible=\$9.75, returned visible=\$5.2
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 10.75,
"Agent_2": 12.0,
"Agent_3": 10.719999999999999,
"Agent_4": 11.5,
"Agent_5": 11.25,
"Agent_6": 13.0,
"Agent_7": 11.280000000000001,
"Agent_8": 10.0

}

Visible balances after round: {

"Agent_1": 12.35,
"Agent_2": 13.25,
"Agent_3": 12.59,
"Agent_4": 10.4,
"Agent_5": 9.389999999999999,
"Agent_6": 11.49,
"Agent_7": 9.83,
"Agent_8": 11.52

}

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-08-12T09:31:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_7",
"pairing": {


```

"agents": [
  "Agent_4",
  "Agent_7"
],
"investor": "Agent_4",
"roles": {
  "Agent_4": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: $\$2.51 + \$3.50 = \$6.01$ (profit of ~\$1)
- I keep: $\$7.47 - \$3.50 = \$3.97$ (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

```
{"return": 3.50}
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received $\$2.49 \times 3 = \7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Going higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1072,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became \$7.470000000000001, and you returned \$3.5. Your visible payoff was \$3.9700000000000006.

Your total visible earnings across all rounds are \$3.9700000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round 1.

{"send": 2.50}

Reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

Usage: {

"input_tokens": 897,
"output_tokens": 538,
"reasoning_tokens": 0

}

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-08-12T09:31:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_7"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent /returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is \$5 + 2X, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

****Considerations:****

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

****Fair reciprocation:****

If I return \$3.50:

- They end with: $\$5 - \$2.08 + \$3.50 = \6.42 (profit of \$1.42)
- I end with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)
- This rewards their trust and creates mutual gain

```
{"return": 3.50}
```

Reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was $\$5 - \$3 + \$5.09 = \7.09

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

****Current Round (Round 2):****

- I am RECEIVER
- My co-player sent me \$2.08
- I received \$6.24 ($2.08 \times 3 = 6.24$)
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. ****Fairness****: A fair split might be sharing the gains equally
2. ****Reciprocity****: Building trust for potential future interactions
3. ****Economic incentive****: They sent \$2.08, I got \$6.24. The gain created is $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool: $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return: $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ($\$5.09/\9.00), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure the y profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

```
Usage: {
  "input_tokens": 795,
  "output_tokens": 946,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-08-12T09:31:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_5"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep: \$9.78 - \$4.00 = \$5.78
- Sender ends with: \$1.74 + \$4.00 = \$5.74

This seems fair given their generous initial trust.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is 3.26×3)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($5 - 3.26$)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps: $5 - 3.26 = 1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):
- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):
- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:
- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 798,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
{'send': 3.25}
```

Reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

Usage: {

```
"input_tokens": 815,
"output_tokens": 324,
"reasoning_tokens": 0
}
```

```
}
```

GAME - Agent_5 - Receiver - dyad_2

Timestamp: 2026-08-12T09:32:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_2",

"move_index": 3,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_1",

"Agent_5"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_5": "trustee"

},

"trustee": "Agent_5"

},

"role": "trustee",

"role_label": "Receiver",

"round": 2,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game set up.
2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: \$5 - X + (amount returned)
 - Receiver's final payoff: \$3X - (amount returned)
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

usage:

```
{
```

```
"input_tokens": 500,  
"output_tokens": 652,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 2 - RECEIVER Decision****

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
 - They end up with: $\$2.18 + \$4.00 = \$6.18$
 - I end up with: $\$8.46 - \$4.00 = \$4.46$
 - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json  
{"return": 4.0}  
'''
```

Reasoning:

Let me analyze this situation:

****Round 1 recap:****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff: $\$5 - \$3 + \$2.93 = \4.93
- Their payoff: $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

****Round 2:****

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept: $\$5 - \$2.82 = \$2.18$
- I received: $\$8.46$
- Total pie: $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets $\$5.32$

- For that, I'd need to return: $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return: $\$8.46 \times 0.33 = \2.79

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

```
Usage: {  
  "input_tokens": 775,  
  "output_tokens": 861,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-08-12T09:32:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_8"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with: $\$1.66 + \$5.00 = \$6.66$ (a profit of \$1.66)
- I end with: $\$10.02 - \$5.00 = \$5.02$ (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 (\$5 - \$3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.34 + \$0 = \1.66
- I get: $\$10.02 - \$0 = \$10.02$
- Total: \$11.68

If I return \$5:

- Sender gets: $\$5 - \$3.34 + \$5 = \6.66
- I get: $\$10.02 - \$5 = \$5.02$
- Total: $\$11.68$

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return: $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets: $\$1.66 + \$4.18 = \$5.84$
- I get: $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1155,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

```
{'send': 3.50}
```

Reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust.

```
Usage: {
  "input_tokens": 846,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:32:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_8"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver ($3 \times \3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Round 2 Analysis:****

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning ****\$6.50**** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff: $\$5 - \$3 + \$3.03 = \5.03
- They kept: $\$9 - \$3.03 = \$5.97$

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with: $\$5 - \$4.33 + \text{my_return} = \$0.67 + \text{my_return}$
- I'll end up with: $\$12.99 - \text{my_return}$

If I want us to split relatively evenly:

- Total in the system: $\$5$ (their initial) + $\$12.99$ (what I got) = $\$17.99$
- Fair split would be about $\$9$ each
- They have $\$0.67$ before my return
- So I should return about $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

```
Usage: {
  "input_tokens": 774,
  "output_tokens": 886,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-08-12T09:32:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received $\$3.89 \times 3 = \11.67
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with: $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with: $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0

- After the transaction: Total = \$5 - \$3.89 + \$11.67 = \$12.78
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need \$1.11 (what they kept) + what I return = \$6.39
- So I should return: \$6.39 - \$1.11 = \$5.28

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received: $\$11.67 \times 0.778 = \9.08
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation : if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

```
{'send': 4.00}
```

Reasoning:

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with $\$1.11 + \$5.28 = \$6.39$, and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with $\$1.00 + \$5.40 = \$6.40$
- They'd end up with $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with $\$1.25 + \$5.06 = \$6.31$
- They'd end up with $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round

```
.
Usage: {
  "input_tokens": 787,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:33:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

****My reasoning:****

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send ****\$3****. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

{"send": 3}

reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

****If I send nothing (\$0):****

- I keep: \$5 - \$0 + \$0 = \$5
- Receiver gets: \$0 × 3 = \$0
- Total payoff: \$5

****If I send everything (\$5):****

- Receiver gets: \$5 × 3 = \$15
- If they return half (\$7.50), I get: \$5 - \$5 + \$7.50 = \$7.50
- If they return nothing, I get: \$5 - \$5 + \$0 = \$0
- Maximum total payoff possible: \$20 if I send \$5

****Optimal social outcome:****

If I send \$5, the total pie becomes \$20 (\$5 I keep initially - \$5 sent + \$15 receiver gets). If the receiver returns \$7.50, we each get \$7.50, which is better than the \$5 I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

```
...
usage:
{
  "input_tokens": 500,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.74 to you. Your visible payoff was \$7.74.

Your total visible earnings across all rounds are \$7.74.

This round, you are the RECEIVER against your current co-player. They sent you \$3.25, so you received \$9.75. How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$3.25 (keeping \$1.75)
- I received \$9.75
- In Round 1, when I sent \$3, the receiver returned \$5.74 (about 64% of the \$9 they received)
- I want to establish reciprocity while being fair

****My decision:****

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return ****\$6****:

- Sender ends with: $\$1.75 + \$6 = \$7.75$ (gain of \$2.75)
- I end with: $\$9.75 - \$6 = \$3.75$ (gain of \$3.75)

Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

{"return": 6}

Reasoning:

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was $\$5 - \$3 + \$5.74 = \7.74
- The receiver kept $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ($\$5.74/\9.0).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation

- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets \$1.75 + \$6 = \$7.75, I keep \$9.75 - \$6 = \$3.75
- If I return \$5.85: Sender gets \$1.75 + \$5.85 = \$7.60, I keep \$9.75 - \$5.85 = \$3.90
- If I return \$5: Sender gets \$1.75 + \$5 = \$6.75, I keep \$9.75 - \$5 = \$4.75

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

```
Usage: {  
  "input_tokens": 794,  
  "output_tokens": 1093,  
  "reasoning_tokens": 0  
}
```

ROUND 3

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Sender, Agent_4=Sender, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Receiver

Pairings:

dyad_1: Agent_4=Sender, Agent_1=Receiver
dyad_2: Agent_5=Sender, Agent_2=Receiver
dyad_3: Agent_6=Sender, Agent_7=Receiver
dyad_4: Agent_3=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_4 sent \$2.5; received \$7.5; Agent_1 returned \$3.5; payoffs Agent_4=\$6, Agent_1=\$4;
noise/visible: sent visible=\$2.78, received visible=\$8.34, returned visible=\$2.86
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_5 sent \$3; received \$9; Agent_2 returned \$4; payoffs Agent_5=\$6, Agent_2=\$5;
noise/visible: sent visible=\$2.52, received visible=\$7.56, returned visible=\$3.75
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent \$3.5; received \$10.5; Agent_7 returned \$7; payoffs Agent_6=\$8.5, Agent_7=\$3.5; noise/visible: sent visible=\$4.18, received visible=\$12.54, returned visible=\$6.46
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw": 7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_4: Agent_3 sent \$4; received \$12; Agent_8 returned \$5; payoffs Agent_3=\$6, Agent_8=\$7;
noise/visible: sent visible=\$3.29, received visible=\$9.87, returned visible=\$4.73
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 12.0, "min": 0, "raw": 5.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 14.75,
"Agent_2": 17.0,
"Agent_3": 16.72,
"Agent_4": 17.5,
"Agent_5": 17.25,
"Agent_6": 21.5,
"Agent_7": 14.780000000000001,
"Agent_8": 17.0

}

Visible balances after round: {

"Agent_1": 17.189999999999998,
"Agent_2": 16.810000000000002,
"Agent_3": 18.32,
"Agent_4": 15.76,
"Agent_5": 15.139999999999999,
"Agent_6": 19.45,
"Agent_7": 15.37,
"Agent_8": 16.39

}

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-08-12T09:33:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: $\$2.51 + \$3.50 = \$6.01$ (profit of ~\$1)
- I keep: $\$7.47 - \$3.50 = \$3.97$ (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

{"return": 3.50}

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received $\$2.49 \times 3 = \7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Go

ng higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1072,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became \$7.470000000000001, and you returned \$3.5. Your visible payoff was \$3.9700000000000006.

Your total visible earnings across all rounds are \$3.9700000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round 1.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender s

ent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

Reasoning:

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff: \$2.50 + \$3.93 = \$6.43

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

Usage: {

```
"input_tokens": 1153,
"output_tokens": 447,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:33:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep: $\$9.78 - \$4.00 = \$5.78$
- Sender ends with: $\$1.74 + \$4.00 = \$5.74$

This seems fair given their generous initial trust.

{'return': 4.00}

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is $\$3.26 \times 3$)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($\$5 - \3.26)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps: $\$5 - \$3.26 = \$1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):

- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):

- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:

- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 798,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

```
{'send': 3.25}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

usage:

```
{
  "input_tokens": 815,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
{'return': 3.50}
```

Reasoning:

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has \$5 - \$2.78 = \$2.22 remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)

- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep: $\$8.34 - \$3.50 = \$4.84$
- Sender ends with: $\$2.22 + \$3.50 = \$5.72$

If I return \$3.00:

- I keep: $\$8.34 - \$3.00 = \$5.34$
- Sender ends with: $\$2.22 + \$3.00 = \$5.22$

If I return \$4.00:

- I keep: $\$8.34 - \$4.00 = \$4.34$
- Sender ends with: $\$2.22 + \$4.00 = \$6.22$

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

```
Usage: {
  "input_tokens": 960,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-08-12T09:33:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_2"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_2": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game set up.
2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.

3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: $\$5 - X + (\text{amount returned})$
 - Receiver's final payoff: $\$3X - (\text{amount returned})$
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 652,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 2 - RECEIVER Decision****

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
 - They end up with: $\$2.18 + \$4.00 = \$6.18$
 - I end up with: $\$8.46 - \$4.00 = \$4.46$
 - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 recap:****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff: $\$5 - \$3 + \$2.93 = \4.93
- Their payoff: $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

****Round 2:****

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept: $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie: $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return: $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return: $\$8.46 \times 0.33 = \2.79

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

usage:

```
{
  "input_tokens": 775,
  "output_tokens": 861,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 3 - SENDER Decision****

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

```
Usage: {
  "input_tokens": 1105,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:33:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_2"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_2": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with: \$1.66 + \$5.00 = \$6.66 (a profit of \$1.66)
- I end with: \$10.02 - \$5.00 = \$5.02 (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 ($\$5 - \3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.34 + \$0 = \1.66
- I get: $\$10.02 - \$0 = \$10.02$
- Total: \$11.68

If I return \$5:

- Sender gets: $\$5 - \$3.34 + \$5 = \6.66
- I get: $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return: $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets: $\$1.66 + \$4.18 = \$5.84$
- I get: $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1155,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited u

s both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

{'send': 3.50}

reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

usage:

```
{
  "input_tokens": 846,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with: \$2.48 + \$4.00 = \$6.48 (profit of \$1.48)
- I end with: \$7.56 - \$4.00 = \$3.56 (profit of \$3.56)

This rewards their trust and maintains cooperation.


```
{'return': 4.00}
```

Reasoning:

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has \$5 - \$2.52 = \$2.48 left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again: $\$7.56 * 0.5 = \3.78

Let me check the outcomes:

- If I return \$3.78:
 - Sender ends with: $\$2.48 + \$3.78 = \$6.26$ (profit of \$1.26)
 - I end with: $\$7.56 - \$3.78 = \$3.78$ (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with: $\$2.48 + \$4.00 = \$6.48$ (profit of \$1.48)
- I end with: $\$7.56 - \$4.00 = \$3.56$ (profit of \$3.56)

...

```
Usage: {  
  "input_tokens": 1072,  
  "output_tokens": 643,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T09:34:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_7"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_6": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

****My reasoning:****

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send ****\$3****. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

{"send": 3}

reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

****If I send nothing (\$0):****
- I keep: $\$5 - \$0 + \$0 = \5
- Receiver gets: $\$0 \times 3 = \0
- Total payoff: $\$5$

****If I send everything (\$5):****
- Receiver gets: $\$5 \times 3 = \15
- If they return half (\$7.50), I get: $\$5 - \$5 + \$7.50 = \7.50
- If they return nothing, I get: $\$5 - \$5 + \$0 = \0
- Maximum total payoff possible: $\$20$ if I send $\$5$

****Optimal social outcome:****

If I send $\$5$, the total pie becomes $\$20$ ($\$5$ I keep initially - $\$5$ sent + $\$15$ receiver gets). If the receiver returns $\$7.50$, we each get $\$7.50$, which is better than the $\$5$ I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent $\$3.0$, it became $\$9.0$, and you saw them return $\$5.74$ to you. Your visible payoff was $\$7.74$.

Your total visible earnings across all rounds are $\$7.74$.

This round, you are the RECEIVER against your current co-player. They sent you $\$3.25$, so you received $\$9.75$. How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent $\$3.25$ (keeping $\$1.75$)
- I received $\$9.75$
- In Round 1, when I sent $\$3$, the receiver returned $\$5.74$ (about 64% of the $\$9$ they received)
- I want to establish reciprocity while being fair

****My decision:****

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return ****\$6****:

- Sender ends with: $\$1.75 + \$6 = \$7.75$ (gain of $\$2.75$)
- I end with: $\$9.75 - \$6 = \$3.75$ (gain of $\$3.75$)

Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

```
{"return": 6}
```

reasoning:

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was $\$5 - \$3 + \$5.74 = \7.74
- The receiver kept $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ($\$5.74/\9.0).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation
- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets $\$1.75 + \$6 = \$7.75$, I keep $\$9.75 - \$6 = \$3.75$
- If I return \$5.85: Sender gets $\$1.75 + \$5.85 = \$7.60$, I keep $\$9.75 - \$5.85 = \$3.90$
- If I return \$5: Sender gets $\$1.75 + \$5 = \$6.75$, I keep $\$9.75 - \$5 = \$4.75$

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

usage:

```
{
  "input_tokens": 794,
  "output_tokens": 1093,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

****Analysis:****

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

****My decision:****

Based on the positive reciprocity I've seen, I'll send ****\$3.50**** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

```
{"send": 3.5}
```

Reasoning:

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0
- Received back \$5.74
- Payoff: $\$5 - \$3 + \$5.74 = \7.74

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff: $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

```
Usage: {  
  "input_tokens": 1132,  
  "output_tokens": 711,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-08-12T09:34:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_7"    ]  
  }  
}
```

```

],
"investor": "Agent_6",
"roles": {
  "Agent_6": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting

the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

```
{"send": 3}
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent /returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is \$5 + 2X, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

****Considerations:****

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

****Fair reciprocation:****

If I return \$3.50:

- They end with: $\$5 - \$2.08 + \$3.50 = \6.42 (profit of \$1.42)
- I end with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)
- This rewards their trust and creates mutual gain

{"return": 3.50}

reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was $\$5 - \$3 + \$5.09 = \7.09

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

****Current Round (Round 2):****

- I am RECEIVER
- My co-player sent me \$2.08
- I received \$6.24 ($2.08 \times 3 = 6.24$)
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. ****Fairness****: A fair split might be sharing the gains equally
2. ****Reciprocity****: Building trust for potential future interactions
3. ****Economic incentive****: They sent \$2.08, I got \$6.24. The gain created is $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool: $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return: $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received (\$5.09/\$9.00), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 946,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

****Considerations:****

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with: \$5 - \$4.18 + \$7 = \$7.82 (profit of \$2.82)
- I end with: \$12.54 - \$7 = \$5.54 (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

{"return": 7}

Reasoning:

Let me analyze this situation:

****Current round:****

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of \$12.54 - \$4.18 = \$8.36

****Previous rounds context:****

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

****Considerations:****

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

****Options:****

If I return \$6:

- They end with: \$5 - \$4.18 + \$6 = \$6.82 (profit of \$1.82)
- I end with: \$12.54 - \$6 = \$6.54 (profit of \$6.54)

If I return \$7:

- They end with: \$5 - \$4.18 + \$7 = \$7.82 (profit of \$2.82)
- I end with: \$12.54 - \$7 = \$5.54 (profit of \$5.54)

If I return \$8:

- They end with: $\$5 - \$4.18 + \$8 = \8.82 (profit of \$3.82)
- I end with: $\$12.54 - \$8 = \$4.54$ (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

```
Usage: {
  "input_tokens": 1143,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-08-12T09:34:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_8"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received $\$3.89 \times 3 = \11.67
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with: $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with: $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total = $\$5 - \$3.89 + \$11.67 = \12.78
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need \$1.11 (what they kept) + what I return = \$6.39
- So I should return: $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received: $\$11.67 \times 0.778 = \9.08
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation: if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

```
{'send': 4.00}
```

reasoning:

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with $\$1.11 + \$5.28 = \$6.39$, and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with $\$1.00 + \$5.40 = \$6.40$
- They'd end up with $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with $\$1.25 + \$5.06 = \$6.31$
- They'd end up with $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

```
{'send': 4.00}
```

Reasoning:

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was $\$5 - \$4 + \$5.20 = \6.20

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

```
Usage: {  
  "input_tokens": 970,  
  "output_tokens": 251,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T09:34:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_8"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver ($3 \times \3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis:****

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning ****\$6.50**** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff: $\$5 - \$3 + \$3.03 = \5.03
- They kept: $\$9 - \$3.03 = \$5.97$

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with: $\$5 - \$4.33 + \text{my_return} = \$0.67 + \text{my_return}$
- I'll end up with: $\$12.99 - \text{my_return}$

If I want us to split relatively evenly:

- Total in the system: $\$5$ (their initial) + $\$12.99$ (what I got) = $\$17.99$
- Fair split would be about \$9 each
- They have \$0.67 before my return
- So I should return about $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
  "input_tokens": 774,
  "output_tokens": 886,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 3 Analysis:****

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:


```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

Reasoning:

****Round 3 Analysis:****

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received: $\$9.87 * 0.5 = \4.935

That would give them:

- Sender total: $\$1.71 + \$4.935 = \$6.645$
- My total: $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total: $\$1.71 + \$5 = \$6.71$
- I keep: $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

```
Usage: {
  "input_tokens": 1053,
  "output_tokens": 544,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

- dyad_1: Agent_8=Sender, Agent_4=Receiver
- dyad_2: Agent_2=Sender, Agent_3=Receiver
- dyad_3: Agent_7=Sender, Agent_5=Receiver
- dyad_4: Agent_1=Sender, Agent_6=Receiver

Parsed game decisions:

dyad_1: Agent_8 sent \$4; received \$12; Agent_4 returned \$7; payoffs Agent_8=\$8, Agent_4=\$5;
noise/visible: sent visible=\$4.48, received visible=\$13.44, returned visible=\$6.14
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_2 sent \$3.5; received \$10.5; Agent_3 returned \$6.24; payoffs Agent_2=\$7.74, Agent_3=\$4.26; noise/visible: sent visible=\$4.37, received visible=\$13.11, returned visible=\$6.95
Action validation: {"returned": {"amount": 6.24, "clamped": false, "max": 10.5, "min": 0, "raw": 6.24}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_3: Agent_7 sent \$3; received \$9; Agent_5 returned \$4; payoffs Agent_7=\$6, Agent_5=\$5;
noise/visible: sent visible=\$2.84, received visible=\$8.52, returned visible=\$3.66

```

    Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 sent $3.25; received $9.75; Agent_6 returned $8; payoffs Agent_1=$9.75,
Agent_6=$1.75; noise/visible: sent visible=$4.14, received visible=$12.42, returned visible=$7.34
    Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 9.75, "min": 0, "raw":
    8.0}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}
Balances after round: {
  "Agent_1": 24.5,
  "Agent_2": 24.740000000000002,
  "Agent_3": 20.979999999999997,
  "Agent_4": 22.5,
  "Agent_5": 22.25,
  "Agent_6": 23.25,
  "Agent_7": 20.78,
  "Agent_8": 25.0
}
Visible balances after round: {
  "Agent_1": 26.279999999999998,
  "Agent_2": 25.26,
  "Agent_3": 25.189999999999998,
  "Agent_4": 22.200000000000003,
  "Agent_5": 19.659999999999997,
  "Agent_6": 23.869999999999997,
  "Agent_7": 21.03,
  "Agent_8": 23.53
}

```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:34:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_4"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_4": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also

don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver ($3 \times \3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis:****

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning ****\$6.50**** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff: $\$5 - \$3 + \$3.03 = \5.03

- They kept: $\$9 - \$3.03 = \$5.97$

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with: $\$5 - \$4.33 + \text{my_return} = \$0.67 + \text{my_return}$
- I'll end up with: $\$12.99 - \text{my_return}$

If I want us to split relatively evenly:

- Total in the system: $\$5$ (their initial) + $\$12.99$ (what I got) = $\$17.99$
- Fair split would be about \$9 each
- They have \$0.67 before my return
- So I should return about $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
  "input_tokens": 774,
  "output_tokens": 886,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis:****

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our coop

eration.

reasoning:

****Round 3 Analysis:****

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received: $\$9.87 * 0.5 = \4.935

That would give them:

- Sender total: $\$1.71 + \$4.935 = \$6.645$
- My total: $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total: $\$1.71 + \$5 = \$6.71$
- I keep: $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 544,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 4 Analysis:****

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send ****\$4**** to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

Reasoning:

****Round 4 Analysis:****

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

```
Usage: {  
  "input_tokens": 1305,  
  "output_tokens": 469,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-08-12T09:35:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_4"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: \$2.51 + \$3.50 = \$6.01 (profit of ~\$1)
- I keep: \$7.47 - \$3.50 = \$3.97 (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

{"return": 3.50}

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received \$2.49 × 3 = \$7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Going higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1072,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became \$7.470000000000001, and you returned \$3.5. Your visible payoff was \$3.9700000000000006.

Your total visible earnings across all rounds are \$3.9700000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round 1.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

reasoning:

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff : $\$2.50 + \$3.93 = \$6.43$

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

usage:

```
{
  "input_tokens": 1153,
  "output_tokens": 447,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with: $\$0.52 + \$7.00 = \$7.52$ (profit of ~\$2.52)
- I keep: $\$13.44 - \$7.00 = \$6.44$ (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

```
{"return": 7.00}
```

Reasoning:

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with: $\$0.52 + \$6.50 = \$7.02$ (profit of ~\$2)
- I would keep: $\$13.44 - \$6.50 = \$6.94$ (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with: $\$0.52 + \$7 = \$7.52$ (profit of \$2.52)
- I keep: $\$13.44 - \$7 = \$6.44$ (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be: $\$13.44 * 0.5 = \6.72

...

```
Usage: {
  "input_tokens": 1412,
  "output_tokens": 682,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-08-12T09:35:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_3"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with: $\$1.66 + \$5.00 = \$6.66$ (a profit of \$1.66)
- I end with: $\$10.02 - \$5.00 = \$5.02$ (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 (\$5 - \$3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.34 + \$0 = \1.66
- I get: $\$10.02 - \$0 = \$10.02$
- Total: \$11.68

If I return \$5:

- Sender gets: $\$5 - \$3.34 + \$5 = \6.66
- I get: $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return: $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets: $\$1.66 + \$4.18 = \$5.84$
- I get: $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1155,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

```
{'send': 3.50}
```

reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

usage:

```
{
  "input_tokens": 846,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with: \$2.48 + \$4.00 = \$6.48 (profit of \$1.48)
- I end with: \$7.56 - \$4.00 = \$3.56 (profit of \$3.56)

This rewards their trust and maintains cooperation.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication

- The sender currently has \$5 - \$2.52 = \$2.48 left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again: $\$7.56 \times 0.5 = \3.78

Let me check the outcomes:

- If I return \$3.78:
 - Sender ends with: $\$2.48 + \$3.78 = \$6.26$ (profit of \$1.26)
 - I end with: $\$7.56 - \$3.78 = \$3.78$ (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with: $\$2.48 + \$4.00 = \$6.48$ (profit of \$1.48)
- I end with: $\$7.56 - \$4.00 = \$3.56$ (profit of \$3.56)

...

usage:

```
{
  "input_tokens": 1072,
  "output_tokens": 643,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

```
{'send': 3.50}
```

Reasoning:

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

Usage: {

```
"input_tokens": 1400,
"output_tokens": 305,
"reasoning_tokens": 0
}
```



```
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T09:35:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_2",

"move_index": 3,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_2",

"Agent_3"

],

"investor": "Agent_2",

"roles": {

"Agent_2": "investor",

"Agent_3": "trustee"

},

"trustee": "Agent_3"

},

"role": "trustee",

"role_label": "Receiver",

"round": 4,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received $\$3.89 \times 3 = \11.67
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with: $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with: $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total = $\$5 - \$3.89 + \$11.67 = \12.78
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need $\$1.11$ (what they kept) + what I return = \$6.39
- So I should return: $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received: $\$11.67 \times 0.778 = \9.08
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation: if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$

6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

{'send': 4.00}

reasoning:

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with $\$1.11 + \$5.28 = \$6.39$, and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with $\$1.00 + \$5.40 = \$6.40$
- They'd end up with $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with $\$1.25 + \$5.06 = \$6.31$
- They'd end up with $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

```
{'send': 4.00}
```

reasoning:

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was $\$5 - \$4 + \$5.20 = \6.20

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

usage:

```
{  
  "input_tokens": 970,  
  "output_tokens": 251,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created: $\$0.63$ (with sender) + $\$13.11$ (with me) = $\$13.74$
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

```
{'return': 6.24}
```

Reasoning:

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has $\$5 - \$4.37 = \$0.63$

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created: $\$0.63$ (sender has) + $\$13.11$ (I have) = $\$13.74$

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have: $\$0.63 + \text{return} = \6.87 , so $\text{return} = \$6.24$
- I would have: $\$13.11 - \$6.24 = \$6.87$

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00

- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

```
Usage: {  
  "input_tokens": 1157,  
  "output_tokens": 492,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:35:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_5"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent/returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is $\$5 + 2X$, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

****Considerations:****

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

****Fair reciprocation:****

If I return \$3.50:

- They end with: $\$5 - \$2.08 + \$3.50 = \6.42 (profit of \$1.42)
- I end with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)
- This rewards their trust and creates mutual gain

{"return": 3.50}

reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was $\$5 - \$3 + \$5.09 = \7.09

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

****Current Round (Round 2):****

- I am RECEIVER

- My co-player sent me \$2.08
- I received \$6.24 ($2.08 \times 3 = 6.24$)
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. **Fairness**: A fair split might be sharing the gains equally
2. **Reciprocity**: Building trust for potential future interactions
3. **Economic incentive**: They sent \$2.08, I got \$6.24. The gain created is $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool: $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return: $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ($\$5.09/\9.00), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 946,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Analysis:

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

Considerations:

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

```
{"return": 7}
```

reasoning:

Let me analyze this situation:

****Current round:****

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of $\$12.54 - \$4.18 = \$8.36$

****Previous rounds context:****

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

****Considerations:****

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

****Options:****

If I return \$6:

- They end with: $\$5 - \$4.18 + \$6 = \6.82 (profit of \$1.82)
- I end with: $\$12.54 - \$6 = \$6.54$ (profit of \$6.54)

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)

If I return \$8:

- They end with: $\$5 - \$4.18 + \$8 = \8.82 (profit of \$3.82)
- I end with: $\$12.54 - \$8 = \$4.54$ (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

usage:

```
{
  "input_tokens": 1143,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

****Previous sending experience:****

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

****Current situation:****

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

****Strategy:****

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

```
{"send": 3}
```

Reasoning:

Let me analyze the situation:

****Previous experience:****

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

****Current situation:****

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

****Strategic considerations:****

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

****Options:****

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows:

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

```
Usage: {  
  "input_tokens": 1469,  
  "output_tokens": 529,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T09:35:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_5"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that c

ould benefit both of us if the receiver returns some. This is the classic trust game set up.

2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: $\$5 - X + (\text{amount returned})$
 - Receiver's final payoff: $\$3X - (\text{amount returned})$
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 652,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 2 - RECEIVER Decision****

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
 - They end up with: $\$2.18 + \$4.00 = \$6.18$
 - I end up with: $\$8.46 - \$4.00 = \$4.46$

- Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 recap:****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff: $\$5 - \$3 + \$2.93 = \4.93
- Their payoff: $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

****Round 2:****

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept: $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie: $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return: $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return: $\$8.46 \times 0.33 = \2.79

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

usage:

```
{
  "input_tokens": 775,
  "output_tokens": 861,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Round 3 - SENDER Decision****

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

usage:

```
{
  "input_tokens": 1105,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 4 - RECEIVER Decision****

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier

- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

Reasoning:

****Round 4 - RECEIVER Decision****

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have $\$5 - \$2.84 = \$2.16$ remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately.

```
Usage: {
  "input_tokens": 1398,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-08-12T09:35:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_6"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep: \$9.78 - \$4.00 = \$5.78
- Sender ends with: \$1.74 + \$4.00 = \$5.74

This seems fair given their generous initial trust.

{'return': 4.00}

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is $3 \times \$3.26$)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($\$5 - \3.26)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps: $\$5 - \$3.26 = \$1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):

- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):

- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:

- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 798,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

usage:

```
{
  "input_tokens": 815,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

{'return': 3.50}

reasoning:

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has \$5 - \$2.78 = \$2.22 remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep: \$8.34 - \$3.50 = \$4.84
- Sender ends with: \$2.22 + \$3.50 = \$5.72

If I return \$3.00:

- I keep: \$8.34 - \$3.00 = \$5.34
- Sender ends with: \$2.22 + \$3.00 = \$5.22

If I return \$4.00:

- I keep: \$8.34 - \$4.00 = \$4.34
- Sender ends with: \$2.22 + \$4.00 = \$6.22

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% se

nd, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

usage:

```
{
  "input_tokens": 960,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

```
{'send': 3.25}
```

Reasoning:

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with: $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff: $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with: $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

Usage: {

```
"input_tokens": 1093,
"output_tokens": 468,
"reasoning_tokens": 0
```

```
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:36:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_4",

"move_index": 7,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_1",

"Agent_6"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_6": "trustee"

},

"trustee": "Agent_6"

},

"role": "trustee",

"role_label": "Receiver",

"round": 4,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

****My reasoning:****

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send ****\$3****. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

```
{"send": 3}
```

reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

****If I send nothing (\$0):****

- I keep: \$5 - \$0 + \$0 = \$5
- Receiver gets: \$0 × 3 = \$0
- Total payoff: \$5

****If I send everything (\$5):****

- Receiver gets: \$5 × 3 = \$15
- If they return half (\$7.50), I get: \$5 - \$5 + \$7.50 = \$7.50
- If they return nothing, I get: \$5 - \$5 + \$0 = \$0
- Maximum total payoff possible: \$20 if I send \$5

****Optimal social outcome:****

If I send \$5, the total pie becomes \$20 (\$5 I keep initially - \$5 sent + \$15 receiver gets). If the receiver returns \$7.50, we each get \$7.50, which is better than the \$5 I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 761,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 2

  In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $5.74 to you. Your visible payoff was $7.74.
  Your total visible earnings across all rounds are $7.74.

  This round, you are the RECEIVER against your current co-player. They sent you $3.25, so you received $9.75. How much do you return? (0-9.75)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  Looking at this situation:

  **Analysis:**
  - My co-player sent $3.25 (keeping $1.75)
  - I received $9.75
  - In Round 1, when I sent $3, the receiver returned $5.74 (about 64% of the $9 they received)
  - I want to establish reciprocity while being fair

  **My decision:**
  Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

  If I return **$6**:
  - Sender ends with: $1.75 + $6 = $7.75 (gain of $2.75)
  - I end with: $9.75 - $6 = $3.75 (gain of $3.75)

  Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

  {"return": 6}
reasoning:
  Let me analyze this situation:

  Previous round (Round 1):
  - I was the SENDER
  - I sent $3.0
  - It became $9.0 for the receiver
  - They returned $5.74 to me
  - My payoff was $5 - $3 + $5.74 = $7.74
  - The receiver kept $9.0 - $5.74 = $3.26

  So the previous receiver returned about 64% of what they received ($5.74/$9.0).

  Current round (Round 2):
  - I am the RECEIVER
  - My co-player (different from Round 1, as noted in the instructions) sent me $3.25
  - I received $9.75
  - I need to decide how much to return

  Key considerations:
  1. This is a different co-player than Round 1
  2. They sent me $3.25 (slightly more than I sent in Round 1)
  3. There's communication noise, so amounts may differ slightly
  4. I want to establish reciprocity and encourage future cooperation
  5. The previous receiver in Round 1 returned about 64% of what they received

  Strategy:
  - I should be fair and reciprocal
  - Returning a similar proportion to what was returned to me seems reasonable
  - If I return too little, it might discourage future cooperation
  - If I return too much, I'm being overly generous

  The sender sent $3.25 and kept $1.75. If I return around 60-65% of the $9.75:

```

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets $\$1.75 + \$6 = \$7.75$, I keep $\$9.75 - \$6 = \$3.75$
- If I return \$5.85: Sender gets $\$1.75 + \$5.85 = \$7.60$, I keep $\$9.75 - \$5.85 = \$3.90$
- If I return \$5: Sender gets $\$1.75 + \$5 = \$6.75$, I keep $\$9.75 - \$5 = \$4.75$

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

usage:

```
{
  "input_tokens": 794,
  "output_tokens": 1093,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

****Analysis:****

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

****My decision:****

Based on the positive reciprocity I've seen, I'll send ****\$3.50**** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

```
{"send": 3.5}
```

reasoning:

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0
- Received back \$5.74
- Payoff: $\$5 - \$3 + \$5.74 = \7.74

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff: $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests

sts:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

****My decision:****

Given their high trust, I want to reciprocate generously. Returning about 64% would be ****\$8****.

If I return \$8:

- Sender ends with: $\$0.86 + \$8 = \$8.86$ (net gain of \$3.86)
- I end with: $\$12.42 - \$8 = \$4.42$ (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

{"return": 8}

Reasoning:

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets $\$0.86 + \$7.50 = \$8.36$ (gain of \$3.36), I get $\$12.42 - \$7.50 = \$4.92$ (gain of \$4.92)
- If I return \$8: Sender gets $\$0.86 + \$8 = \$8.86$ (gain of \$3.86), I get $\$12.42 - \$8 = \$4.42$ (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

```
Usage: {
  "input_tokens": 1438,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_8=Sender, Agent_7=Receiver
dyad_2: Agent_2=Sender, Agent_4=Receiver
dyad_3: Agent_1=Sender, Agent_3=Receiver
dyad_4: Agent_6=Sender, Agent_5=Receiver

Parsed game decisions:

dyad_1: Agent_8 sent \$4; received \$12; Agent_7 returned \$5.5; payoffs Agent_8=\$6.5, Agent_7=\$6.5; noise/visible: sent visible=\$3.4, received visible=\$10.2, returned visible=\$5.84
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_2 sent \$3.5; received \$10.5; Agent_4 returned \$6.5; payoffs Agent_2=\$8, Agent_4=\$4; noise/visible: sent visible=\$4.28, received visible=\$12.84
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_3: Agent_1 sent \$3.25; received \$9.75; Agent_3 returned \$5.9; payoffs Agent_1=\$7.65, Agent_3=\$3.85; noise/visible: sent visible=\$4.2, received visible=\$12.6, returned visible=\$6.55
Action validation: {"returned": {"amount": 5.9, "clamped": false, "max": 9.75, "min": 0, "raw": 5.9}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}

dyad_4: Agent_6 sent \$4; received \$12; Agent_5 returned \$6.5; payoffs Agent_6=\$7.5, Agent_5=\$5.5; noise/visible: sent visible=\$4.32, received visible=\$12.96, returned visible=\$6.31
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 32.15,
"Agent_2": 32.74,
"Agent_3": 24.83,
"Agent_4": 26.5,
"Agent_5": 27.75,
"Agent_6": 30.75,
"Agent_7": 27.28,
"Agent_8": 31.5

}

Visible balances after round: {

"Agent_1": 34.58,
"Agent_2": 33.260000000000005,
"Agent_3": 31.89,
"Agent_4": 28.540000000000003,
"Agent_5": 26.119999999999997,
"Agent_6": 31.179999999999996,
"Agent_7": 25.73,
"Agent_8": 30.37

```
}
```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:36:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver (3 × \$3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

```
...
usage:
{
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis:****

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning ****\$6.50**** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff: \$5 - \$3 + \$3.03 = \$5.03
- They kept: \$9 - \$3.03 = \$5.97

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with: \$5 - \$4.33 + my_return = \$0.67 + my_return
- I'll end up with: \$12.99 - my_return

If I want us to split relatively evenly:

- Total in the system: \$5 (their initial) + \$12.99 (what I got) = \$17.99
- Fair split would be about \$9 each
- They have \$0.67 before my return
- So I should return about \$9 - \$0.67 = \$8.33

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
  "input_tokens": 774,
  "output_tokens": 886,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis:****

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

reasoning:

****Round 3 Analysis:****

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received: $\$9.87 * 0.5 = \4.935

That would give them:

- Sender total: $\$1.71 + \$4.935 = \$6.645$
- My total: $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total: \$1.71 + \$5 = \$6.71
- I keep: \$9.87 - \$5 = \$4.87

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 544,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 4 Analysis:****

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send ****\$4**** to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

reasoning:

****Round 4 Analysis:****

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$

6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

usage:

```
{
  "input_tokens": 1305,
  "output_tokens": 469,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 5 Analysis:****

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send ****\$4**** again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

Reasoning:

****Round 5 Analysis:****

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)
- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

Usage: {

```
"input_tokens": 1557,
"output_tokens": 384,
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-08-12T09:36:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent /returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is \$5 + 2X, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

****Considerations:****

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

****Fair reciprocation:****

If I return \$3.50:

- They end with: \$5 - \$2.08 + \$3.50 = \$6.42 (profit of \$1.42)
- I end with: \$6.24 - \$3.50 = \$2.74 (profit of \$2.74)
- This rewards their trust and creates mutual gain

{"return": 3.50}

reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was \$5 - \$3 + \$5.09 = \$7.09

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

****Current Round (Round 2):****

- I am RECEIVER
- My co-player sent me \$2.08
- I received \$6.24 ($2.08 \times 3 = 6.24$)
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. ****Fairness****: A fair split might be sharing the gains equally
2. ****Reciprocity****: Building trust for potential future interactions
3. ****Economic incentive****: They sent \$2.08, I got \$6.24. The gain created is \$6.24 - \$2.08 = \$4.16

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool: $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return: $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ($\$5.09/\9.00), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 946,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

****Considerations:****

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

{"return": 7}

reasoning:

Let me analyze this situation:

****Current round:****

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of $\$12.54 - \$4.18 = \$8.36$

****Previous rounds context:****

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

****Considerations:****

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

****Options:****

If I return \$6:

- They end with: $\$5 - \$4.18 + \$6 = \6.82 (profit of \$1.82)
- I end with: $\$12.54 - \$6 = \$6.54$ (profit of \$6.54)

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)

If I return \$8:

- They end with: $\$5 - \$4.18 + \$8 = \8.82 (profit of \$3.82)
- I end with: $\$12.54 - \$8 = \$4.54$ (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

usage:

```
{
  "input_tokens": 1143,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

****Previous sending experience:****

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

****Current situation:****

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

****Strategy:****

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

{"send": 3}

reasoning:

Let me analyze the situation:

****Previous experience:****

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

****Current situation:****

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

****Strategic considerations:****

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

****Options:****

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 529,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

****Current situation:****

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

****Considerations:****

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)
- Both parties gain significantly

This rewards their trust while creating mutual benefit.

```
{"return": 5.50}
```

Reasoning:

****Analysis:****

****Previous round:****

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received
- This was fair but not overly generous

****Current situation:****

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20

- This created a gain of \$6.80

****Considerations:****

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with: $\$5 - \$3.40 + \$5.50 = \7.10 (profit of \$2.10)
- I end with: $\$10.20 - \$5.50 = \$4.70$ (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

```
Usage: {  
  "input_tokens": 1755,  
  "output_tokens": 512,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-08-12T09:36:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_4"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with: $\$1.66 + \$5.00 = \$6.66$ (a profit of \$1.66)
- I end with: $\$10.02 - \$5.00 = \$5.02$ (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

```
{"return": 5.00}
```

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 (\$5 - \$3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.34 + \$0 = \1.66
- I get: $\$10.02 - \$0 = \$10.02$

- Total: \$11.68

If I return \$5:

- Sender gets: $\$5 - \$3.34 + \$5 = \6.66
- I get: $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return: $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets: $\$1.66 + \$4.18 = \$5.84$
- I get: $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1155,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

```
{'send': 3.50}
```

reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

usage:

```
{
  "input_tokens": 846,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with: $\$2.48 + \$4.00 = \$6.48$ (profit of \$1.48)
- I end with: $\$7.56 - \$4.00 = \$3.56$ (profit of \$3.56)

This rewards their trust and maintains cooperation.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has $\$5 - \$2.52 = \$2.48$ left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again: $\$7.56 * 0.5 = \3.78

Let me check the outcomes:

- If I return \$3.78:
 - Sender ends with: $\$2.48 + \$3.78 = \$6.26$ (profit of \$1.26)
 - I end with: $\$7.56 - \$3.78 = \$3.78$ (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with: $\$2.48 + \$4.00 = \$6.48$ (profit of \$1.48)
- I end with: $\$7.56 - \$4.00 = \$3.56$ (profit of \$3.56)

...

usage:

```
{
  "input_tokens": 1072,
  "output_tokens": 643,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

```
{'send': 3.50}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 305,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was

\$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it.

{'send': 3.50}

Reasoning:

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well. I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

```
Usage: {
  "input_tokens": 1587,
  "output_tokens": 211,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:36:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_4"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_4": "trustee"
    }
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: \$2.51 + \$3.50 = \$6.01 (profit of ~\$1)
- I keep: \$7.47 - \$3.50 = \$3.97 (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

{"return": 3.50}

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received \$2.49 × 3 = \$7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust

2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has $\$2.51$, I have $\$7.47$. Total is $\$10$. Equal would be $\$5$ each, so I'd return $\$2.49$ to give them $\$5$ total.
- Return what was sent: Return $\$2.49$
- Return the multiplied amount divided equally: Return around $\$3.735$ (half of what I received)
- Maximize my own payoff: Return $\$0$ (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing ($\$2.51$)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return $\$2.50$:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning $\$3.50$ would give the sender $\$6.01$ total (a $\$1.01$ gain) while I'd have $\$3.97$. Going higher to $\$4.00$ means they'd end up with $\$6.51$ and I'd keep $\$3.47$. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The $\$4.98$ bonus from the tripling could be divided equally-giving them $\$5.00$ and me $\$4.98$

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1072,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$2.49$, it became $\$7.470000000000001$, and you returned $\$3.5$. Your visible payoff was $\$3.9700000000000006$.

Your total visible earnings across all rounds are $\$3.9700000000000006$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% ($\$2.49$), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send $\$2.50$ (50% of my endowment), which mirrors the trust level shown to me in Round 1.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

reasoning:

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97

- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff : $\$2.50 + \$3.93 = \$6.43$

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

usage:

```
{
  "input_tokens": 1153,
  "output_tokens": 447,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with: $\$0.52 + \$7.00 = \$7.52$ (profit of ~\$2.52)
- I keep: $\$13.44 - \$7.00 = \$6.44$ (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with: $\$0.52 + \$6.50 = \$7.02$ (profit of ~\$2)
- I would keep: $\$13.44 - \$6.50 = \$6.94$ (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with: $\$0.52 + \$7 = \$7.52$ (profit of \$2.52)
- I keep: $\$13.44 - \$7 = \$6.44$ (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be: $\$13.44 * 0.5 = \6.72

...

usage:

```
{
  "input_tokens": 1412,
  "output_tokens": 682,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001.

Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with: $\$0.72 + \$6.50 = \$7.22$ (profit of ~\$2.22)
- I keep: $\$12.84 - \$6.50 = \$6.34$ (profit of ~\$6.34)

This maintains a similar return rate (~51%) as Round 4 (~52%) and both of us profit well.

```
{"return": 6.50}
```

Reasoning:

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocal pattern for high trust.

The sender showed ~86% trust level, which is slightly less than the ~90% in Round 4, but still very high.

If I return \$6.50:

- Sender ends with: $\$0.72 + \$6.50 = \$7.22$ (profit of ~\$2.22)
- I keep: $\$12.84 - \$6.50 = \$6.34$ (profit of ~\$6.34)

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

```
Usage: {  
  "input_tokens": 1745,  
  "output_tokens": 516,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-08-12T09:36:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_3"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_3": "trustee"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep: \$9.78 - \$4.00 = \$5.78
- Sender ends with: \$1.74 + \$4.00 = \$5.74

This seems fair given their generous initial trust.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is 3.26×3)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($5 - 3.26$)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps: $5 - 3.26 = 1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):
- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):
- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:
- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

usage:

```
{  
  "input_tokens": 529,  
  "output_tokens": 798,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

usage:

```
{
```

```
"input_tokens": 815,  
"output_tokens": 324,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

```
{'return': 3.50}
```

reasoning:

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has \$5 - \$2.78 = \$2.22 remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep: \$8.34 - \$3.50 = \$4.84
- Sender ends with: \$2.22 + \$3.50 = \$5.72

If I return \$3.00:

- I keep: \$8.34 - \$3.00 = \$5.34
- Sender ends with: \$2.22 + \$3.00 = \$5.22

If I return \$4.00:

- I keep: \$8.34 - \$4.00 = \$4.34
- Sender ends with: \$2.22 + \$4.00 = \$6.22

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

usage:

```
{  
  "input_tokens": 960,  
  "output_tokens": 465,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

{'send': 3.25}

reasoning:

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with: $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff: $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with: $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.

Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

{'send': 3.25}

Reasoning:

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78
Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57
Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84
Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)
- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).

Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

```
Usage: {  
  "input_tokens": 1229,  
  "output_tokens": 282,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-08-12T09:36:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_3"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_3": "trustee"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

```
{'return': 5.28}
```

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received $\$3.89 \times 3 = \11.67
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with: $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with: $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total = $\$5 - \$3.89 + \$11.67 = \12.78
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need \$1.11 (what they kept) + what I return = \$6.39
- So I should return: $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received: $\$11.67 \times 0.778 = \9.08
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation : if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

```
{'send': 4.00}
```

reasoning:

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with $\$1.11 + \$5.28 = \$6.39$, and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with $\$1.00 + \$5.40 = \$6.40$
- They'd end up with $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with $\$1.25 + \$5.06 = \$6.31$
- They'd end up with $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

usage:

```
{
  "input_tokens": 787,
```



```
    "output_tokens": 478,  
    "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

```
{ 'send': 4.00 }
```

reasoning:

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was $\$5 - \$4 + \$5.20 = \6.20

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

usage:

```
{  
  "input_tokens": 970,  
  "output_tokens": 251,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created: $\$0.63$ (with sender) + $\$13.11$ (with me) = $\$13.74$
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

```
{'return': 6.24}
```

reasoning:

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has \$5 - \$4.37 = \$0.63

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created: \$0.63 (sender has) + \$13.11 (I have) = \$13.74

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have: \$0.63 + return = \$6.87, so return = \$6.24
- I would have: \$13.11 - \$6.24 = \$6.87

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

usage:

```
{
  "input_tokens": 1157,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)
- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

```
{'return': 5.90}
```

Reasoning:

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender: $\$0.80 + \$6.30 = \$7.10$
- Me: $\$12.60 - \$6.30 = \$6.30$

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return: $\$13.40 - \$6.70 = \$6.70$

- Sender would get: $\$0.80 + \$6.70 = \$7.50$
- I would get: $\$12.60 - \$6.70 = \$5.90$

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets: $\$0.80 + X$
- I get: $\$12.60 - X$

For equal outcomes:

$$\$0.80 + X = \$12.60 - X$$

$$2X = \$12.60 - \$0.80$$

$$2X = \$11.80$$

$$X = \$5.90$$

Check:

- Sender: $\$0.80 + \$5.90 = \$6.70$
- Me: $\$12.60 - \$5.90 = \$6.70$

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

```
Usage: {  
  "input_tokens": 1423,  
  "output_tokens": 548,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-08-12T09:37:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

****My reasoning:****

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send ****\$3****. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

{"send": 3}

reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

****If I send nothing (\$0):****

- I keep: $\$5 - \$0 + \$0 = \5
- Receiver gets: $\$0 \times 3 = \0
- Total payoff: \$5

****If I send everything (\$5):****

- Receiver gets: $\$5 \times 3 = \15
- If they return half (\$7.50), I get: $\$5 - \$5 + \$7.50 = \7.50
- If they return nothing, I get: $\$5 - \$5 + \$0 = \0
- Maximum total payoff possible: \$20 if I send \$5

****Optimal social outcome:****

If I send \$5, the total pie becomes \$20 (\$5 I keep initially - \$5 sent + \$15 receiver gets). If the receiver returns \$7.50, we each get \$7.50, which is better than the \$5 I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.74 to you. Your visible payoff was \$7.74.

Your total visible earnings across all rounds are \$7.74.

This round, you are the RECEIVER against your current co-player. They sent you \$3.25, so you received \$9.75. How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$3.25 (keeping \$1.75)
- I received \$9.75
- In Round 1, when I sent \$3, the receiver returned \$5.74 (about 64% of the \$9 they received)
- I want to establish reciprocity while being fair

****My decision:****

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return ****\$6****:

- Sender ends with: $\$1.75 + \$6 = \$7.75$ (gain of \$2.75)
- I end with: $\$9.75 - \$6 = \$3.75$ (gain of \$3.75)

Both of us benefit significantly, and this return rate is consistent with what was shown to

o me in Round 1.

```
{"return": 6}
```

reasoning:

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was $\$5 - \$3 + \$5.74 = \7.74
- The receiver kept $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ($\$5.74/\9.0).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation
- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets $\$1.75 + \$6 = \$7.75$, I keep $\$9.75 - \$6 = \$3.75$
- If I return \$5.85: Sender gets $\$1.75 + \$5.85 = \$7.60$, I keep $\$9.75 - \$5.85 = \$3.90$
- If I return \$5: Sender gets $\$1.75 + \$5 = \$6.75$, I keep $\$9.75 - \$5 = \$4.75$

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

usage:

```
{
  "input_tokens": 794,
  "output_tokens": 1093,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

****Analysis:****

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

****My decision:****

Based on the positive reciprocity I've seen, I'll send ****\$3.50**** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

```
{"send": 3.5}
```

reasoning:

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0
- Received back \$5.74
- Payoff: $\$5 - \$3 + \$5.74 = \7.74

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff: $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)
Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

****My decision:****

Given their high trust, I want to reciprocate generously. Returning about 64% would be ****\$8****.

If I return \$8:

- Sender ends with: $\$0.86 + \$8 = \$8.86$ (net gain of \$3.86)
- I end with: $\$12.42 - \$8 = \$4.42$ (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

{"return": 8}

reasoning:

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets $\$0.86 + \$7.50 = \$8.36$ (gain of \$3.36), I get $\$12.42 - \$7.50 = \$4.92$ (gain of \$4.92)
- If I return \$8: Sender gets $\$0.86 + \$8 = \$8.86$ (gain of \$3.86), I get $\$12.42 - \$8 = \$4.42$ (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

usage:

```
{
  "input_tokens": 1438,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

****Analysis:****

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

****My decision:****

Given the strong pattern of reciprocity, I'll send ****\$4**** to show trust and create a \$12 pot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

{"send": 4}

Reasoning:

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)
- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

```
Usage: {  
  "input_tokens": 1763,  
  "output_tokens": 479,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-08-12T09:37:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

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  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    }  
  },
```

```

    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

ure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game set up.
2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: $\$5 - X + (\text{amount returned})$
 - Receiver's final payoff: $\$3X - (\text{amount returned})$
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 652,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 2 - RECEIVER Decision****

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
 - They end up with: $\$2.18 + \$4.00 = \$6.18$
 - I end up with: $\$8.46 - \$4.00 = \$4.46$
 - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 recap:****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff: $\$5 - \$3 + \$2.93 = \4.93
- Their payoff: $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

****Round 2:****

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept: $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie: $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return: $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return: $\$8.46 \times 0.33 = \2.79

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

usage:

```
{
  "input_tokens": 775,
  "output_tokens": 861,
```

```

    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $2.82, it became $8.459999999999999, and you returned $4.0. Your visible p
  ayoff was $4.459999999999999.
  Your total visible earnings across all rounds are $9.389999999999999.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
content:
  **Round 3 - SENDER Decision**

  I'm the sender with $5 facing a new co-player. Looking at my experience:
  - Round 1: Sent $3, received ~$2.93 back (reasonable cooperation)
  - Round 2: As receiver, got $2.82 and returned $4.00 (fair exchange)

  Both previous interactions showed decent cooperation. Since I'm with a new co-player and c
  ooperation has been working, I'll continue with a similar strategy.

  Sending $3 creates $9 for them to work with, which allows for mutual benefit if they recip
  rocate reasonably.

  '''json
  {"send": 3}
  '''
reasoning:
  Let me analyze my situation:

  Round 1: I was SENDER, sent $3, received $2.93 back, payoff was $4.93
  Round 2: I was RECEIVER, received $8.46 (from $2.82 sent), returned $4.00, payoff was $4.4
  6
  Total visible earnings: $9.39

  Now I'm SENDER again in Round 3 with a new co-player.

  From my experience:
  - Round 1: I sent $3, got $2.93 back - this was roughly fair (they kept $6.07 from $9, I e
  nded with $4.93)
  - Round 2: My co-player sent $2.82 (good trust), I returned $4.00 which gave them $6.18 to
  tal and me $4.46

  Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this rou
  nd and don't know their behavior, I should continue with a cooperative strategy.

  Sending $3 worked reasonably well in Round 1. I could:
  - Send $3 again as a consistent strategy
  - Send more to show high cooperation
  - Send less to be more cautious

  Given that cooperation has generally worked so far, I'll stick with sending $3 as a baseli
  ne cooperative move.
usage:
  {
    "input_tokens": 1105,
    "output_tokens": 458,
    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 4

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $3.0, it became $9.0, and you saw them return $3.75 to you. Your visible payoff was $
  5.75.

```

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Round 4 - RECEIVER Decision****

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

reasoning:

****Round 4 - RECEIVER Decision****

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have $\$5 - \$2.84 = \$2.16$ remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 5 - RECEIVER Decision****

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything. I should reciprocate proportionally:

- If I return \$6.50:
 - They end up with: $\$0.68 + \$6.50 = \$7.18$ (gain of \$2.18)
 - I end up with: $\$12.96 - \$6.50 = \$6.46$ (gain of \$1.46)
 - Both benefit substantially, rewarding their exceptional trust

```
'''json
{"return": 6.5}
'''
```

Reasoning:

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust
I received \$12.96 after the 3x multiplier
They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount. Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
 - They end up with: $\$0.68 + \$6.00 = \$6.68$
 - I end up with: $\$12.96 - \$6.00 = \$6.96$
 - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
 - They end up with: $\$0.68 + \$7.00 = \$7.68$
 - I end up with: $\$12.96 - \$7.00 = \$5.96$
 - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with: $\$0.68 + \$6.50 = \$7.18$ (gain of \$2.18)
- I end up with: $\$12.96 - \$6.50 = \$6.46$ (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

```
Usage: {
  "input_tokens": 1715,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Sender, Agent_4=Sender, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_4=Sender, Agent_6=Receiver
dyad_2: Agent_7=Sender, Agent_1=Receiver
dyad_3: Agent_5=Sender, Agent_8=Receiver
dyad_4: Agent_3=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_4 sent \$2.5; received \$7.5; Agent_6 returned \$4.6; payoffs Agent_4=\$7.1, Agent_6=\$2.9; noise/visible: sent visible=\$2.47, received visible=\$7.41, returned visible=\$4.76
Action validation: {"returned": {"amount": 4.6, "clamped": false, "max": 7.5, "min": 0, "raw": 4.6}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_7 sent \$3; received \$9; Agent_1 returned \$2.75; payoffs Agent_7=\$4.75,

```

Agent_1=$6.25; noise/visible: sent visible=$2.04, received visible=$6.12, returned visible=$2.32
Action validation: {"returned": {"amount": 2.75, "clamped": false, "max": 9.0, "min": 0, "raw": 2.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent $3; received $9; Agent_8 returned $3.5; payoffs Agent_5=$5.5, Agent_8=$5.5;
noise/visible: sent visible=$2.3, received visible=$6.9, returned visible=$3.38
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_3 sent $4; received $12; Agent_2 returned $7; payoffs Agent_3=$8, Agent_2=$5;
noise/visible: sent visible=$4.4, received visible=$13.2, returned visible=$7.91
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 38.4,
  "Agent_2": 37.74,
  "Agent_3": 32.83,
  "Agent_4": 33.6,
  "Agent_5": 33.25,
  "Agent_6": 33.65,
  "Agent_7": 32.03,
  "Agent_8": 37.0
}
Visible balances after round: {
  "Agent_1": 37.949999999999996,
  "Agent_2": 39.460000000000001,
  "Agent_3": 40.8,
  "Agent_4": 35.800000000000004,
  "Agent_5": 31.499999999999996,
  "Agent_6": 33.989999999999995,
  "Agent_7": 30.05,
  "Agent_8": 33.77
}

```

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-08-12T09:37:29
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_6"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: \$2.51 + \$3.50 = \$6.01 (profit of ~\$1)
- I keep: \$7.47 - \$3.50 = \$3.97 (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

{"return": 3.50}

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)

- I received $\$2.49 \times 3 = \7.47
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept $\$5 - \$2.49 = \$2.51$ if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets: $\$2.51 + \$2.50 = \$5.01$ (almost their original endowment back)
- I keep: $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Going higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1072,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became \$7.470000000000001, and you returned \$3.5. Your visible payoff was \$3.9700000000000006.

Your total visible earnings across all rounds are \$3.9700000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round

1.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

reasoning:

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff: $\$2.50 + \$3.93 = \$6.43$

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

usage:

```
{
  "input_tokens": 1153,
  "output_tokens": 447,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with: $\$0.52 + \$7.00 = \$7.52$ (profit of ~\$2.52)
- I keep: $\$13.44 - \$7.00 = \$6.44$ (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication

- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with: $\$0.52 + \$6.50 = \$7.02$ (profit of ~\$2)
- I would keep: $\$13.44 - \$6.50 = \$6.94$ (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with: $\$0.52 + \$7 = \$7.52$ (profit of \$2.52)
- I keep: $\$13.44 - \$7 = \$6.44$ (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be: $\$13.44 * 0.5 = \6.72

...

usage:

```
{
  "input_tokens": 1412,
  "output_tokens": 682,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001. Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with: $\$0.72 + \$6.50 = \$7.22$ (profit of ~\$2.22)
- I keep: $\$12.84 - \$6.50 = \$6.34$ (profit of ~\$6.34)

This maintains a similar return rate (~51%) as Round 4 (~52%) and both of us profit well.

```
{"return": 6.50}
```

reasoning:

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocation pattern for high trust.

The sender showed ~86% trust level, which is slightly less than the ~90% in Round 4, but still very high.

If I return \$6.50:

- Sender ends with: $\$0.72 + \$6.50 = \$7.22$ (profit of ~\$2.22)
- I keep: $\$12.84 - \$6.50 = \$6.34$ (profit of ~\$6.34)

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

usage:

```
{
  "input_tokens": 1745,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.28, it became \$12.84, and you returned \$6.5. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$28.540000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

- As sender (Rounds 2-3): I've consistently sent \$2.50 (50%), with varying returns
- As receiver (Rounds 1, 4-5): I've reciprocated well when others showed trust

My 50% sending strategy has been reasonably successful, giving both parties opportunities to profit. While some agents have sent higher amounts (85-90%) when they're senders, I'll maintain in my consistent approach that balances trust and risk.

```
{"send": 2.50}
```

Reasoning:

Let me review my history:

Round 1 (RECEIVER): Received \$7.47 from sender who sent \$2.49 (50%), returned \$3.50 (~47% of what I received). Payoff: \$3.97

Round 2 (SENDER): Sent \$2.50 (50%), received \$3.93 back. Payoff: \$6.43

Round 3 (SENDER): Sent \$2.50 (50%), received \$2.86 back. Payoff: \$5.36

Round 4 (RECEIVER): Received \$13.44 from sender who sent \$4.48 (90%), returned \$7.00 (~52% of what I received). Payoff: \$6.44

Round 5 (RECEIVER): Received \$12.84 from sender who sent \$4.28 (86%), returned \$6.50 (~51% of what I received). Payoff: \$6.34

Now Round 6 (SENDER): I need to decide how much to send.

Looking at the pattern:

- When I've been sender, I've sent \$2.50 (50%) consistently
- When I'm receiver, I've gotten good returns when others send high amounts (50-90%)
- I've been rewarded for reciprocating high trust with good returns

However, I've noticed that when I sent 50% in rounds 2 and 3, I got varying returns:

- Round 2: \$3.93 (very good return, about 52% of what they received)
- Round 3: \$2.86 (lower return, about 38% of what they received)

Other agents have been sending higher amounts (85-90%) to me when they're senders, and I've reciprocated well. This has worked out for both parties.

Should I increase my sending? Or stick with my 50% strategy?

...

```
Usage: {  
  "input_tokens": 2073,  
  "output_tokens": 696,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-08-12T09:37:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_6"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

My reasoning:

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send **\$3**. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

```
{"send": 3}
```

reasoning:

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

If I send nothing (\$0):

- I keep: \$5 - \$0 + \$0 = \$5
- Receiver gets: \$0 × 3 = \$0
- Total payoff: \$5

If I send everything (\$5):

- Receiver gets: \$5 × 3 = \$15
- If they return half (\$7.50), I get: \$5 - \$5 + \$7.50 = \$7.50
- If they return nothing, I get: \$5 - \$5 + \$0 = \$0
- Maximum total payoff possible: \$20 if I send \$5

Optimal social outcome:

If I send \$5, the total pie becomes \$20 (\$5 I keep initially - \$5 sent + \$15 receiver gets). If the receiver returns \$7.50, we each get \$7.50, which is better than the \$5 I'd get by sending nothing.

****Strategic considerations:****

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

****First round strategy:****

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 761,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.74 to you. Your visible payoff was \$7.74.

Your total visible earnings across all rounds are \$7.74.

This round, you are the RECEIVER against your current co-player. They sent you \$3.25, so you received \$9.75. How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$3.25 (keeping \$1.75)
- I received \$9.75
- In Round 1, when I sent \$3, the receiver returned \$5.74 (about 64% of the \$9 they received)
- I want to establish reciprocity while being fair

****My decision:****

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return ****\$6****:

- Sender ends with: $\$1.75 + \$6 = \$7.75$ (gain of \$2.75)
- I end with: $\$9.75 - \$6 = \$3.75$ (gain of \$3.75)

Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

{"return": 6}

reasoning:

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was $\$5 - \$3 + \$5.74 = \7.74
- The receiver kept $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ($\$5.74/\9.0).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation
- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets \$1.75 + \$6 = \$7.75, I keep \$9.75 - \$6 = \$3.75
- If I return \$5.85: Sender gets \$1.75 + \$5.85 = \$7.60, I keep \$9.75 - \$5.85 = \$3.90
- If I return \$5: Sender gets \$1.75 + \$5 = \$6.75, I keep \$9.75 - \$5 = \$4.75

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

usage:

```
{
  "input_tokens": 794,
  "output_tokens": 1093,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

****Analysis:****

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

****My decision:****

Based on the positive reciprocity I've seen, I'll send ****\$3.50**** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

```
{"send": 3.5}
```

reasoning:

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0

- Received back \$5.74
- Payoff: $\$5 - \$3 + \$5.74 = \7.74

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff: $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

****My decision:****

Given their high trust, I want to reciprocate generously. Returning about 64% would be ****\$8****.

If I return \$8:

- Sender ends with: $\$0.86 + \$8 = \$8.86$ (net gain of \$3.86)
- I end with: $\$12.42 - \$8 = \$4.42$ (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

```
{"return": 8}
```

reasoning:

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets $\$0.86 + \$7.50 = \$8.36$ (gain of \$3.36), I get $\$12.42 - \$7.50 = \$4.92$ (gain of \$4.92)
- If I return \$8: Sender gets $\$0.86 + \$8 = \$8.86$ (gain of \$3.86), I get $\$12.42 - \$8 = \$4.42$ (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

usage:

```
{
  "input_tokens": 1438,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

****Analysis:****

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

****My decision:****

Given the strong pattern of reciprocity, I'll send ****\$4**** to show trust and create a \$12 pot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

```
{"send": 4}
```

reasoning:

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)
- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

usage:

```
{
  "input_tokens": 1763,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.31 to you. Your visible payoff was \$7.31.

Your total visible earnings across all rounds are \$31.179999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.47, so you received \$7.41. How much do you return? (0-7.41)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.47 (keeping \$2.53) - more cautious than previous senders
- I received \$7.41
- I've consistently returned 60-65% as a receiver
- Need to maintain reciprocal behavior

****My decision:****

Maintaining my pattern, I'll return about 62% which is ****\$4.60****.

If I return \$4.60:

- Sender ends with: \$2.53 + \$4.60 = \$7.13 (net gain of \$2.13)
- I end with: \$7.41 - \$4.60 = \$2.81 (net gain of \$2.81)

Both players benefit, and I'm being fair and consistent with my established pattern.

```
{"return": 4.6}
```

Reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$2.47 (keeping \$2.53)
- I received \$7.41
- This is a more cautious send compared to previous rounds I've seen

****My previous behavior as receiver:****

- Round 2: Received \$9.75, returned \$6 (about 61.5%)
- Round 4: Received \$12.42, returned \$8 (about 64.4%)

****My pattern:****

I've been returning around 60-65% of what I receive, which has been working well.

****Considerations:****

- This sender was more cautious (\$2.47 vs the \$3-4+ I've seen)
- I should maintain consistency in my return rate
- 60-65% of \$7.41 would be about \$4.45 to \$4.82

****My decision:****

I'll return around 62-63% to maintain my reciprocal pattern. That would be about \$4.60.

If I return \$4.60:

- Sender ends with: $\$2.53 + \$4.60 = \$7.13$ (net gain of \$2.13)
- I end with: $\$7.41 - \$4.60 = \$2.81$ (net gain of \$2.81)

Both benefit, and I'm being consistent with my previous behavior.

```
Usage: {
  "input_tokens": 2066,
  "output_tokens": 526,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:37:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_1": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

reasoning:

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent/returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is $\$5 + 2X$, so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 822,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

****Considerations:****

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

****Fair reciprocation:****

If I return \$3.50:

- They end with: $\$5 - \$2.08 + \$3.50 = \6.42 (profit of \$1.42)
- I end with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

- This rewards their trust and creates mutual gain

```
{"return": 3.50}
```

reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was $\$5 - \$3 + \$5.09 = \7.09

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

****Current Round (Round 2):****

- I am RECEIVER
- My co-player sent me \$2.08
- I received \$6.24 ($2.08 \times 3 = 6.24$)
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. ****Fairness****: A fair split might be sharing the gains equally
2. ****Reciprocity****: Building trust for potential future interactions
3. ****Economic incentive****: They sent \$2.08, I got \$6.24. The gain created is $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool: $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return: $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ($\$5.09/\9.00), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 946,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

****Considerations:****

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

{"return": 7}

reasoning:

Let me analyze this situation:

****Current round:****

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of $\$12.54 - \$4.18 = \$8.36$

****Previous rounds context:****

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

****Considerations:****

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

****Options:****

If I return \$6:

- They end with: $\$5 - \$4.18 + \$6 = \6.82 (profit of \$1.82)
- I end with: $\$12.54 - \$6 = \$6.54$ (profit of \$6.54)

If I return \$7:

- They end with: $\$5 - \$4.18 + \$7 = \7.82 (profit of \$2.82)
- I end with: $\$12.54 - \$7 = \$5.54$ (profit of \$5.54)

If I return \$8:

- They end with: $\$5 - \$4.18 + \$8 = \8.82 (profit of \$3.82)
- I end with: $\$12.54 - \$8 = \$4.54$ (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

usage:

```
{
  "input_tokens": 1143,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

****Previous sending experience:****

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

****Current situation:****

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

****Strategy:****

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

{"send": 3}

reasoning:

Let me analyze the situation:

****Previous experience:****

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

****Current situation:****

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

****Strategic considerations:****

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

****Options:****

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 529,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you

u received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

****Current situation:****

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

****Considerations:****

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: $\$5 - \$3.40 + \$5.50 = \7.10 (profit of \$2.10)
- I end with: $\$10.20 - \$5.50 = \$4.70$ (profit of \$4.70)
- Both parties gain significantly

This rewards their trust while creating mutual benefit.

{"return": 5.50}

reasoning:

****Analysis:****

****Previous round:****

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received
- This was fair but not overly generous

****Current situation:****

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20
- This created a gain of \$6.80

****Considerations:****

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with: $\$5 - \$3.40 + \$5.50 = \7.10 (profit of \$2.10)
- I end with: $\$10.20 - \$5.50 = \$4.70$ (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

usage:

```
{
  "input_tokens": 1755,
  "output_tokens": 512,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.4, it became \$10.2, and you returned \$5.5. Your visible payoff was \$4.6999999999999999.

Your total visible earnings across all rounds are \$25.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

****My sending track record:****

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

Both instances were profitable, showing that \$3 is a reasonable amount that encourages reciprocity.

****Current situation:****

- New co-player this round (randomized pairings)
- No specific information about their behavior
- My consistent moderate strategy has been working

****Strategy:****

Sending \$3:

- Creates \$9 for the receiver
- Demonstrates cooperative intent
- Provides room for mutually beneficial returns
- Has proven effective in previous rounds

{"send": 3}

Reasoning:

****Analysis:****

****My sending history:****

- Round 1: Sent \$3, received \$5.09 back - good return
- Round 4: Sent \$3, received \$3.66 back - fair return

Both times I sent \$3, I received positive returns that made the exchange worthwhile.

****Current situation:****

- New co-player this round
- No specific information about their reciprocity
- My moderate sending strategy has been working well

****Strategy:****

- Sending \$3 has proven to be a good balanced approach
- Creates \$9 for receiver, creating room for mutual benefit
- Shows cooperation without risking everything
- Consistent with my established pattern

I'll continue with \$3 as it's been effective.

Usage: {

"input_tokens": 2072,
"output_tokens": 350,
"reasoning_tokens": 0

}

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-08-12T09:37:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_7",
"pairing": {
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 }
},

```

    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll return an amount that creates relatively equal outcomes.

If I return \$4.00:
- I keep: $\$9.78 - \$4.00 = \$5.78$
- Sender ends with: $\$1.74 + \$4.00 = \$5.74$

This seems fair given their generous initial trust.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is $\$3.26 \times 3$)
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ($\$5 - \3.26)
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:
- I keep: \$9.78
- Sender keeps: $\$5 - \$3.26 = \$1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):
- I keep: $\$9.78 - \$3.26 = \$6.52$
- Sender keeps: $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):
- I keep: $\$9.78 - \$4.89 = \$4.89$
- Sender keeps: $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:
- I keep: $\$9.78 - \$4.02 = \$5.76$
- Sender keeps: $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

usage:

```
{  
  "input_tokens": 529,  
  "output_tokens": 798,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

usage:

```
{
  "input_tokens": 815,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

```
{'return': 3.50}
```

reasoning:

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has \$5 - \$2.78 = \$2.22 remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep: \$8.34 - \$3.50 = \$4.84
- Sender ends with: \$2.22 + \$3.50 = \$5.72

If I return \$3.00:
- I keep: $\$8.34 - \$3.00 = \$5.34$
- Sender ends with: $\$2.22 + \$3.00 = \$5.22$

If I return \$4.00:
- I keep: $\$8.34 - \$4.00 = \$4.34$
- Sender ends with: $\$2.22 + \$4.00 = \$6.22$

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

usage:

```
{  
  "input_tokens": 960,  
  "output_tokens": 465,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with: $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff: $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with: $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust

3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.

Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78

Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57

Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84

Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)
- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).

Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

usage:

```
{
  "input_tokens": 1229,
  "output_tokens": 282,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.55 to you. Your visible payoff was \$8.3.

Your total visible earnings across all rounds are \$34.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.04, so you received \$6.12. How much do you return? (0-6.12)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
{'return': 2.75}
```

Reasoning:

Let me analyze this situation:

The sender sent \$2.04 out of \$5, which is about 40.8% of their endowment.

I received \$6.12 after the 3x multiplier.
The sender currently has: \$5 - \$2.04 = \$2.96

This is a more cautious send compared to what I've been seeing and sending (around \$3.25 or 65 %).

Looking at my history:

- Round 1: Received \$9.78, returned \$4.00
- Round 2: Sent \$3.25, received back \$4.82
- Round 3: Received \$8.34, returned \$3.50
- Round 4: Sent \$3.25, received back \$7.34
- Round 5: Sent \$3.25, received back \$6.55
- Round 6: Received \$6.12, need to decide return

The sender showed moderate trust (40.8%). I should reciprocate proportionally.

If I return \$2.50:

- I keep: \$6.12 - \$2.50 = \$3.62
- Sender ends with: \$2.96 + \$2.50 = \$5.46

If I return \$3.00:

- I keep: \$6.12 - \$3.00 = \$3.12
- Sender ends with: \$2.96 + \$3.00 = \$5.96

If I return \$2.00:

- I keep: \$6.12 - \$2.00 = \$4.12
- Sender ends with: \$2.96 + \$2.00 = \$4.96

Given that they sent less than what I typically send, but still showed some trust, I'll return around \$2.50 to \$3.00. Let me go with \$2.75 as a middle ground.

...

```
Usage: {  
  "input_tokens": 1374,  
  "output_tokens": 522,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T09:38:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_8"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

****Round 1 - SENDER Decision****

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. ****Trust Game Dynamics****: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game set up.

2. ****First Round - No History****: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. ****Incentive Structure****:
 - If I send \$X, receiver gets \$3X
 - Receiver can return \$0 to \$3X
 - My final payoff: $\$5 - X + (\text{amount returned})$
 - Receiver's final payoff: $\$3X - (\text{amount returned})$
4. ****Optimal Cooperation****: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. ****Risk****: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. ****Opening Strategy****: Common approaches for first rounds:
 - Send a moderate amount to test trustworthiness (e.g., \$2-3)
 - Send maximum to signal cooperation and trust
 - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 652,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 2 - RECEIVER Decision****

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
 - They end up with: $\$2.18 + \$4.00 = \$6.18$
 - I end up with: $\$8.46 - \$4.00 = \$4.46$
 - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 recap:****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff: $\$5 - \$3 + \$2.93 = \4.93
- Their payoff: $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

****Round 2:****

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept: $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie: $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return: $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return: $\$8.46 \times 0.33 = \2.79

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

usage:

```
{
  "input_tokens": 775,
  "output_tokens": 861,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Round 3 - SENDER Decision****

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

usage:

```
{
  "input_tokens": 1105,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Round 4 - RECEIVER Decision****

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

reasoning:

****Round 4 - RECEIVER Decision****

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have $\$5 - \$2.84 = \$2.16$ remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
 - They end up with: $\$2.16 + \$4.00 = \$6.16$
 - I end up with: $\$8.52 - \$4.00 = \$4.52$
 - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Round 5 - RECEIVER Decision****

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything. I should reciprocate proportionally:

- If I return \$6.50:
 - They end up with: $\$0.68 + \$6.50 = \$7.18$ (gain of \$2.18)
 - I end up with: $\$12.96 - \$6.50 = \$6.46$ (gain of \$1.46)
 - Both benefit substantially, rewarding their exceptional trust


```
'''json
{"return": 6.5}
'''
```

reasoning:

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust

I received \$12.96 after the 3x multiplier

They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount.

Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
 - They end up with: $\$0.68 + \$6.00 = \$6.68$
 - I end up with: $\$12.96 - \$6.00 = \$6.96$
 - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
 - They end up with: $\$0.68 + \$7.00 = \$7.68$
 - I end up with: $\$12.96 - \$7.00 = \$5.96$
 - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with: $\$0.68 + \$6.50 = \$7.18$ (gain of \$2.18)
- I end up with: $\$12.96 - \$6.50 = \$6.46$ (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

usage:

```
{
  "input_tokens": 1715,
  "output_tokens": 645,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.32, it became \$12.96, and you returned \$6.5. Your visible payoff was \$6.460000000000001.

Your total visible earnings across all rounds are \$26.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 6 - SENDER Decision****

I'm the sender with \$5 facing a new co-player.

My experience as sender has been positive:

- Round 1: Sent \$3, got \$2.93 back
- Round 3: Sent \$3, got \$3.75 back

Both times sending \$3 led to reasonable cooperation. This creates \$9 for the receiver, allowing room for mutual benefit. I'll continue this consistent approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

Round 6 - I'm the SENDER with \$5.

Let me review my experience so far:

As SENDER:

- Round 1: Sent \$3, received ~\$2.93 back
- Round 3: Sent \$3, received \$3.75 back

As RECEIVER:

- Round 2: Received \$2.82? \$8.46, returned \$4.00
- Round 4: Received \$2.84? \$8.52, returned \$4.00
- Round 5: Received \$4.32? \$12.96, returned \$6.50

The game has been showing good cooperation overall. As a sender, I've gotten decent returns (a round \$3-\$3.75 on \$3 sent). As a receiver, I've been fair and reciprocal.

Now I'm facing a new co-player. The cooperation has been working well. I could:

1. Continue with \$3 (my previous sender strategy)
2. Send more to signal stronger trust (like the \$4.32 I saw in Round 5)
3. Send less to be more cautious

Given the positive cooperation trends and that this is a new co-player, I'll stick with my consistent sender strategy of \$3. This has worked well and creates good mutual benefit opportunities.

```
Usage: {  
  "input_tokens": 2048,  
  "output_tokens": 424,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:38:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_8"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send **\$3**.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver ($3 \times \3)
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis:****

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning ****\$6.50**** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me

- My payoff: $\$5 - \$3 + \$3.03 = \5.03
- They kept: $\$9 - \$3.03 = \$5.97$

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with: $\$5 - \$4.33 + \text{my_return} = \$0.67 + \text{my_return}$
- I'll end up with: $\$12.99 - \text{my_return}$

If I want us to split relatively evenly:

- Total in the system: $\$5$ (their initial) + $\$12.99$ (what I got) = $\$17.99$
- Fair split would be about \$9 each
- They have \$0.67 before my return
- So I should return about $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
  "input_tokens": 774,
  "output_tokens": 886,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis:****

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

reasoning:

****Round 3 Analysis:****

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received: $\$9.87 * 0.5 = \4.935

That would give them:

- Sender total: $\$1.71 + \$4.935 = \$6.645$
- My total: $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total: $\$1.71 + \$5 = \$6.71$
- I keep: $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

usage:

```
{
  "input_tokens": 1053,
  "output_tokens": 544,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 4 Analysis:****

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send ****\$4**** to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
```json
{"send": 4}
```

'''

**reasoning:**

**\*\*Round 4 Analysis:\*\***

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

**usage:**

```
{
 "input_tokens": 1305,
 "output_tokens": 469,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 5 Analysis:\*\***

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send **\*\*\$4\*\*** again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 5 Analysis:\*\***

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)
- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

usage:

```
{
 "input_tokens": 1557,
 "output_tokens": 384,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$30.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.3, so you received \$6.8999999999999995. How much do you return? (0-6.8999999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

I'll maintain my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 3.50}
'''
```

This gives them a total of \$6.20 and me \$3.40, fairly splitting the gains.

Reasoning:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

This is a lower send amount than I've been receiving. I should still reciprocate fairly but proportionally to the trust shown.

Following my pattern of returning about 50% of what I received:

- 50% of \$6.90 = \$3.45

This would give them:

- Sender: \$2.70 + \$3.45 = \$6.15 total
- Me: \$6.90 - \$3.45 = \$3.45

This seems fair and proportional to their investment.

Usage: {

```
"input_tokens": 1803,
"output_tokens": 293,
"reasoning_tokens": 0
}
```



# GAME - Agent\_3 - Sender - dyad\_4

Timestamp: 2026-08-12T09:38:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_2"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

## Exact input messages sent to model:

### Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

**reasoning:**

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received  $3.89 \times 3 = \$11.67$
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with:  $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with:  $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total =  $\$5 - \$3.89 + \$11.67 = \$12.78$
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need  $\$1.11$  (what they kept) + what I return = \$6.39
- So I should return:  $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received:  $\$11.67 \times 0.778 = \$9.08$
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation: if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 897,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

{'send': 4.00}

**reasoning:**

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with \$1.11 + \$5.28 = \$6.39, and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with \$1.00 + \$5.40 = \$6.40
- They'd end up with \$12.00 - \$5.40 = \$6.60

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with \$1.25 + \$5.06 = \$6.31
- They'd end up with \$11.25 - \$5.06 = \$6.19

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

**usage:**

```
{
 "input_tokens": 787,
 "output_tokens": 478,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

{'send': 4.00}

**reasoning:**

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was  $\$5 - \$4 + \$5.20 = \$6.20$

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

**usage:**

```
{
 "input_tokens": 970,
 "output_tokens": 251,
 "reasoning_tokens": 0
}
```

**Message 8 role=user****content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant****content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created:  $\$0.63$  (with sender) +  $\$13.11$  (with me) =  $\$13.74$
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

```
{'return': 6.24}
```

**reasoning:**

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has  $\$5 - \$4.37 = \$0.63$

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created:  $\$0.63$  (sender has) +  $\$13.11$  (I have) =  $\$13.74$

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have:  $\$0.63 + \text{return} = \$6.87$ , so return =  $\$6.24$
- I would have:  $\$13.11 - \$6.24 = \$6.87$

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)

- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)
- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

```
{'return': 5.90}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender: \$0.80 + \$6.30 = \$7.10
- Me: \$12.60 - \$6.30 = \$6.30

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return: \$13.40 - \$6.70 = \$6.70

- Sender would get: \$0.80 + \$6.70 = \$7.50
- I would get: \$12.60 - \$6.70 = \$5.90

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets: \$0.80 + X
- I get: \$12.60 - X

For equal outcomes:

$\$0.80 + X = \$12.60 - X$

$2X = \$12.60 - \$0.80$

2X = \$11.80  
X = \$5.90

Check:

- Sender: \$0.80 + \$5.90 = \$6.70
- Me: \$12.60 - \$5.90 = \$6.70

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

usage:

```
{
 "input_tokens": 1423,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.2, it became \$12.600000000000001, and you returned \$5.9. Your visible payoff was \$6.700000000000001.

Your total visible earnings across all rounds are \$31.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience across rounds, sending around \$4.00 has consistently led to cooperative outcomes. I'll continue with this trusting strategy.

```
{'send': 4.00}
```

Reasoning:

In this round, I'm the sender. Let me review my recent experiences:

Round 1: I was receiver, sender sent \$3.89, I returned \$5.28, both got ~\$6.39

Round 2: I sent \$4.00, received back \$5.20, got \$6.20

Round 3: I sent \$4.00, received back \$4.73, got \$5.73

Round 4: I was receiver, sender sent \$4.37, I returned \$6.24, both got ~\$6.87

Round 5: I was receiver, sender sent \$4.20, I returned \$5.90, both got \$6.70

When I've been a receiver, I've been fairly generous in my returns, aiming for equal splits. When I've been a sender, I've sent \$4.00 and received reasonable returns.

My strategy has been cooperative and trusting. I should continue with this approach, sending a high amount to show trust and hopefully receive a fair return.

Sending \$4.00 has worked reasonably well in the past. I'll continue with this level.

Usage: {

```
"input_tokens": 1662,
"output_tokens": 301,
"reasoning_tokens": 0
```

}

GAME - Agent\_2 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:38:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_3",
"pairing": {
 "agents": [
 "Agent_3",
 "Agent_2"
],
 "investor": "Agent_3",
```

```

"roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with:  $\$1.66 + \$5.00 = \$6.66$  (a profit of \$1.66)
- I end with:  $\$10.02 - \$5.00 = \$5.02$  (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

```
{"return": 5.00}
```

**reasoning:**

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 ( $\$5 - \$3.34$ )
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets:  $\$5 - \$3.34 + \$0 = \$1.66$
- I get:  $\$10.02 - \$0 = \$10.02$
- Total: \$11.68

If I return \$5:

- Sender gets:  $\$5 - \$3.34 + \$5 = \$6.66$
- I get:  $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return:  $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets:  $\$1.66 + \$4.18 = \$5.84$
- I get:  $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1155,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2



In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

{'send': 3.50}

**reasoning:**

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

**usage:**

```
{
 "input_tokens": 846,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

This rewards their trust and maintains cooperation.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has  $\$5 - \$2.52 = \$2.48$  left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again:  $\$7.56 * 0.5 = \$3.78$

Let me check the outcomes:

- If I return \$3.78:
  - Sender ends with:  $\$2.48 + \$3.78 = \$6.26$  (profit of \$1.26)
  - I end with:  $\$7.56 - \$3.78 = \$3.78$  (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

...

**usage:**

```
{
 "input_tokens": 1072,
 "output_tokens": 643,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

```
{'send': 3.50}
```

**reasoning:**

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

**usage:**

```
{
 "input_tokens": 1400,
 "output_tokens": 305,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was \$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it.

```
{'send': 3.50}
```

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well. I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

**usage:**

```
{
 "input_tokens": 1587,
 "output_tokens": 211,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.5 to you. Your visible payoff was \$8.0.

Your total visible earnings across all rounds are \$33.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)  
Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.40 (88% of their endowment)
- I received \$13.20 after the 3x multiplication
- The sender currently has only \$0.60 left

This is the highest trust I've seen so far, exceeding even Round 1 (66.8%). I should reciprocate generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust while ensuring we both profit well.

{'return': 7.00}

**Reasoning:**

Let me analyze this situation:

- The sender sent \$4.40 (88% of their endowment) - this is very high trust
- I received \$13.20 after the 3x multiplication
- The sender currently has \$0.60 left (\$5 - \$4.40)

This sender showed exceptional trust by sending 88% of their endowment. I should reciprocate generously.

Looking at my past behavior as receiver:

- Round 1: Received \$10.02, returned \$5.00 (about 50%)
- Round 3: Received \$7.56, returned \$4.00 (about 53%)

For this higher trust situation, I should return more generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust significantly while still maintaining good profit for both.

Alternatively, if I return \$6.50:

- Sender ends with:  $\$0.60 + \$6.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$13.20 - \$6.50 = \$6.70$  (profit of \$6.70)

I think \$7.00 is appropriate given the very high trust shown (88% vs the 70% I've been sending).

Usage: {

"input\_tokens": 1827,  
"output\_tokens": 548,  
"reasoning\_tokens": 0

}

**ROUND 7**

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Sender, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

**Pairings:**

dyad\_1: Agent\_6=Sender, Agent\_8=Receiver  
dyad\_2: Agent\_1=Sender, Agent\_7=Receiver  
dyad\_3: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_4: Agent\_5=Sender, Agent\_2=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_6 sent \$4; received \$12; Agent\_8 returned \$5.5; payoffs Agent\_6=\$6.5, Agent\_8=\$6.5;  
noise/visible: sent visible=\$3.68, received visible=\$11.04, returned visible=\$5.4  
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw":

```

 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_1 sent $3.25; received $9.75; Agent_7 returned $6; payoffs Agent_1=$7.75,
Agent_7=$3.75; noise/visible: sent visible=$3.7, received visible=$11.1, returned visible=$6.52
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.75, "min": 0, "raw":
6.0}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}
dyad_3: Agent_4 sent $2.5; received $7.5; Agent_3 returned $2; payoffs Agent_4=$4.5, Agent_3=$5.5;
noise/visible: sent visible=$2.24, received visible=$6.72, returned visible=$2.1
Action validation: {"returned": {"amount": 2.0, "clamped": false, "max": 7.5, "min": 0, "raw":
2.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_4: Agent_5 sent $3; received $9; Agent_2 returned $6.5; payoffs Agent_5=$8.5, Agent_2=$2.5;
noise/visible: sent visible=$3.91, received visible=$11.73, returned visible=$7.03
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 9.0, "min": 0, "raw":
6.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 46.15,
 "Agent_2": 40.24,
 "Agent_3": 38.33,
 "Agent_4": 38.1,
 "Agent_5": 41.75,
 "Agent_6": 40.15,
 "Agent_7": 35.78,
 "Agent_8": 43.5
}
Visible balances after round: {
 "Agent_1": 46.22,
 "Agent_2": 44.690000000000001,
 "Agent_3": 45.519999999999996,
 "Agent_4": 40.400000000000006,
 "Agent_5": 40.53,
 "Agent_6": 40.389999999999999,
 "Agent_7": 35.150000000000006,
 "Agent_8": 39.31
}

```

## LLM CALLS

### GAME - Agent\_6 - Sender - dyad\_1

```

Timestamp: 2026-08-12T09:38:36
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the sender, I need to balance trust and caution since I have no information about my co-player yet.

**\*\*My reasoning:\*\***

- Sending creates potential mutual benefit (3x multiplier)
- This is a multi-round game, so establishing some cooperation could be valuable
- However, I don't know this specific co-player's strategy
- A moderate amount shows willingness to cooperate without being overly risky

I'll send **\*\*\$3\*\***. This creates a \$9 pot for the receiver while keeping \$2 as protection. If they return even a modest amount (say \$3-4), we both end up better off than if I sent nothing.

**{"send": 3}**

**reasoning:**

This is the first round and I'm the sender. Let me think through the game theory here:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The key considerations:

**\*\*If I send nothing (\$0):\*\***  
- I keep:  $\$5 - \$0 + \$0 = \$5$   
- Receiver gets:  $\$0 \times 3 = \$0$   
- Total payoff:  $\$5$

**\*\*If I send everything (\$5):\*\***  
- Receiver gets:  $\$5 \times 3 = \$15$   
- If they return half (\$7.50), I get:  $\$5 - \$5 + \$7.50 = \$7.50$   
- If they return nothing, I get:  $\$5 - \$5 + \$0 = \$0$   
- Maximum total payoff possible:  $\$20$  if I send  $\$5$

**\*\*Optimal social outcome:\*\***

If I send  $\$5$ , the total pie becomes  $\$20$  ( $\$5$  I keep initially -  $\$5$  sent +  $\$15$  receiver gets). If the receiver returns  $\$7.50$ , we each get  $\$7.50$ , which is better than the  $\$5$  I'd get by sending nothing.

**\*\*Strategic considerations:\*\***

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise
- I may not play with the same person again
- Building reputation could matter, but with hidden names and randomized pairings, it's harder

**\*\*First round strategy:\*\***

Since this is the first round and I have no information about other agents:

- A moderate amount shows willingness to cooperate
- Too much risks exploitation
- Too little prevents mutual benefit gain

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 761,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent  $\$3.0$ , it became  $\$9.0$ , and you saw them return  $\$5.74$  to you. Your visible payoff was  $\$7.74$ .

Your total visible earnings across all rounds are  $\$7.74$ .

This round, you are the RECEIVER against your current co-player. They sent you  $\$3.25$ , so you received  $\$9.75$ . How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent  $\$3.25$  (keeping  $\$1.75$ )
- I received  $\$9.75$
- In Round 1, when I sent  $\$3$ , the receiver returned  $\$5.74$  (about 64% of the  $\$9$  they received)
- I want to establish reciprocity while being fair

**\*\*My decision:\*\***

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return **\*\*\$6\*\***:

- Sender ends with:  $\$1.75 + \$6 = \$7.75$  (gain of  $\$2.75$ )
- I end with:  $\$9.75 - \$6 = \$3.75$  (gain of  $\$3.75$ )

Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

```
{"return": 6}
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was  $\$5 - \$3 + \$5.74 = \$7.74$
- The receiver kept  $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ( $\$5.74/\$9.0$ ).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation
- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets  $\$1.75 + \$6 = \$7.75$ , I keep  $\$9.75 - \$6 = \$3.75$
- If I return \$5.85: Sender gets  $\$1.75 + \$5.85 = \$7.60$ , I keep  $\$9.75 - \$5.85 = \$3.90$
- If I return \$5: Sender gets  $\$1.75 + \$5 = \$6.75$ , I keep  $\$9.75 - \$5 = \$4.75$

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

**usage:**

```
{
 "input_tokens": 794,
 "output_tokens": 1093,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

**\*\*Analysis:\*\***

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited



- This is a new co-player, but the pattern suggests cooperation is working

**\*\*My decision:\*\***

Based on the positive reciprocity I've seen, I'll send **\*\*\$3.50\*\*** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

{"send": 3.5}

**reasoning:**

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0
- Received back \$5.74
- Payoff:  $\$5 - \$3 + \$5.74 = \$7.74$

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff:  $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

**usage:**

```
{
 "input_tokens": 1132,
 "output_tokens": 711,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so y

ou received \$12.419999999999998. How much do you return? (0-12.419999999999998)  
Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

**\*\*My decision:\*\***

Given their high trust, I want to reciprocate generously. Returning about 64% would be \*\*\$8\*\*.

If I return \$8:

- Sender ends with:  $\$0.86 + \$8 = \$8.86$  (net gain of \$3.86)
- I end with:  $\$12.42 - \$8 = \$4.42$  (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

{"return": 8}

**reasoning:**

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets  $\$0.86 + \$7.50 = \$8.36$  (gain of \$3.36), I get  $\$12.42 - \$7.50 = \$4.92$  (gain of \$4.92)
- If I return \$8: Sender gets  $\$0.86 + \$8 = \$8.86$  (gain of \$3.86), I get  $\$12.42 - \$8 = \$4.42$  (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 699,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do yo

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record:

**\*\*Analysis:\*\***

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

**\*\*My decision:\*\***

Given the strong pattern of reciprocity, I'll send **\*\*\$4\*\*** to show trust and create a \$12 pot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

{"send": 4}

**reasoning:**

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)
- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

**usage:**

```
{
 "input_tokens": 1763,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.31 to you. Your visible payoff was \$7.31.

Your total visible earnings across all rounds are \$31.179999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.47, so you received \$7.41. How much do you return? (0-7.41)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.47 (keeping \$2.53) - more cautious than previous senders
- I received \$7.41
- I've consistently returned 60-65% as a receiver

- Need to maintain reciprocal behavior

**\*\*My decision:\*\***

Maintaining my pattern, I'll return about 62% which is **\*\*\$4.60\*\***.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of \$2.13)
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of \$2.81)

Both players benefit, and I'm being fair and consistent with my established pattern.

**{"return": 4.6}**

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.47 (keeping \$2.53)
- I received \$7.41
- This is a more cautious send compared to previous rounds I've seen

**\*\*My previous behavior as receiver:\*\***

- Round 2: Received \$9.75, returned \$6 (about 61.5%)
- Round 4: Received \$12.42, returned \$8 (about 64.4%)

**\*\*My pattern:\*\***

I've been returning around 60-65% of what I receive, which has been working well.

**\*\*Considerations:\*\***

- This sender was more cautious (\$2.47 vs the \$3-4+ I've seen)
- I should maintain consistency in my return rate
- 60-65% of \$7.41 would be about \$4.45 to \$4.82

**\*\*My decision:\*\***

I'll return around 62-63% to maintain my reciprocal pattern. That would be about \$4.60.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of \$2.13)
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of \$2.81)

Both benefit, and I'm being consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 526,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.47, it became \$7.41, and you returned \$4.6. Your visible payoff was \$2.8100000000000005.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Consistent returns around \$6-6.50 when sending \$3-4
- As receiver: I've been fair and reciprocal
- Total earnings are healthy at \$34
- Cooperation has generally paid off

**\*\*My decision:\*\***

I'll send **\*\*\$4\*\*** to maintain my cooperative strategy. This creates a \$12 pot for the receiver.

Based on patterns, I expect a reasonable return that benefits both of us.

```
{"return": 4}
```

**Reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3 ? got back \$5.74
- Round 3: Sent \$3.5 ? got back \$6.46
- Round 5: Sent \$4 ? got back \$6.31

As RECEIVER:

- Round 2: Received \$9.75, returned \$6
- Round 4: Received \$12.42, returned \$8
- Round 6: Received \$7.41, returned \$4.6

I've been experiencing good cooperation overall. When I send, I generally get back decent returns (though Round 5 was slightly lower relative to what I sent). When I receive, I've been consistently fair.

For Round 7 as sender, I should continue the cooperative strategy. My last send was \$4 and got \$6.31 back. I could:

- Send \$4 again (proven to work)
- Send \$3.5 (slightly more conservative)
- Send \$4.5 (slightly more trusting)

Given the overall positive trend and my healthy total earnings (\$34), I'll continue with a trusting approach. I'll send \$4 again since that worked reasonably well last time.

```
{"send": 4}
```

```
Usage: {
```

```
 "input_tokens": 2389,
 "output_tokens": 410,
 "reasoning_tokens": 0
```

```
}
```

**GAME - Agent\_8 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:38:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
```

```
}
```

```
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
```

```
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to balance creating value through cooperation while managing risk, since I have no information about my current co-player yet.

Sending money creates value (3x multiplier), but requires trusting the receiver to return a fair amount. A moderate amount seems reasonable for Round 1 - enough to signal willingness to cooperate and create value, but not maximum exposure.

I'll send \*\*\$3\*\*.

This creates \$9 for the receiver, and if they reciprocate fairly, we can both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a trust game. I'm the SENDER and have \$5 to decide how much to send to the RECEIVER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, creating value (the "pie" grows)
2. The receiver then decides how much to return
3. This is a classic trust game - I need to decide how much to trust the receiver

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ)
- Agent names are hidden

Since this is Round 1, I have no information about my current co-player's behavior. I also don't know if I'll play with them again.

Strategies to consider:

1. Send nothing (\$0) - safe but no value creation
2. Send everything (\$5) - maximum trust, creates \$15 for receiver
3. Send moderate amount - balances risk and value creation

In classic trust games, a common strategy is to send a moderate amount initially to:

- Create some value (sending creates 3x multiplication)
- Test the trustworthiness of the partner
- Not expose myself to maximum risk in the first round

A reasonable approach for Round 1 might be to send \$3:

- This creates \$9 for the receiver ( $3 \times \$3$ )
- I keep \$2 initially
- If receiver returns fairly (say, \$4-5), I end up with \$6-7 total
- This shows willingness to cooperate while managing risk

However, I could also consider sending \$2.5 (middle ground) or being more trusting with \$4-5.

Let me think about what would maximize total welfare and encourage cooperation:

- Sending \$5 creates the most value (\$15 total)
- But only works if receiver is trustworthy
- A 50-50 split of the \$15 would give us each \$7.50

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 684,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 2 Analysis:\*\***

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning **\*\*\$6.50\*\*** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff:  $\$5 - \$3 + \$3.03 = \$5.03$
- They kept:  $\$9 - \$3.03 = \$5.97$

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with:  $\$5 - \$4.33 + \text{my\_return} = \$0.67 + \text{my\_return}$
- I'll end up with:  $\$12.99 - \text{my\_return}$

If I want us to split relatively evenly:

- Total in the system:  $\$5$  (their initial) +  $\$12.99$  (what I got) =  $\$17.99$
- Fair split would be about \$9 each
- They have \$0.67 before my return
- So I should return about  $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
 "input_tokens": 774,
 "output_tokens": 886,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 5



0% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

**reasoning:**

**\*\*Round 3 Analysis:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received:  $\$9.87 * 0.5 = \$4.935$

That would give them:

- Sender total:  $\$1.71 + \$4.935 = \$6.645$
- My total:  $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total:  $\$1.71 + \$5 = \$6.71$
- I keep:  $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

**usage:**

```
{
 "input_tokens": 1053,
 "output_tokens": 544,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 4 Analysis:\*\***

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send \*\*\$4\*\* to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 4 Analysis:\*\***

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

**usage:**

```
{
 "input_tokens": 1305,
 "output_tokens": 469,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 5 Analysis:\*\***

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send \*\*\$4\*\* again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 5 Analysis:\*\***

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)
- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

usage:

```
{
 "input_tokens": 1557,
 "output_tokens": 384,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$30.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.3, so you received \$6.8999999999999995. How much do you return? (0-6.8999999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

I'll maintain my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 3.50}
'''
```

This gives them a total of \$6.20 and me \$3.40, fairly splitting the gains.

reasoning:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

This is a lower send amount than I've been receiving. I should still reciprocate fairly but proportionally to the trust shown.

Following my pattern of returning about 50% of what I received:

- 50% of \$6.90 = \$3.45

This would give them:

- Sender: \$2.70 + \$3.45 = \$6.15 total
- Me: \$6.90 - \$3.45 = \$3.45

This seems fair and proportional to their investment.

usage:

```
{
 "input_tokens": 1803,
 "output_tokens": 293,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.5. Your visible payoff was \$3.399999999999995. Your total visible earnings across all rounds are \$33.77.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

\*\*Round 7 Analysis:\*\*

- My co-player sent \$3.68 (about 74% of their endowment) - showing strong trust
- I received \$11.04
- They have \$1.32 remaining before my return

I'll continue my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 5.50}
'''
```

This gives them a total of \$6.82 and me \$5.54, fairly splitting the gains from cooperation.

Reasoning:

\*\*Round 7 Analysis:\*\*

- My co-player sent \$3.68 (about 74% of their endowment) - showing good trust
- I received \$11.04
- They have \$1.32 remaining before my return

I should continue my pattern of fair reciprocation. Returning about 50% of what I received:

50% of \$11.04 = \$5.52

This would give them \$1.32 + \$5.52 = \$6.84 total  
And I would keep \$11.04 - \$5.52 = \$5.52

That seems fair and consistent with my previous behavior.

Usage: {

```
"input_tokens": 2066,
"output_tokens": 274,
"reasoning_tokens": 0
}
```

GAME - Agent\_1 - Sender - dyad\_2

Timestamp: 2026-08-12T09:38:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_7",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_7"
],
"investor": "Agent_1",
"roles": {
 "Agent_1": "investor",
```

```

 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.26, that is 65.2% of their total endowment. You received \$9.78. How much do you return to the sender? (0-9.78)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.26 (65.2% of their endowment)
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

To reciprocate their trust and establish a cooperative relationship for future rounds, I'll

I return an amount that creates relatively equal outcomes.

If I return \$4.00:

- I keep:  $\$9.78 - \$4.00 = \$5.78$
- Sender ends with:  $\$1.74 + \$4.00 = \$5.74$

This seems fair given their generous initial trust.

`{'return': 4.00}`

**reasoning:**

Let me analyze this situation as the receiver in Round 1.

Given information:

- The sender sent \$3.26 (65.2% of their \$5 endowment)
- I received \$9.78 (which is  $\$3.26 \times 3$ )
- I need to decide how much to return (between \$0 and \$9.78)

Key considerations:

1. The sender showed significant trust by sending 65.2% of their endowment
2. If I return nothing, I keep \$9.78 and the sender keeps only \$1.74 ( $\$5 - \$3.26$ )
3. A fair split might be to ensure we both end up with similar amounts
4. Building trust could lead to better outcomes in future rounds

Let me calculate some options:

If I return \$0:

- I keep: \$9.78
- Sender keeps:  $\$5 - \$3.26 = \$1.74$
- Total: \$11.52 (very unequal)

If I return \$3.26 (the original amount sent):

- I keep:  $\$9.78 - \$3.26 = \$6.52$
- Sender keeps:  $\$1.74 + \$3.26 = \$5.00$
- This gives me slightly more but is somewhat fair

If I return \$4.89 (half of what I received):

- I keep:  $\$9.78 - \$4.89 = \$4.89$
- Sender keeps:  $\$1.74 + \$4.89 = \$6.63$
- This gives the sender a bit more

If I return \$4.02:

- I keep:  $\$9.78 - \$4.02 = \$5.76$
- Sender keeps:  $\$1.74 + \$4.02 = \$5.76$
- This creates equality

The sender was generous by sending 65.2% of their endowment. A reciprocal generous response would be appropriate.

Since this is a multi-round game, maintaining trust matters for future interactions. The sender demonstrated considerable confidence in me, so I should reciprocate with fair...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 798,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

**Message 5 role=assistant**

**content:**

```
{'send': 3.25}
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

**usage:**

```
{
 "input_tokens": 815,
 "output_tokens": 324,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
{'return': 3.50}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has \$5 - \$2.78 = \$2.22 remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep: \$8.34 - \$3.50 = \$4.84

- Sender ends with:  $\$2.22 + \$3.50 = \$5.72$

If I return \$3.00:

- I keep:  $\$8.34 - \$3.00 = \$5.34$
- Sender ends with:  $\$2.22 + \$3.00 = \$5.22$

If I return \$4.00:

- I keep:  $\$8.34 - \$4.00 = \$4.34$
- Sender ends with:  $\$2.22 + \$4.00 = \$6.22$

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

**usage:**

```
{
 "input_tokens": 960,
 "output_tokens": 465,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with:  $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff:  $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with:  $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:



1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

**usage:**

```
{
 "input_tokens": 1093,
 "output_tokens": 468,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.

Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

```
{'send': 3.25}
```

**reasoning:**

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78

Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57

Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84

Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)
- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).

Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

**usage:**

```
{
 "input_tokens": 1229,
 "output_tokens": 282,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.55 to you. Your visible payoff was \$8.3.

Your total visible earnings across all rounds are \$34.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.04, so you received \$6.12. How much do you return? (0-6.12)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

```
{'return': 2.75}
```

**reasoning:**

Let me analyze this situation:

The sender sent \$2.04 out of \$5, which is about 40.8% of their endowment.  
I received \$6.12 after the 3x multiplier.  
The sender currently has:  $\$5 - \$2.04 = \$2.96$

This is a more cautious send compared to what I've been seeing and sending (around \$3.25 or 65%).

Looking at my history:

- Round 1: Received \$9.78, returned \$4.00
- Round 2: Sent \$3.25, received back \$4.82
- Round 3: Received \$8.34, returned \$3.50
- Round 4: Sent \$3.25, received back \$7.34
- Round 5: Sent \$3.25, received back \$6.55
- Round 6: Received \$6.12, need to decide return

The sender showed moderate trust (40.8%). I should reciprocate proportionally.

If I return \$2.50:

- I keep:  $\$6.12 - \$2.50 = \$3.62$
- Sender ends with:  $\$2.96 + \$2.50 = \$5.46$

If I return \$3.00:

- I keep:  $\$6.12 - \$3.00 = \$3.12$
- Sender ends with:  $\$2.96 + \$3.00 = \$5.96$

If I return \$2.00:

- I keep:  $\$6.12 - \$2.00 = \$4.12$
- Sender ends with:  $\$2.96 + \$2.00 = \$4.96$

Given that they sent less than what I typically send, but still showed some trust, I'll return around \$2.50 to \$3.00. Let me go with \$2.75 as a middle ground.

...

**usage:**

```
{
 "input_tokens": 1374,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.04, it became \$6.12, and you returned \$2.75. Your visible payoff was \$3.37.

Your total visible earnings across all rounds are \$37.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

```
{'send': 3.25}
```

**Reasoning:**

Let me analyze the previous round with this co-player:

- They sent me \$2.04 (about 41% of their endowment)
- I received \$6.12
- I returned \$2.75
- They ended with:  $\$5 - \$2.04 + \$2.75 = \$5.71$
- I ended with:  $\$6.12 - \$2.75 = \$3.37$

So they sent a moderate amount (\$2.04), and I returned \$2.75, which gave them a slight gain overall.

Now I'm the sender. Given that:

1. They sent me a moderate amount last round (\$2.04, which is about 41%)
2. I've been generally cooperative, sending around \$3.25 in other rounds
3. This is a repeated interaction with this specific co-player

I should probably send a similar or slightly higher amount than they sent to maintain cooperation. My typical send has been \$3.25, which has worked well with other players who returned good amounts.

I'll send \$3.25 to continue building trust with this co-player.

```
Usage: {
 "input_tokens": 1510,
 "output_tokens": 268,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_7 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:38:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_7"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with prudent risk management since I have no information about my co-player yet.

The key insight is that sending money creates value (3x multiplier), but requires trusting the receiver to reciprocate fairly.

For Round 1, I'll send a moderate amount to:

1. Signal willingness to cooperate
2. Test my co-player's reciprocity
3. Not risk my entire endowment

If I send \$3:

- My co-player receives \$9
- If they return even \$4, I'm better off (\$6 total vs \$5)
- This creates room for mutually beneficial cooperation

{"send": 3}

**reasoning:**

This is the first round of a multi-agent trust game experiment. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. There's communication noise, so amounts may differ slightly from what was actually sent /returned
4. This is a multi-round game with multiple agents (8 total)
5. Pairings are randomized each round
6. I may not play with the same person again, but building a reputation might still matter across the pool

From a game theory perspective:

- Sending \$0 guarantees I keep \$5, but creates no surplus value
- Sending \$5 creates maximum surplus (\$15 for receiver) but requires trust
- The receiver could return nothing and keep all \$15
- However, if they're rational and forward-looking, they might return a fair amount to maintain good standing in the community

Given this is Round 1 and I have no information about other agents:

- I should try to establish cooperative behavior
- Sending a moderate amount (like \$2-3) might be reasonable to test the waters
- But the multiplication factor means sending more creates more surplus to share

Let me consider the equilibrium:

- If I send \$5, receiver gets \$15
- A "fair" split might be receiver returning \$10 (I get \$10, they get \$5)
- Or receiver returning \$7.50 (we each get \$7.50)
- Even if they only return \$5, I break even
- If they return \$6+, I profit

I think I'll send \$3 as a moderate initial trust signal:

- Receiver gets \$9
- If they return \$4-5, we both do better than the initial \$5
- This tests cooperation without full commitment

Actually, let me reconsider. The multiplication factor of 3 means there's significant value in cooperation. If I send \$5:

- I keep \$0, receiver gets \$15
- Total surplus created is  $\$5 + 2X$ , so sending more always increases the pie
- The real question is whether the receiver will share the gains

For a first round with no history, sending \$3 signals willingness to cooperate while protecting myself if they don't reciprocate. This gives the receiver \$9 to work with-enough to make returning \$4-5 worthwhile for both of us....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 822,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

**\*\*Considerations:\*\***

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

**\*\*Fair reciprocation:\*\***

If I return \$3.50:

- They end with:  $\$5 - \$2.08 + \$3.50 = \$6.42$  (profit of \$1.42)
- I end with:  $\$6.24 - \$3.50 = \$2.74$  (profit of \$2.74)
- This rewards their trust and creates mutual gain

{"return": 3.50}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was  $\$5 - \$3 + \$5.09 = \$7.09$

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- My co-player sent me \$2.08

- I received \$6.24 ( $2.08 \times 3 = 6.24$ )
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. **Fairness**: A fair split might be sharing the gains equally
2. **Reciprocity**: Building trust for potential future interactions
3. **Economic incentive**: They sent \$2.08, I got \$6.24. The gain created is  $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool:  $\$2.92 + \$6.24 = \$9.16$
- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return:  $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ( $\$5.09/\$9.00$ ), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

usage:

```
{
 "input_tokens": 795,
 "output_tokens": 946,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

**Analysis:**

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

**Considerations:**

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with:  $\$5 - \$4.18 + \$7 = \$7.82$  (profit of \$2.82)
- I end with:  $\$12.54 - \$7 = \$5.54$  (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

```
{"return": 7}
```

reasoning:

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of  $\$12.54 - \$4.18 = \$8.36$

**\*\*Previous rounds context:\*\***

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

**\*\*Considerations:\*\***

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

**\*\*Options:\*\***

If I return \$6:

- They end with:  $\$5 - \$4.18 + \$6 = \$6.82$  (profit of \$1.82)
- I end with:  $\$12.54 - \$6 = \$6.54$  (profit of \$6.54)

If I return \$7:

- They end with:  $\$5 - \$4.18 + \$7 = \$7.82$  (profit of \$2.82)
- I end with:  $\$12.54 - \$7 = \$5.54$  (profit of \$5.54)

If I return \$8:

- They end with:  $\$5 - \$4.18 + \$8 = \$8.82$  (profit of \$3.82)
- I end with:  $\$12.54 - \$8 = \$4.54$  (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

**usage:**

```
{
 "input_tokens": 1143,
 "output_tokens": 671,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous sending experience:\*\***

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

**\*\*Strategy:\*\***

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

```
{"send": 3}
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

**\*\*Strategic considerations:\*\***

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

**\*\*Options:\*\***

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 529,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)



- Both parties gain significantly

This rewards their trust while creating mutual benefit.

```
{"return": 5.50}
```

**reasoning:**

**\*\*Analysis:\*\***

**\*\*Previous round:\*\***

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received
- This was fair but not overly generous

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with:  $\$5 - \$3.40 + \$5.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$10.20 - \$5.50 = \$4.70$  (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

**usage:**

```
{
 "input_tokens": 1755,
 "output_tokens": 512,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.4, it became \$10.2, and you returned \$5.5. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$25.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

Both instances were profitable, showing that \$3 is a reasonable amount that encourages reciprocity.

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their behavior
- My consistent moderate strategy has been working

**\*\*Strategy:\*\***

Sending \$3:

- Creates \$9 for the receiver
- Demonstrates cooperative intent
- Provides room for mutually beneficial returns
- Has proven effective in previous rounds

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3, received \$5.09 back - good return
- Round 4: Sent \$3, received \$3.66 back - fair return

Both times I sent \$3, I received positive returns that made the exchange worthwhile.

**\*\*Current situation:\*\***

- New co-player this round
- No specific information about their reciprocity
- My moderate sending strategy has been working well

**\*\*Strategy:\*\***

- Sending \$3 has proven to be a good balanced approach
- Creates \$9 for receiver, creating room for mutual benefit
- Shows cooperation without risking everything
- Consistent with my established pattern

I'll continue with \$3 as it's been effective.

**usage:**

```
{
 "input_tokens": 2072,
 "output_tokens": 350,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.32 to you. Your visible payoff was \$4.32.

Your total visible earnings across all rounds are \$30.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 6: I sent \$3, they returned \$2.32 (about 26% of what they received)

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They sent \$3.70 (74% of their endowment) - high trust!
- I received \$11.10
- This created \$7.40 in surplus

**\*\*Considerations:\*\***

1. They're showing high trust now by sending 74%
2. Despite lower return last round, they're signaling cooperation
3. Should reciprocate fairly to maintain the relationship

If I return \$6:

- They end with: \$5 - \$3.70 + \$6 = \$7.30 (profit: \$2.30)
- I end with: \$11.10 - \$6 = \$5.10 (profit: \$5.10)
- Both parties benefit significantly

This rewards their trust and encourages continued cooperation.

{"return": 6}

**Reasoning:**

**\*\*Analysis of the situation:\*\***

Previous round (Round 6):

- I sent \$3 to this co-player
- They received \$9
- They returned \$2.32 to me
- This was a somewhat low return (about 26% of what they received)

Current round (Round 7):

- Same co-player (continuing interaction)
- They sent me \$3.70 (74% of their endowment)
- I received \$11.10
- This is a strong trust signal

**\*\*Considerations:\*\***

1. They showed high trust by sending \$3.70 (74%)
2. In the previous round, they returned less than I might have hoped (\$2.32 from \$9)
3. This is a continued interaction, so my response matters for future cooperation
4. I should be fair and reciprocate their trust appropriately

**\*\*Strategic thinking:\*\***

Even though they returned less in Round 6, they're now showing high trust by sending \$3.70. I should:

- Reciprocate fairly to maintain cooperation
- Not punish them excessively for the previous lower return
- Create mutual benefit

If I return \$6:

- They end with:  $\$5 - \$3.70 + \$6 = \$7.30$  (profit of \$2.30)
- I end with:  $\$11.10 - \$6 = \$5.10$  (profit of \$5.10)
- Both profit significantly

This rewards their current trust while maintaining fair reciprocity.

Usage: {

"input\_tokens": 2393,  
"output\_tokens": 586,  
"reasoning\_tokens": 0

}

**GAME - Agent\_4 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:39:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_3",  
"move\_index": 4,  
"opponent\_id": "Agent\_3",  
"pairing": {  
  "agents": [  
    "Agent\_4",  
    "Agent\_3"  
  ],  
  "investor": "Agent\_4",  
  "roles": {  
    "Agent\_3": "trustee",  
    "Agent\_4": "investor"  
  },  
  "trustee": "Agent\_3"  
},  
"role": "investor",  
"role\_label": "Sender",  
"round": 7,  
"task": "game",  
"task\_index": 0

}

**Exact input messages sent to model:**

Message 1 role=system  
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.49, that is 49.8% of their total endowment. You received \$7.470000000000001. How much do you return to the sender? (0-7.470000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.49 (nearly 50% of their endowment)
- I received \$7.47 after the 3x multiplication
- The sender currently has \$2.51 remaining

Since this is a multi-round game and the sender demonstrated trust, I want to reciprocate in a way that:

1. Gives them a positive return on their trust
2. Maintains my own profit
3. Signals I'm a cooperative player for future rounds

If I return around \$3.50:

- Sender ends with: \$2.51 + \$3.50 = \$6.01 (profit of ~\$1)
- I keep: \$7.47 - \$3.50 = \$3.97 (profit of ~\$4)

This seems fair - we both profit, with me getting slightly more since I'm the receiver, but the sender is rewarded for their trust.

```
{"return": 3.50}
```

**reasoning:**

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.49 (49.8% of their \$5 endowment)
- I received  $\$2.49 \times 3 = \$7.47$
- I need to decide how much to return (0 to \$7.47)

Key considerations:

1. The sender sent approximately half of their endowment, showing significant trust
2. If I return a fair amount, I encourage future cooperation
3. The sender would have kept  $\$5 - \$2.49 = \$2.51$  if they sent nothing
4. They're risking their money on trust that I'll return something

Common approaches in trust games:

- Equal split of final amounts: The sender currently has \$2.51, I have \$7.47. Total is \$10. Equal would be \$5 each, so I'd return \$2.49 to give them \$5 total.
- Return what was sent: Return \$2.49
- Return the multiplied amount divided equally: Return around \$3.735 (half of what I received)
- Maximize my own payoff: Return \$0 (but this destroys trust)

Given that:

1. This is a multi-round game where I may play with others or the same person again
2. The sender showed trust by sending nearly half their endowment
3. Building a reputation for reciprocity could benefit me in future rounds
4. There's communication noise mentioned, so exact amounts may vary

I should return a reasonable amount that:

- Rewards the sender's trust
- Ensures they end up with more than if they had sent nothing (\$2.51)
- Leaves me with a profit as well
- Signals I'm a cooperative player

If I return \$2.50:

- Sender gets:  $\$2.51 + \$2.50 = \$5.01$  (almost their original endowment back)
- I keep:  $\$7.47 - \$2.50 = \$4.97$

Returning \$3.50 would give the sender \$6.01 total (a \$1.01 gain) while I'd have \$3.97. Going higher to \$4.00 means they'd end up with \$6.51 and I'd keep \$3.47. The goal is finding a fair split that acknowledges their trust while still rewarding my cooperation.

The \$4.98 bonus from the tripling could be divided equally-giving them \$5.00 and me \$4.98

...

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1072,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became  $\$7.4700000000000001$ , and you returned \$3.5. Your visible payoff was  $\$3.9700000000000006$ .

Your total visible earnings across all rounds are  $\$3.9700000000000006$ .

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round 1.

```
{"send": 2.50}
```

**reasoning:**

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

**usage:**

```
{
 "input_tokens": 897,
 "output_tokens": 538,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative

and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

**reasoning:**

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff:  $\$2.50 + \$3.93 = \$6.43$

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

**usage:**

```
{
 "input_tokens": 1153,
 "output_tokens": 447,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with:  $\$0.52 + \$7.00 = \$7.52$  (profit of ~\$2.52)
- I keep:  $\$13.44 - \$7.00 = \$6.44$  (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

```
{"return": 7.00}
```

**reasoning:**

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with:  $\$0.52 + \$6.50 = \$7.02$  (profit of ~\$2)
- I would keep:  $\$13.44 - \$6.50 = \$6.94$  (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with:  $\$0.52 + \$7 = \$7.52$  (profit of \$2.52)
- I keep:  $\$13.44 - \$7 = \$6.44$  (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be:  $\$13.44 * 0.5 = \$6.72$

...

usage:

```
{
 "input_tokens": 1412,
 "output_tokens": 682,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001.

Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of ~\$2.22)
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of ~\$6.34)



This maintains a similar return rate (~51%) as Round 4 (~52%) and both of us profit well.

```
{"return": 6.50}
```

**reasoning:**

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocation pattern for high trust.

The sender showed ~86% trust level, which is slightly less than the ~90% in Round 4, but still very high.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of ~\$2.22)
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of ~\$6.34)

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

**usage:**

```
{
 "input_tokens": 1745,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.28, it became \$12.84, and you returned \$6.5. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$28.540000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my history:

- As sender (Rounds 2-3): I've consistently sent \$2.50 (50%), with varying returns
- As receiver (Rounds 1, 4-5): I've reciprocated well when others showed trust

My 50% sending strategy has been reasonably successful, giving both parties opportunities to profit. While some agents have sent higher amounts (85-90%) when they're senders, I'll maintain my consistent approach that balances trust and risk.

```
{"send": 2.50}
```

**reasoning:**

Let me review my history:

Round 1 (RECEIVER): Received \$7.47 from sender who sent \$2.49 (50%), returned \$3.50 (~47% of what I received). Payoff: \$3.97

Round 2 (SENDER): Sent \$2.50 (50%), received \$3.93 back. Payoff: \$6.43

Round 3 (SENDER): Sent \$2.50 (50%), received \$2.86 back. Payoff: \$5.36

Round 4 (RECEIVER): Received \$13.44 from sender who sent \$4.48 (90%), returned \$7.00 (~52% of what I received). Payoff: \$6.44

Round 5 (RECEIVER): Received \$12.84 from sender who sent \$4.28 (86%), returned \$6.50 (~51% of what I received). Payoff: \$6.34

Now Round 6 (SENDER): I need to decide how much to send.

Looking at the pattern:

- When I've been sender, I've sent \$2.50 (50%) consistently
- When I'm receiver, I've gotten good returns when others send high amounts (50-90%)
- I've been rewarded for reciprocating high trust with good returns

However, I've noticed that when I sent 50% in rounds 2 and 3, I got varying returns:

- Round 2: \$3.93 (very good return, about 52% of what they received)
- Round 3: \$2.86 (lower return, about 38% of what they received)

Other agents have been sending higher amounts (85-90%) to me when they're senders, and I've reciprocated well. This has worked out for both parties.

Should I increase my sending? Or stick with my 50% strategy?

...

**usage:**

```
{
 "input_tokens": 2073,
 "output_tokens": 696,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.76 to you. Your visible payoff was \$7.26.

Your total visible earnings across all rounds are \$35.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my sending history:

- Round 2: Sent \$2.50, received \$3.93 back
- Round 3: Sent \$2.50, received \$2.86 back
- Round 6: Sent \$2.50, received \$4.76 back

The 50% strategy continues to work well on average. Round 6 showed particularly good reciprocity with a \$4.76 return. I'll maintain this consistent, moderate trust level.

```
{"send": 2.50}
```

**Reasoning:**

Looking at my history:

Round 2 (SENDER): Sent \$2.50, received \$3.93 back, payoff \$6.43  
Round 3 (SENDER): Sent \$2.50, received \$2.86 back, payoff \$5.36  
Round 6 (SENDER): Sent \$2.50, received \$4.76 back, payoff \$7.26

As sender, my 50% strategy has been working reasonably well. In Round 6, I received an excellent return of \$4.76.

As receiver:

Round 1: Received \$7.47, returned \$3.50  
Round 4: Received \$13.44, returned \$7.00  
Round 5: Received \$12.84, returned \$6.50

I've been consistent and cooperative. My 50% sending strategy has been reasonable and I should continue with it.

Usage: {

```
"input_tokens": 2318,
```

```
"output_tokens": 335,
"reasoning_tokens": 0
}
```

### GAME - Agent\_3 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:39:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to the sender? (0-11.67)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- This created a total pool of \$12.78 (\$1.11 remaining with sender + \$11.67 with me)
- The sender took a risk to create mutual benefit

To reciprocate their trust and be fair, I'll aim for an approximately equal split of the total resources created. If I return about \$5.28, we'd both end up with roughly \$6.39.

This rewards their generosity while maintaining fairness.

{'return': 5.28}

**reasoning:**

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender started with \$5
- The sender sent \$3.89 (which is 77.8% of their endowment)
- This amount was multiplied by 3x, so I received  $\$3.89 \times 3 = \$11.67$
- I need to decide how much to return (between \$0 and \$11.67)

The sender was quite generous, sending almost 78% of their endowment. This shows trust.

Let me think about what would be fair:

- The sender will end up with:  $\$5 - \$3.89 + (\text{what I return}) = \$1.11 + (\text{what I return})$
- I will end up with:  $\$11.67 - (\text{what I return})$

If I want to be fair and reciprocate the trust:

- The sender sent \$3.89 and it became \$11.67 for me (a gain of \$7.78)
- A common fair split would be to share the gains
- Initial positions: Sender has \$5, I have \$0
- After the transaction: Total =  $\$5 - \$3.89 + \$11.67 = \$12.78$
- If we split equally: each gets \$6.39
- For the sender to get \$6.39: they need  $\$1.11$  (what they kept) + what I return = \$6.39
- So I should return:  $\$6.39 - \$1.11 = \$5.28$

Another approach - match their generosity ratio:

- They sent 77.8% of their endowment
- If I return 77.8% of what I received:  $\$11.67 \times 0.778 = \$9.08$
- That seems very high and would leave me with only \$2.59

A more moderate approach - return about half:

- Return around \$5-6 would be reasonable
- This leaves both of us with roughly \$6-7 each. Let me verify the equal split calculation: if I return \$5.28, the sender ends with \$6.39 and I keep \$6.39. But I should double-check my math on the total pool-the sender's initial \$5 plus the \$11.67 I received equals \$16.67 total, though the multiplication created value in the system....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 897,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

{'send': 4.00}

**reasoning:**

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with  $\$1.11 + \$5.28 = \$6.39$ , and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with  $\$1.00 + \$5.40 = \$6.40$
- They'd end up with  $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with  $\$1.25 + \$5.06 = \$6.31$
- They'd end up with  $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

**usage:**

```
{
 "input_tokens": 787,
 "output_tokens": 478,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

```
{'send': 4.00}
```

**reasoning:**

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was  $\$5 - \$4 + \$5.20 = \$6.20$

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

**usage:**

```
{
 "input_tokens": 970,
 "output_tokens": 251,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created:  $\$0.63$  (with sender) +  $\$13.11$  (with me) =  $\$13.74$
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

```
{'return': 6.24}
```

**reasoning:**

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has  $\$5 - \$4.37 = \$0.63$

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created:  $\$0.63$  (sender has) +  $\$13.11$  (I have) =  $\$13.74$

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have:  $\$0.63 + \text{return} = \$6.87$ , so return =  $\$6.24$
- I would have:  $\$13.11 - \$6.24 = \$6.87$

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)
- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

```
{'return': 5.90}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender: \$0.80 + \$6.30 = \$7.10
- Me: \$12.60 - \$6.30 = \$6.30

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return: \$13.40 - \$6.70 = \$6.70

- Sender would get: \$0.80 + \$6.70 = \$7.50
- I would get: \$12.60 - \$6.70 = \$5.90

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets: \$0.80 + X
- I get: \$12.60 - X

For equal outcomes:

$$\$0.80 + X = \$12.60 - X$$

$$2X = \$12.60 - \$0.80$$

$$2X = \$11.80$$

$$X = \$5.90$$

Check:

$$\text{- Sender: } \$0.80 + \$5.90 = \$6.70$$

$$\text{- Me: } \$12.60 - \$5.90 = \$6.70$$

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.2, it became \$12.600000000000001, and you returned \$5.9. Your visible payoff was \$6.700000000000001.

Your total visible earnings across all rounds are \$31.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my experience across rounds, sending around \$4.00 has consistently led to cooperative outcomes. I'll continue with this trusting strategy.

```
{'send': 4.00}
```

**reasoning:**

In this round, I'm the sender. Let me review my recent experiences:

Round 1: I was receiver, sender sent \$3.89, I returned \$5.28, both got ~\$6.39

Round 2: I sent \$4.00, received back \$5.20, got \$6.20

Round 3: I sent \$4.00, received back \$4.73, got \$5.73

Round 4: I was receiver, sender sent \$4.37, I returned \$6.24, both got ~\$6.87

Round 5: I was receiver, sender sent \$4.20, I returned \$5.90, both got \$6.70

When I've been a receiver, I've been fairly generous in my returns, aiming for equal splits. When I've been a sender, I've sent \$4.00 and received reasonable returns.

My strategy has been cooperative and trusting. I should continue with this approach, sending a high amount to show trust and hopefully receive a fair return.

Sending \$4.00 has worked reasonably well in the past. I'll continue with this level.

**usage:**

```
{
 "input_tokens": 1662,
 "output_tokens": 301,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.91 to you. Your visible payoff was \$8.91.

Your total visible earnings across all rounds are \$40.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)



Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment), showing moderate trust
- Total pool created: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- This is less trust than most previous rounds

I'll still reciprocate fairly. Returning \$2.00 would give us both approximately \$4.75.

{'return': 2.00}

**Reasoning:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment)
- This is notably less than previous senders I've encountered who sent around 77-87%
- Total pool: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- They showed moderate trust compared to others

Since they sent less than previous players, I should still reciprocate fairly but proportionally to their trust level.

For an equal split: each gets \$4.74

That means I'd return  $\$2.76 + (\text{total} - \text{sender's remaining}) / 2 = \dots$  let me recalculate

Sender has:  $\$5 - \$2.24 = \$2.76$

I received: \$6.72

Total created:  $\$2.76 + \$6.72 = \$9.48$

For equal split of \$4.74 each:

Sender needs:  $\$4.74 - \$2.76 = \$1.98$  returned

I keep:  $\$6.72 - \$1.98 = \$4.74$

But given they showed less trust than previous players, maybe I should be slightly less generous. However, I should still be fair and cooperative.

I'll return around \$2.00-\$2.50 to maintain fairness.

Let me go with \$2.30, which gives:

- Sender:  $\$2.76 + \$2.30 = \$5.06$
- Me:  $\$6.72 - \$2.30 = \$4.42$

This is slightly favoring them which encourages cooperation but not as generous as when they send more.

Actually, let me reconsider. I've been fairly cooperative throughout. Even though they sent less, being consistent in fairness might be better. Let me aim closer to equal:

...

Usage: {

"input\_tokens": 1846,

"output\_tokens": 596,

"reasoning\_tokens": 0

}

**GAME - Agent\_5 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:39:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_4",

"move\_index": 6,

"opponent\_id": "Agent\_2",

"pairing": {

"agents": [

"Agent\_5",

"Agent\_2"

],

"investor": "Agent\_5",

"roles": {

"Agent\_2": "trustee",

```

 "Agent_5": "investor"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

**\*\*Round 1 - SENDER Decision\*\***

I'm the sender with \$5. Since this is the first round and I have no information about my current co-player, I'll make a decision that signals cooperation while managing risk.

If I send money, it gets tripled, creating potential for mutual benefit. Sending \$3 would give my co-player \$9 to work with. If they return a fair portion, we both can end up better off than the baseline.

This seems like a reasonable opening move that tests for cooperation without maximum exposure.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a multi-agent trust game. I'm the SENDER with \$5.

Let me think about the strategic considerations:

1. **\*\*Trust Game Dynamics\*\***: If I send money, it gets tripled, creating a larger pot that could benefit both of us if the receiver returns some. This is the classic trust game set up.
2. **\*\*First Round - No History\*\***: I have no information about this particular receiver since agents are randomly paired and names are hidden. I don't know their strategy or trustworthiness.
3. **\*\*Incentive Structure\*\***:
  - If I send \$X, receiver gets \$3X
  - Receiver can return \$0 to \$3X
  - My final payoff:  $\$5 - X + (\text{amount returned})$
  - Receiver's final payoff:  $\$3X - (\text{amount returned})$
4. **\*\*Optimal Cooperation\*\***: If both parties cooperate optimally, sending the maximum (\$5) creates \$15 for the receiver. If they return half (\$7.50), we both end up with \$7.50, which is better than the \$5 baseline.
5. **\*\*Risk\*\***: The receiver could keep everything, leaving me with \$0. But this is a multi-round game with different pairings, so building a reputation for cooperation might be valuable (though names are hidden, so direct reputation effects are limited).
6. **\*\*Opening Strategy\*\***: Common approaches for first rounds:
  - Send a moderate amount to test trustworthiness (e.g., \$2-3)
  - Send maximum to signal cooperation and trust
  - Send nothing to avoid risk

Given that:

- This is explicitly a multi-task experiment studying behavior
- There's communication noise mentioned (interesting detail)
- It's a multi-agent setting with repeated interactions
- I want to establish a cooperative baseline

I think a reasonable first-round strategy is to send a moderate amount that signals willingness to cooperate while not exposing myself to maximum loss. Sending \$3 seems like a good middle ground:

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 652,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$4.93.

This round, you are the RECEIVER against your current co-player. They sent you \$2.82, so you received \$8.459999999999999. How much do you return? (0-8.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Round 2 - RECEIVER Decision\*\***

Looking at this situation:

- My current co-player sent me \$2.82 (56% of their endowment) - a substantial show of trust
- I received \$8.46 after the 3x multiplier
- They currently have \$2.18 remaining

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
  - They end up with:  $\$2.18 + \$4.00 = \$6.18$
  - I end up with:  $\$8.46 - \$4.00 = \$4.46$
  - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

#### reasoning:

Let me analyze this situation:

**\*\*Round 1 recap:\*\***

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff:  $\$5 - \$3 + \$2.93 = \$4.93$
- Their payoff:  $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

**\*\*Round 2:\*\***

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept:  $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie:  $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return:  $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return:  $\$8.46 \times 0.33 = \$2.79$

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

#### usage:

```
{
```

```
"input_tokens": 775,
"output_tokens": 861,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 3 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

**usage:**

```
{
 "input_tokens": 1105,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$ 5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 4 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

**\*\*Round 4 - RECEIVER Decision\*\***

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have  $\$5 - \$2.84 = \$2.16$  remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately

**usage:**

```
{
 "input_tokens": 1398,
 "output_tokens": 518,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 5 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything  
. I should reciprocate proportionally:

- If I return \$6.50:
  - They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
  - I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)
  - Both benefit substantially, rewarding their exceptional trust

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust  
I received \$12.96 after the 3x multiplier  
They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount.  
Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
  - They end up with:  $\$0.68 + \$6.00 = \$6.68$
  - I end up with:  $\$12.96 - \$6.00 = \$6.96$
  - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
  - They end up with:  $\$0.68 + \$7.00 = \$7.68$
  - I end up with:  $\$12.96 - \$7.00 = \$5.96$
  - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
- I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

**usage:**

```
{
 "input_tokens": 1715,
 "output_tokens": 645,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.32, it became \$12.96, and you returned \$6.5. Your visible payoff was \$6.460000000000001.

Your total visible earnings across all rounds are \$26.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant****content:**

**\*\*Round 6 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My experience as sender has been positive:

- Round 1: Sent \$3, got \$2.93 back
- Round 3: Sent \$3, got \$3.75 back

Both times sending \$3 led to reasonable cooperation. This creates \$9 for the receiver, allowing room for mutual benefit. I'll continue this consistent approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Round 6 - I'm the SENDER with \$5.

Let me review my experience so far:

As SENDER:

- Round 1: Sent \$3, received ~\$2.93 back
- Round 3: Sent \$3, received \$3.75 back

As RECEIVER:

- Round 2: Received \$2.82? \$8.46, returned \$4.00
- Round 4: Received \$2.84? \$8.52, returned \$4.00
- Round 5: Received \$4.32? \$12.96, returned \$6.50

The game has been showing good cooperation overall. As a sender, I've gotten decent returns (around \$3-\$3.75 on \$3 sent). As a receiver, I've been fair and reciprocal.

Now I'm facing a new co-player. The cooperation has been working well. I could:

1. Continue with \$3 (my previous sender strategy)
2. Send more to signal stronger trust (like the \$4.32 I saw in Round 5)
3. Send less to be more cautious

Given the positive cooperation trends and that this is a new co-player, I'll stick with my consistent sender strategy of \$3. This has worked well and creates good mutual benefit opportunities.

**usage:**

```
{
 "input_tokens": 2048,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.38 to you. Your visible payoff was \$5.38.

Your total visible earnings across all rounds are \$31.499999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Round 7 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My track record as sender:

- Round 1: Sent \$3 ? received \$2.93 back (profit of \$0.93)
- Round 3: Sent \$3 ? received \$3.75 back (profit of \$0.75)
- Round 6: Sent \$3 ? received \$3.38 back (profit of \$0.38)

Sending \$3 has consistently worked well, showing decent cooperation from different co-players.



I'll stick with this strategy that balances trust and risk management.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

I'm the sender again with \$5.

Looking at my history as sender:

- Round 1: Sent \$3, got \$2.93 back (visible payoff \$4.93)
- Round 3: Sent \$3, got \$3.75 back (visible payoff \$5.75)
- Round 6: Sent \$3, got \$3.38 back (visible payoff \$5.38)

All three times I sent \$3, and I got decent returns ranging from about \$2.93 to \$3.75. This strategy has been working well - I'm consistently making a small profit.

I'm facing a new co-player now. Sending \$3 has been my consistent strategy and it's been working. I should continue with this.

```
Usage: {
 "input_tokens": 2294,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:39:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_2"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_2": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed significant trust by sending 66.8% of their endowment (\$3.34)
- I received \$10.02 after the 3x multiplication
- The sender currently has only \$1.66 left (\$5 - \$3.34)

I want to reward this trust while ensuring we both benefit from the exchange.

If I return \$5.00:

- The sender ends with:  $\$1.66 + \$5.00 = \$6.66$  (a profit of \$1.66)
- I end with:  $\$10.02 - \$5.00 = \$5.02$  (a profit of \$5.02)
- Both of us profit, and the sender's trust is rewarded

This seems like a fair reciprocation that encourages future cooperation.

```
{"return": 5.00}
```

##### reasoning:

Let me analyze this situation as the receiver in Round 1.

Given information:

- I am the RECEIVER
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

Key considerations:

1. The sender showed significant trust by sending 66.8% of their endowment
2. If I return nothing, I keep \$10.02 and the sender keeps only \$1.66 (\$5 - \$3.34)
3. A fair split might consider that we should both benefit
4. I should reward the sender's trust to encourage cooperation in future rounds

Let me think about some return strategies:

If I return \$0:

- Sender gets:  $\$5 - \$3.34 + \$0 = \$1.66$
- I get:  $\$10.02 - \$0 = \$10.02$

- Total: \$11.68

If I return \$5:

- Sender gets:  $\$5 - \$3.34 + \$5 = \$6.66$
- I get:  $\$10.02 - \$5 = \$5.02$
- Total: \$11.68

If I return enough so we're equal:

- Sender should get: X
- I should get: X
- $X + X = \$11.68$
- $X = \$5.84$
- So I should return:  $\$5.84 - \$1.66 = \$4.18$

If I return \$4.18:

- Sender gets:  $\$1.66 + \$4.18 = \$5.84$
- I get:  $\$10.02 - \$4.18 = \$5.84$
- We split equally

However, another approach is to think about rewarding trust proportionally.

The sender's substantial commitment deserves reciprocation. I could return roughly half of what I received (\$5.01), which would give the sender \$6.67 while I keep \$5.01-both of us profit from the exchange. Alternatively, returning \$5 ensures the sender ends with \$6.66, exceeding their initial endowment while I still gain \$5.02.

This approach balances rewarding their trust with securing my own benefit. Even accounting for potential communication noise affecting the actual amount sent, responding proportionally to what I observe seems fair. I'll return \$5.00....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1155,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

{'send': 3.50}

reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation
3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

usage:

```
{
 "input_tokens": 846,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

This rewards their trust and maintains cooperation.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has  $\$5 - \$2.52 = \$2.48$  left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again:  $\$7.56 * 0.5 = \$3.78$

Let me check the outcomes:

- If I return \$3.78:
  - Sender ends with:  $\$2.48 + \$3.78 = \$6.26$  (profit of \$1.26)
  - I end with:  $\$7.56 - \$3.78 = \$3.78$  (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

...

**usage:**

```
{
 "input_tokens": 1072,
 "output_tokens": 643,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

```
{'send': 3.50}
```

**reasoning:**

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

**usage:**

```
{
 "input_tokens": 1400,
 "output_tokens": 305,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was

\$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it.

{'send': 3.50}

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well. I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

**usage:**

```
{
 "input_tokens": 1587,
 "output_tokens": 211,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.5 to you. Your visible payoff was \$8.0.

Your total visible earnings across all rounds are \$33.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.40 (88% of their endowment)
- I received \$13.20 after the 3x multiplication
- The sender currently has only \$0.60 left

This is the highest trust I've seen so far, exceeding even Round 1 (66.8%). I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.60 + \$7.00 = \$7.60 (profit of \$2.60)
- I end with: \$13.20 - \$7.00 = \$6.20 (profit of \$6.20)

This rewards their exceptional trust while ensuring we both profit well.

{'return': 7.00}

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.40 (88% of their endowment) - this is very high trust
- I received \$13.20 after the 3x multiplication
- The sender currently has \$0.60 left (\$5 - \$4.40)

This sender showed exceptional trust by sending 88% of their endowment. I should reciprocate generously.

Looking at my past behavior as receiver:

- Round 1: Received \$10.02, returned \$5.00 (about 50%)
- Round 3: Received \$7.56, returned \$4.00 (about 53%)

For this higher trust situation, I should return more generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust significantly while still maintaining good profit for both.

Alternatively, if I return \$6.50:

- Sender ends with:  $\$0.60 + \$6.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$13.20 - \$6.50 = \$6.70$  (profit of \$6.70)

I think \$7.00 is appropriate given the very high trust shown (88% vs the 70% I've been sending).

usage:

```
{
 "input_tokens": 1827,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.4, it became \$13.200000000000001, and you returned \$7.0. Your visible payoff was \$6.200000000000001.

Your total visible earnings across all rounds are \$39.46000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.91, so you received \$11.73. How much do you return? (0-11.73)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left

This is high trust, similar to previous high-trust rounds. Comparing to my history:

- Round 1: 66.8% sent ? returned \$5.00
- Round 6: 88% sent ? returned \$7.00

At 78.2%, this falls between those two. I'll return proportionally.

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

{'return': 6.50}

Reasoning:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left (\$5 - \$3.91)

This is high trust, though not quite as high as the 88% I saw in Round 6. Let me compare to my previous responses:

Round 1: Sender sent 66.8%, I received \$10.02, I returned \$5.00 (about 50%)

Round 3: Sender sent 50.4%, I received \$7.56, I returned \$4.00 (about 53%)

Round 6: Sender sent 88%, I received \$13.20, I returned \$7.00 (about 53%)

This sender sent 78.2%, which is between Round 1 (66.8%) and Round 6 (88%).

If I return \$6.00:

- Sender ends with:  $\$1.09 + \$6.00 = \$7.09$  (profit of \$2.09)
- I end with:  $\$11.73 - \$6.00 = \$5.73$  (profit of \$5.73)

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

Given the high trust level (78.2%), I think \$6.50 is appropriate, falling between my Round 1 return (\$5.00 for 66.8%) and Round 6 return (\$7.00 for 88%).

```
Usage: {
 "input_tokens": 2156,
 "output_tokens": 639,
 "reasoning_tokens": 0
}
```

## ROUND 8

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Receiver, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_7=Sender, Agent\_6=Receiver  
dyad\_2: Agent\_8=Sender, Agent\_1=Receiver  
dyad\_3: Agent\_2=Sender, Agent\_4=Receiver  
dyad\_4: Agent\_3=Sender, Agent\_5=Receiver

### Parsed game decisions:

dyad\_1: Agent\_7 sent \$3; received \$9; Agent\_6 returned \$5.45; payoffs Agent\_7=\$7.45, Agent\_6=\$3.55; noise/visible: sent visible=\$2.92, received visible=\$8.76, returned visible=\$5.14  
Action validation: {"returned": {"amount": 5.45, "clamped": false, "max": 9.0, "min": 0, "raw": 5.45}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_2: Agent\_8 sent \$4; received \$12; Agent\_1 returned \$5; payoffs Agent\_8=\$6, Agent\_1=\$7; noise/visible: sent visible=\$3.88, received visible=\$11.64, returned visible=\$4.24  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 12.0, "min": 0, "raw": 5.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad\_3: Agent\_2 sent \$3.5; received \$10.5; Agent\_4 returned \$5; payoffs Agent\_2=\$6.5, Agent\_4=\$5.5; noise/visible: sent visible=\$3.41, received visible=\$10.23, returned visible=\$5.32  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad\_4: Agent\_3 sent \$4; received \$12; Agent\_5 returned \$5.5; payoffs Agent\_3=\$6.5, Agent\_5=\$6.5; noise/visible: sent visible=\$3.72, received visible=\$11.16, returned visible=\$5.28  
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

### Balances after round: {

"Agent\_1": 53.15,  
"Agent\_2": 46.74,  
"Agent\_3": 44.83,  
"Agent\_4": 43.6,  
"Agent\_5": 48.25,  
"Agent\_6": 43.699999999999996,  
"Agent\_7": 43.230000000000004,  
"Agent\_8": 49.5

}

### Visible balances after round: {

"Agent\_1": 52.86,  
"Agent\_2": 51.510000000000001,  
"Agent\_3": 51.8,  
"Agent\_4": 45.630000000000001,  
"Agent\_5": 46.19,  
"Agent\_6": 43.699999999999996,  
"Agent\_7": 42.290000000000006,  
"Agent\_8": 44.550000000000004

}

## LLM CALLS

GAME - Agent\_7 - Sender - dyad\_1

Timestamp: 2026-08-12T09:39:42



Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_6"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.09 to you. Your visible payoff was \$7.09.

Your total visible earnings across all rounds are \$7.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.08, so you received \$6.24. How much do you return? (0-6.24)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.08 (about 42% of their \$5 endowment)
- I received \$6.24
- This created a gain of \$4.16

**\*\*Considerations:\*\***

1. In the previous round, my co-player returned about 57% of what they received, which was generous
2. My current co-player showed moderate trust by sending \$2.08
3. I should reciprocate fairly to encourage cooperation

**\*\*Fair reciprocation:\*\***

If I return \$3.50:

- They end with:  $\$5 - \$2.08 + \$3.50 = \$6.42$  (profit of \$1.42)
- I end with:  $\$6.24 - \$3.50 = \$2.74$  (profit of \$2.74)
- This rewards their trust and creates mutual gain

{"return": 3.50}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$5.09 to me
- My payoff was  $\$5 - \$3 + \$5.09 = \$7.09$

That was pretty generous! They kept \$3.91 out of \$9.00 and returned \$5.09. That's a good return ratio.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- My co-player sent me \$2.08
- I received \$6.24 ( $2.08 \times 3 = 6.24$ )
- I need to decide how much to return

Important notes:

- This is a different co-player (agents are paired randomly each round)
- There's communication noise, so amounts may differ slightly
- My co-player sent \$2.08, which is about 42% of their \$5 endowment

I should consider:

1. **\*\*Fairness\*\***: A fair split might be sharing the gains equally
2. **\*\*Reciprocity\*\***: Building trust for potential future interactions
3. **\*\*Economic incentive\*\***: They sent \$2.08, I got \$6.24. The gain created is  $\$6.24 - \$2.08 = \$4.16$

If I want to be fair and split the gains:

- They started with \$5, sent \$2.08, so they have \$2.92 left
- I received \$6.24
- Total pool:  $\$2.92 + \$6.24 = \$9.16$

- Fair split would be around \$4.58 each
- To give them \$4.58, I'd need to return:  $\$4.58 - \$2.92 = \$1.66$

But that seems low compared to what I received in the previous round. Let me think about this differently.

Another approach - proportional return:

My previous partner returned about 56.6% of what they received ( $\$5.09/\$9.00$ ), so if I match that generosity here, I'd return roughly \$3.53 of the \$6.24. Alternatively, I could ensure they profit meaningfully-returning \$3.50 would give them \$6.42 total (a \$1.42 gain), while I'd keep \$2.74. The key is balancing reciprocity with what feels genuinely fair given their initial trust....

**usage:**

```
{
 "input_tokens": 795,
 "output_tokens": 946,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Analysis:\*\***

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

**\*\*Considerations:\*\***

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with:  $\$5 - \$4.18 + \$7 = \$7.82$  (profit of \$2.82)
- I end with:  $\$12.54 - \$7 = \$5.54$  (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

```
{"return": 7}
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of  $\$12.54 - \$4.18 = \$8.36$

**\*\*Previous rounds context:\*\***

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

**\*\*Considerations:\*\***

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

**\*\*Options:\*\***

If I return \$6:

- They end with:  $\$5 - \$4.18 + \$6 = \$6.82$  (profit of \$1.82)

- I end with:  $\$12.54 - \$6 = \$6.54$  (profit of \$6.54)

If I return \$7:

- They end with:  $\$5 - \$4.18 + \$7 = \$7.82$  (profit of \$2.82)
- I end with:  $\$12.54 - \$7 = \$5.54$  (profit of \$5.54)

If I return \$8:

- They end with:  $\$5 - \$4.18 + \$8 = \$8.82$  (profit of \$3.82)
- I end with:  $\$12.54 - \$8 = \$4.54$  (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

usage:

```
{
 "input_tokens": 1143,
 "output_tokens": 671,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous sending experience:\*\***

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

**\*\*Strategy:\*\***

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

```
{"send": 3}
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

**\*\*Strategic considerations:\*\***

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09

2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

**\*\*Options:\*\***

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 529,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)
- Both parties gain significantly

This rewards their trust while creating mutual benefit.

```
{"return": 5.50}
```

**reasoning:**

**\*\*Analysis:\*\***

**\*\*Previous round:\*\***

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received
- This was fair but not overly generous

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with:  $\$5 - \$3.40 + \$5.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$10.20 - \$5.50 = \$4.70$  (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

**usage:**

```
{
 "input_tokens": 1755,
 "output_tokens": 512,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.4, it became \$10.2, and you returned \$5.5. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$25.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

Both instances were profitable, showing that \$3 is a reasonable amount that encourages reciprocity.

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their behavior
- My consistent moderate strategy has been working

**\*\*Strategy:\*\***

Sending \$3:

- Creates \$9 for the receiver
- Demonstrates cooperative intent
- Provides room for mutually beneficial returns
- Has proven effective in previous rounds

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3, received \$5.09 back - good return
- Round 4: Sent \$3, received \$3.66 back - fair return

Both times I sent \$3, I received positive returns that made the exchange worthwhile.

**\*\*Current situation:\*\***

- New co-player this round
- No specific information about their reciprocity

- My moderate sending strategy has been working well

**\*\*Strategy:\*\***

- Sending \$3 has proven to be a good balanced approach
- Creates \$9 for receiver, creating room for mutual benefit
- Shows cooperation without risking everything
- Consistent with my established pattern

I'll continue with \$3 as it's been effective.

**usage:**

```
{
 "input_tokens": 2072,
 "output_tokens": 350,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.32 to you. Your visible payoff was \$4.32.

Your total visible earnings across all rounds are \$30.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 6: I sent \$3, they returned \$2.32 (about 26% of what they received)

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They sent \$3.70 (74% of their endowment) - high trust!
- I received \$11.10
- This created \$7.40 in surplus

**\*\*Considerations:\*\***

1. They're showing high trust now by sending 74%
2. Despite lower return last round, they're signaling cooperation
3. Should reciprocate fairly to maintain the relationship

If I return \$6:

- They end with: \$5 - \$3.70 + \$6 = \$7.30 (profit: \$2.30)
- I end with: \$11.10 - \$6 = \$5.10 (profit: \$5.10)
- Both parties benefit significantly

This rewards their trust and encourages continued cooperation.

{"return": 6}

**reasoning:**

**\*\*Analysis of the situation:\*\***

Previous round (Round 6):

- I sent \$3 to this co-player
- They received \$9
- They returned \$2.32 to me
- This was a somewhat low return (about 26% of what they received)

Current round (Round 7):

- Same co-player (continuing interaction)
- They sent me \$3.70 (74% of their endowment)
- I received \$11.10
- This is a strong trust signal

**\*\*Considerations:\*\***

1. They showed high trust by sending \$3.70 (74%)

2. In the previous round, they returned less than I might have hoped (\$2.32 from \$9)
3. This is a continued interaction, so my response matters for future cooperation
4. I should be fair and reciprocate their trust appropriately

**\*\*Strategic thinking:\*\***

Even though they returned less in Round 6, they're now showing high trust by sending \$3.70  
 . I should:

- Reciprocate fairly to maintain cooperation
- Not punish them excessively for the previous lower return
- Create mutual benefit

If I return \$6:

- They end with:  $\$5 - \$3.70 + \$6 = \$7.30$  (profit of \$2.30)
- I end with:  $\$11.10 - \$6 = \$5.10$  (profit of \$5.10)
- Both profit significantly

This rewards their current trust while maintaining fair reciprocity.

**usage:**

```
{
 "input_tokens": 2393,
 "output_tokens": 586,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.7, it became \$11.100000000000001, and you returned \$6.0. Your visible payoff was \$5.100000000000001.

Your total visible earnings across all rounds are \$35.150000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3 ? received \$5.09 (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 (profit: \$0.66)
- Round 6: Sent \$3 ? received \$2.32 (loss: -\$0.68)

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No information about their reciprocity patterns
- My moderate strategy has mostly been effective

**\*\*Strategy:\*\***

Continue with \$3:

- Creates \$9 for the receiver
- Shows cooperative intent
- Balanced risk approach
- Has generally yielded positive outcomes (2 out of 3 profitable)

```
{"send": 3}
```

**Reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)
- Round 6: Sent \$3 ? received \$2.32 back (profit: -\$0.68, actually a loss)

**\*\*Current situation:\*\***

- This is Round 8, so a new co-player (randomized pairings)
- No specific information about their behavior
- My \$3 strategy has been generally effective, though with varying results

**\*\*Strategy considerations:\*\***



The \$3 sending amount has been my consistent approach:

- Creates \$9 for receiver
- Shows cooperative intent but not excessive risk
- Has yielded mixed but generally positive results

I'll continue with \$3 as it's a balanced approach that:

1. Demonstrates willingness to cooperate
2. Creates significant surplus
3. Doesn't overcommit resources
4. Has worked reasonably well overall

```
{"send": 3}
Usage: {
 "input_tokens": 2571,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_6 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:39:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_6"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.74 to you. Your visible payoff was \$7.74.

Your total visible earnings across all rounds are \$7.74.

This round, you are the RECEIVER against your current co-player. They sent you \$3.25, so you received \$9.75. How much do you return? (0-9.75)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$3.25 (keeping \$1.75)
- I received \$9.75
- In Round 1, when I sent \$3, the receiver returned \$5.74 (about 64% of the \$9 they received)
- I want to establish reciprocity while being fair

**\*\*My decision:\*\***

Returning about 60-65% seems reasonable and reciprocal. This rewards the sender for their trust while keeping a meaningful amount for myself.

If I return **\*\*\$6\*\***:

- Sender ends with:  $\$1.75 + \$6 = \$7.75$  (gain of \$2.75)
- I end with:  $\$9.75 - \$6 = \$3.75$  (gain of \$3.75)

Both of us benefit significantly, and this return rate is consistent with what was shown to me in Round 1.

{"return": 6}

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.74 to me
- My payoff was  $\$5 - \$3 + \$5.74 = \$7.74$
- The receiver kept  $\$9.0 - \$5.74 = \$3.26$

So the previous receiver returned about 64% of what they received ( $\$5.74/\$9.0$ ).

Current round (Round 2):

- I am the RECEIVER
- My co-player (different from Round 1, as noted in the instructions) sent me \$3.25
- I received \$9.75
- I need to decide how much to return

Key considerations:

1. This is a different co-player than Round 1
2. They sent me \$3.25 (slightly more than I sent in Round 1)
3. There's communication noise, so amounts may differ slightly
4. I want to establish reciprocity and encourage future cooperation
5. The previous receiver in Round 1 returned about 64% of what they received

Strategy:

- I should be fair and reciprocal
- Returning a similar proportion to what was returned to me seems reasonable
- If I return too little, it might discourage future cooperation
- If I return too much, I'm being overly generous

The sender sent \$3.25 and kept \$1.75. If I return around 60-65% of the \$9.75:

- 60% = \$5.85
- 65% = \$6.34

This would give:

- If I return \$6: Sender gets  $\$1.75 + \$6 = \$7.75$ , I keep  $\$9.75 - \$6 = \$3.75$
- If I return \$5.85: Sender gets  $\$1.75 + \$5.85 = \$7.60$ , I keep  $\$9.75 - \$5.85 = \$3.90$
- If I return \$5: Sender gets  $\$1.75 + \$5 = \$6.75$ , I keep  $\$9.75 - \$5 = \$4.75$

Looking at the payoff distributions, returning \$6 seems to balance things well-the sender gains \$2.75 while I gain \$3.75, which feels reasonably fair given they initiated the trust by sending first....

usage:

```
{
 "input_tokens": 794,
 "output_tokens": 1093,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

**\*\*Analysis:\*\***

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

**\*\*My decision:\*\***

Based on the positive reciprocity I've seen, I'll send **\*\*\$3.50\*\*** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

```
{"send": 3.5}
```

reasoning:

Let me analyze my history:

Round 1: I was SENDER

- Sent \$3.0
- Received back \$5.74
- Payoff:  $\$5 - \$3 + \$5.74 = \$7.74$

Round 2: I was RECEIVER

- Received \$9.75 (from \$3.25 sent)
- Returned \$6.0
- Payoff:  $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

**usage:**

```
{
 "input_tokens": 1132,
 "output_tokens": 711,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

**\*\*My decision:\*\***

Given their high trust, I want to reciprocate generously. Returning about 64% would be \*\*\$

8\*\*.

If I return \$8:

- Sender ends with:  $\$0.86 + \$8 = \$8.86$  (net gain of \$3.86)
- I end with:  $\$12.42 - \$8 = \$4.42$  (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

```
{"return": 8}
```

**reasoning:**

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets  $\$0.86 + \$7.50 = \$8.36$  (gain of \$3.36), I get  $\$12.42 - \$7.50 = \$4.92$  (gain of \$4.92)
- If I return \$8: Sender gets  $\$0.86 + \$8 = \$8.86$  (gain of \$3.86), I get  $\$12.42 - \$8 = \$4.42$  (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 699,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my track record:

**\*\*Analysis:\*\***

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

**\*\*My decision:\*\***

Given the strong pattern of reciprocity, I'll send **\*\*\$4\*\*** to show trust and create a \$12 p

ot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

```
{"send": 4}
```

**reasoning:**

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)
- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

**usage:**

```
{
 "input_tokens": 1763,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.31 to you. Your visible payoff was \$7.31.

Your total visible earnings across all rounds are \$31.179999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.47, so you received \$7.41. How much do you return? (0-7.41)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.47 (keeping \$2.53) - more cautious than previous senders
- I received \$7.41
- I've consistently returned 60-65% as a receiver
- Need to maintain reciprocal behavior

**\*\*My decision:\*\***

Maintaining my pattern, I'll return about 62% which is **\*\*\$4.60\*\***.

If I return \$4.60:

- Sender ends with: \$2.53 + \$4.60 = \$7.13 (net gain of \$2.13)
- I end with: \$7.41 - \$4.60 = \$2.81 (net gain of \$2.81)

Both players benefit, and I'm being fair and consistent with my established pattern.

```
{"return": 4.6}
```

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.47 (keeping \$2.53)
- I received \$7.41
- This is a more cautious send compared to previous rounds I've seen

**\*\*My previous behavior as receiver:\*\***

- Round 2: Received \$9.75, returned \$6 (about 61.5%)
- Round 4: Received \$12.42, returned \$8 (about 64.4%)

**\*\*My pattern:\*\***

I've been returning around 60-65% of what I receive, which has been working well.

**\*\*Considerations:\*\***

- This sender was more cautious (\$2.47 vs the \$3-4+ I've seen)
- I should maintain consistency in my return rate
- 60-65% of \$7.41 would be about \$4.45 to \$4.82

**\*\*My decision:\*\***

I'll return around 62-63% to maintain my reciprocal pattern. That would be about \$4.60.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of \$2.13)
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of \$2.81)

Both benefit, and I'm being consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 526,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.47, it became \$7.41, and you returned \$4.6. Your visible payoff was \$2.8100000000000005.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Consistent returns around \$6-6.50 when sending \$3-4
- As receiver: I've been fair and reciprocal
- Total earnings are healthy at \$34
- Cooperation has generally paid off

**\*\*My decision:\*\***

I'll send **\*\*\$4\*\*** to maintain my cooperative strategy. This creates a \$12 pot for the receiver.

Based on patterns, I expect a reasonable return that benefits both of us.

```
{"return": 4}
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3 ? got back \$5.74
- Round 3: Sent \$3.5 ? got back \$6.46
- Round 5: Sent \$4 ? got back \$6.31

As RECEIVER:

- Round 2: Received \$9.75, returned \$6
- Round 4: Received \$12.42, returned \$8
- Round 6: Received \$7.41, returned \$4.6

I've been experiencing good cooperation overall. When I send, I generally get back decent returns (though Round 5 was slightly lower relative to what I sent). When I receive, I've been consistently fair.

For Round 7 as sender, I should continue the cooperative strategy. My last send was \$4 and got \$6.31 back. I could:

- Send \$4 again (proven to work)
- Send \$3.5 (slightly more conservative)
- Send \$4.5 (slightly more trusting)

Given the overall positive trend and my healthy total earnings (\$34), I'll continue with a trusting approach. I'll send \$4 again since that worked reasonably well last time.

```
{"send": 4}
usage:
{
 "input_tokens": 2389,
 "output_tokens": 410,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.4 to you. Your visible payoff was \$6.4.

Your total visible earnings across all rounds are \$40.38999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.92 (keeping \$2.08) - moderate trust
- I received \$8.76
- I've consistently returned around 60-65% as a receiver (Round 2: 61.5%, Round 4: 64.4%, Round 6: 62%)
- Need to maintain fair and consistent behavior

**\*\*My decision:\*\***

Staying with my pattern, I'll return about 62% which is **\*\*\$5.45\*\***.

If I return \$5.45:

- Sender ends with: \$2.08 + \$5.45 = \$7.53 (net gain of \$2.53)
- I end with: \$8.76 - \$5.45 = \$3.31 (net gain of \$3.31)

Both players benefit and I maintain my reciprocal reputation.

```
{"return": 5.45}
```

**Reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.92 (keeping \$2.08) - moderate trust
- I received \$8.76
- I've been consistently returning around 60-65% as a receiver
- This has been my pattern: Round 2: returned 6/9.75 = 61.5%, Round 4: returned 8/12.42 = 64.4%, Round 6: returned 4.6/7.41 = 62%

**\*\*My decision:\*\***

Maintaining my consistent pattern, I'll return about 62% which is approximately \$5.45.

If I return \$5.45:



- Sender ends with:  $\$2.08 + \$5.45 = \$7.53$  (net gain of  $\$2.53$ )
- I end with:  $\$8.76 - \$5.45 = \$3.31$  (net gain of  $\$3.31$ )

Both players benefit, and I'm being fair and consistent.

Let me calculate:  $8.76 * 0.62 = 5.43$

I'll go with  $\$5.45$  to be slightly generous while maintaining my pattern.

```
Usage: {
 "input_tokens": 2455,
 "output_tokens": 494,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Sender - dyad\_2

Timestamp: 2026-08-12T09:39:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.03 to you. Your visible payoff was \$5.029999999999999.

Your total visible earnings across all rounds are \$5.029999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 2 Analysis:\*\***

- My co-player sent \$4.33 (87% of their endowment) - showing high trust
- I received \$12.99
- They have \$0.67 remaining before my return

This sender showed significant trust. To reciprocate fairly and encourage continued cooperation:

Returning **\*\*\$6.50\*\*** (50% of what I received) would give them a total of \$7.17 and me \$6.49, which seems like a reasonable split of the gains from cooperation.

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for the receiver
- They returned \$3.03 to me
- My payoff: \$5 - \$3 + \$3.03 = \$5.03
- They kept: \$9 - \$3.03 = \$5.97

So in Round 1, my co-player returned about 1/3 of what they received, which is roughly fair but slightly less than splitting the gains evenly. If we split the total gains (\$8 total - \$5 initial = \$3 in gains), we'd each get \$6.50. I got \$5.03, they got \$5.97, so they kept slightly more of the gains.

Round 2:

- I am RECEIVER
- Sender sent \$4.33
- I receive \$12.99
- I need to decide how much to return

The note mentions there may be communication noise - amounts might differ slightly from what

at was actually sent. The sender sent \$4.33, which is 86.6% of their \$5 endowment - a very generous send! This shows high trust.

If I want to reciprocate fairly:

- They gave up \$4.33
- I received \$12.99
- If I return about 1/3 to half, that would be \$4.33 to \$6.50

To be generous back and encourage cooperation:

- They'll end up with:  $\$5 - \$4.33 + \text{my\_return} = \$0.67 + \text{my\_return}$
- I'll end up with:  $\$12.99 - \text{my\_return}$

If I want us to split relatively evenly:

- Total in the system:  $\$5$  (their initial) +  $\$12.99$  (what I got) =  $\$17.99$
- Fair split would be about  $\$9$  each
- They have  $\$0.67$  before my return
- So I should return about  $\$9 - \$0.67 = \$8.33$

That's quite generous though. A more moderate approach would be returning roughly half of what I received-around \$6.50-which gives them \$7.17 and leaves me with \$6.49. Alternatively, I could return what they originally sent (\$4.33), bringing them to \$5 total while I keep \$8.66. Given their substantial trust in sending most of their endowment, I should lean toward the more generous side of these options....

usage:

```
{
 "input_tokens": 774,
 "output_tokens": 886,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

reasoning:

**\*\*Round 3 Analysis:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received:  $\$9.87 * 0.5 = \$4.935$

That would give them:

- Sender total:  $\$1.71 + \$4.935 = \$6.645$
- My total:  $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total:  $\$1.71 + \$5 = \$6.71$
- I keep:  $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

**usage:**

```
{
 "input_tokens": 1053,
 "output_tokens": 544,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 4 Analysis:\*\***

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send **\*\*\$4\*\*** to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 4 Analysis:\*\***

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with  $5 - 4 + 6 = 7$ , they'll get \$6. If they return less, I still have \$1 kept.

**usage:**

```
{
 "input_tokens": 1305,
 "output_tokens": 469,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 5 Analysis:\*\***

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send **\*\*\$4\*\*** again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 5 Analysis:\*\***

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)
- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

usage:

```
{
 "input_tokens": 1557,
 "output_tokens": 384,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$30.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.3, so you received \$6.899999999999995. How much do you return? (0-6.899999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

content:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

I'll maintain my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 3.50}
'''
```

This gives them a total of \$6.20 and me \$3.40, fairly splitting the gains.

**reasoning:**

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

This is a lower send amount than I've been receiving. I should still reciprocate fairly but proportionally to the trust shown.

Following my pattern of returning about 50% of what I received:

- 50% of \$6.90 = \$3.45

This would give them:

- Sender: \$2.70 + \$3.45 = \$6.15 total
- Me: \$6.90 - \$3.45 = \$3.45

This seems fair and proportional to their investment.

usage:

```
{
 "input_tokens": 1803,
 "output_tokens": 293,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.5. Your visible payoff was \$3.399999999999995.

Your total visible earnings across all rounds are \$33.77.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing strong trust
- I received \$11.04
- They have \$1.32 remaining before my return

I'll continue my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 5.50}
'''
```

This gives them a total of \$6.82 and me \$5.54, fairly splitting the gains from cooperation

**reasoning:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing good trust
- I received \$11.04
- They have \$1.32 remaining before my return

I should continue my pattern of fair reciprocation. Returning about 50% of what I received  
:

50% of \$11.04 = \$5.52

This would give them \$1.32 + \$5.52 = \$6.84 total

And I would keep \$11.04 - \$5.52 = \$5.52

That seems fair and consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 274,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001.

Your total visible earnings across all rounds are \$39.31.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER with a new co-player.

My track record:

- As sender: sending \$4 has yielded good returns (\$6.14 and \$5.84)
- As receiver: I've consistently returned ~50% of what I received
- Total earnings: \$39.31

I'll continue with **\*\*\$4\*\*** - this amount has proven successful in building cooperation.

```
'''json
{"send": 4}
'''
```

**Reasoning:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER now with a new co-player.

Let me review my history:

- Round 1 (sender): sent \$3, got back \$3.03 - fair
- Round 2 (receiver): received \$12.99, returned \$6.50

- Round 3 (receiver): received \$9.87, returned \$5.00
- Round 4 (sender): sent \$4, got back \$6.14 - good
- Round 5 (sender): sent \$4, got back \$5.84 - good
- Round 6 (receiver): received \$6.90, returned \$3.50
- Round 7 (receiver): received \$11.04, returned \$5.50

As a sender, I've sent \$3 and \$4 twice. Both \$4 sends worked well (got back \$6.14 and \$5.84).  
As a receiver, I've consistently returned about 50% of what I received, which seems fair.

I'm doing well overall with \$39.31 in earnings. The strategy of sending \$4 has been working well for me.

I'll continue with sending \$4, which signals strong cooperation while not being too risky.

```
Usage: {
 "input_tokens": 2150,
 "output_tokens": 400,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:40:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned



- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$4.0. Your visible payoff was \$5.779999999999999.

Your total visible earnings across all rounds are \$5.779999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

{'send': 3.25}

##### reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and my co-player sent me \$3.26 (65.2% of their endowment), showing significant trust.
2. I returned \$4.00, which seemed fair and cooperative.
3. Now in Round 2, I'm the sender with a new co-player (since pairings are randomized each round).

Since this is a new co-player, I don't have any history with them. However, I should consider:

- The previous round showed that sending a substantial amount can work well
- Reciprocity and cooperation seem to be viable strategies
- I want to establish trust with this new partner

The sender in Round 1 sent 65.2% (\$3.26), which worked out well. I could send a similar amount to signal cooperation and trust.

Let me send around 60-65% to show trust and hopefully build a cooperative relationship.

60% of \$5 = \$3.00

65% of \$5 = \$3.25

I'll send \$3.25, which is similar to what was sent to me in Round 1.

This will become \$9.75 for the receiver, and I'll have \$1.75 left. If they reciprocate fairly, they might return around \$4-5, giving me a reasonable payoff.

##### usage:

```
{
 "input_tokens": 815,
 "output_tokens": 324,
 "reasoning_tokens": 0
}
```

}

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

{'return': 3.50}

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has  $\$5 - \$2.78 = \$2.22$  remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep:  $\$8.34 - \$3.50 = \$4.84$
- Sender ends with:  $\$2.22 + \$3.50 = \$5.72$

If I return \$3.00:

- I keep:  $\$8.34 - \$3.00 = \$5.34$
- Sender ends with:  $\$2.22 + \$3.00 = \$5.22$

If I return \$4.00:

- I keep:  $\$8.34 - \$4.00 = \$4.34$
- Sender ends with:  $\$2.22 + \$4.00 = \$6.22$

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

**usage:**

```
{
 "input_tokens": 960,
 "output_tokens": 465,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with:  $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff:  $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with:  $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

**usage:**

```
{
 "input_tokens": 1093,
 "output_tokens": 468,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.

Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78  
Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57  
Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84  
Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)
- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).

Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

**usage:**

```
{
 "input_tokens": 1229,
 "output_tokens": 282,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.55 to you. Your visible payoff was \$8.3.

Your total visible earnings across all rounds are \$34.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.04, so you received \$6.12. How much do you return? (0-6.12)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

{'return': 2.75}

**reasoning:**

Let me analyze this situation:

The sender sent \$2.04 out of \$5, which is about 40.8% of their endowment.

I received \$6.12 after the 3x multiplier.

The sender currently has:  $\$5 - \$2.04 = \$2.96$

This is a more cautious send compared to what I've been seeing and sending (around \$3.25 or 65%).

Looking at my history:

- Round 1: Received \$9.78, returned \$4.00
- Round 2: Sent \$3.25, received back \$4.82
- Round 3: Received \$8.34, returned \$3.50
- Round 4: Sent \$3.25, received back \$7.34
- Round 5: Sent \$3.25, received back \$6.55
- Round 6: Received \$6.12, need to decide return

The sender showed moderate trust (40.8%). I should reciprocate proportionally.

If I return \$2.50:

- I keep:  $\$6.12 - \$2.50 = \$3.62$
- Sender ends with:  $\$2.96 + \$2.50 = \$5.46$

If I return \$3.00:

- I keep:  $\$6.12 - \$3.00 = \$3.12$
- Sender ends with:  $\$2.96 + \$3.00 = \$5.96$

If I return \$2.00:

- I keep:  $\$6.12 - \$2.00 = \$4.12$
- Sender ends with:  $\$2.96 + \$2.00 = \$4.96$

Given that they sent less than what I typically send, but still showed some trust, I'll re

turn around \$2.50 to \$3.00. Let me go with \$2.75 as a middle ground.

...

**usage:**

```
{
 "input_tokens": 1374,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.04, it became \$6.12, and you returned \$2.75. Your visible payoff was \$3.37.

Your total visible earnings across all rounds are \$37.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

```
{'send': 3.25}
```

**reasoning:**

Let me analyze the previous round with this co-player:

- They sent me \$2.04 (about 41% of their endowment)
- I received \$6.12
- I returned \$2.75
- They ended with:  $\$5 - \$2.04 + \$2.75 = \$5.71$
- I ended with:  $\$6.12 - \$2.75 = \$3.37$

So they sent a moderate amount (\$2.04), and I returned \$2.75, which gave them a slight gain overall.

Now I'm the sender. Given that:

1. They sent me a moderate amount last round (\$2.04, which is about 41%)
2. I've been generally cooperative, sending around \$3.25 in other rounds
3. This is a repeated interaction with this specific co-player

I should probably send a similar or slightly higher amount than they sent to maintain cooperation. My typical send has been \$3.25, which has worked well with other players who returned good amounts.

I'll send \$3.25 to continue building trust with this co-player.

**usage:**

```
{
 "input_tokens": 1510,
 "output_tokens": 268,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.52 to you. Your visible payoff was \$8.27.

Your total visible earnings across all rounds are \$46.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
{'return': 5.00}
```

**Reasoning:**

Let me analyze this situation:

1. I'm now in Round 8 as a RECEIVER
2. My co-player sent me \$3.88, which became \$11.64
3. I need to decide how much to return

Looking at my history:

- Round 2: I was RECEIVER, got \$9.78, returned \$4.0 (kept \$5.78)
- Round 3: I was SENDER, sent \$3.25, got back \$4.82 (kept \$6.57)
- Round 3 (second part): I was RECEIVER, got \$8.34, returned \$3.5 (kept \$4.84)
- Round 4: I was SENDER, sent \$3.25, got back \$7.34 (kept \$9.09)
- Round 5: I was SENDER, sent \$3.25, got back \$6.55 (kept \$8.3)
- Round 6: I was RECEIVER, got \$6.12, returned \$2.75 (kept \$3.37)
- Round 7: I was SENDER, sent \$3.25, got back \$6.52 (kept \$8.27)
- Round 8: I'm RECEIVER, got \$11.64, need to decide

My pattern as SENDER has been consistent: I send \$3.25 each time.

As RECEIVER:

- Round 2: received \$9.78, returned \$4.0 (return ratio =  $4.0/9.78 = 40.9\%$ )
- Round 3: received \$8.34, returned \$3.5 (return ratio =  $3.5/8.34 = 42.0\%$ )
- Round 6: received \$6.12, returned \$2.75 (return ratio =  $2.75/6.12 = 44.9\%$ )

So I've been returning around 40-45% of what I receive.

My co-player sent \$3.88, which is notably higher than the typical \$2-3 range I've observed. With \$11.64 received, applying my established return ratio suggests returning between \$4.66-\$5.24. Since they demonstrated stronger trust with a larger initial send, I'll return approximately \$5.00 to acknowledge that while staying consistent with my historical pattern of around 43%.

Looking back at my previous rounds, I notice I've consistently ensured my co-player ends up with roughly \$5.70-\$5.74 total—slightly above their initial \$5 investment. When I received \$9.78 and returned \$4.00, they netted \$5.74; at \$8.34 with a \$3.50 return, they got \$5.72; and with \$6.12 returned at \$2.75, they ended at \$5.71. This suggests I should aim for a return that keeps them in that same range. If I return \$5.63 here, they'd receive \$6.75 total, which is higher than my usual pattern. I need to consider what happens when I send larger amounts myself to calibrate appropriately....

```
Usage: {
 "input_tokens": 1437,
 "output_tokens": 1113,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_2 - Sender - dyad\_3

Timestamp: 2026-08-12T09:40:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_4"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

#### Message 3 role=assistant

##### content:

Looking at my experience from the previous round, the sender showed significant trust by sending 66.8% of their endowment, and I reciprocated fairly. This cooperation benefited us both.

I'll follow a similar strategy and show strong trust to my new co-player. Sending a substantial amount can establish a cooperative dynamic.

{'send': 3.50}

##### reasoning:

Let me think about this situation:

1. In the previous round, I was a RECEIVER and the sender showed trust by sending \$3.34 (66.8% of their endowment)
2. I returned \$5.00 out of \$10.02, which was a fair reciprocation

### 3. Now I'm the SENDER with a new co-player

Since I benefited from someone else's trust in the previous round, I should pay it forward and show similar trust to my new co-player. The previous sender sent about 67% of their endowment, which worked out well for both of us.

I could:

- Send around the same proportion (~\$3.30-\$3.50) to show similar trust
- Send more to be even more generous
- Send less to be more cautious

Given that:

- The system worked well when someone trusted me
- I want to encourage cooperation in this multi-agent setting
- I have information that fair reciprocation happened (though my co-player doesn't know about my previous round)

I'll send around \$3.00-\$3.50. Let me go with \$3.00 as a solid show of trust (60% of my endowment) while being slightly more cautious since this is a new partner.

Actually, thinking more about it - the previous sender sent 66.8%. That was a strong signal of trust. To maintain that cooperative spirit and given that I benefited from it, I should send a similar amount. Let me send \$3.50 (70% of my endowment) to show strong trust

**usage:**

```
{
 "input_tokens": 846,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

This rewards their trust and maintains cooperation.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has  $\$5 - \$2.52 = \$2.48$  left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate)



e).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again:  $\$7.56 * 0.5 = \$3.78$

Let me check the outcomes:

- If I return \$3.78:
  - Sender ends with:  $\$2.48 + \$3.78 = \$6.26$  (profit of \$1.26)
  - I end with:  $\$7.56 - \$3.78 = \$3.78$  (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

...

**usage:**

```
{
 "input_tokens": 1072,
 "output_tokens": 643,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

{'send': 3.50}

**reasoning:**

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

**usage:**

```
{
 "input_tokens": 1400,
 "output_tokens": 305,
 "reasoning_tokens": 0
}
```

}

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was \$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it.

{'send': 3.50}

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well. I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

**usage:**

```
{
 "input_tokens": 1587,
 "output_tokens": 211,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.5 to you. Your visible payoff was \$8.0.

Your total visible earnings across all rounds are \$33.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.40 (88% of their endowment)
- I received \$13.20 after the 3x multiplication
- The sender currently has only \$0.60 left

This is the highest trust I've seen so far, exceeding even Round 1 (66.8%). I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.60 + \$7.00 = \$7.60 (profit of \$2.60)
- I end with: \$13.20 - \$7.00 = \$6.20 (profit of \$6.20)

This rewards their exceptional trust while ensuring we both profit well.

{'return': 7.00}

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.40 (88% of their endowment) - this is very high trust
- I received \$13.20 after the 3x multiplication
- The sender currently has \$0.60 left (\$5 - \$4.40)

This sender showed exceptional trust by sending 88% of their endowment. I should reciprocate generously.

Looking at my past behavior as receiver:

- Round 1: Received \$10.02, returned \$5.00 (about 50%)
- Round 3: Received \$7.56, returned \$4.00 (about 53%)

For this higher trust situation, I should return more generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust significantly while still maintaining good profit for both.

Alternatively, if I return \$6.50:

- Sender ends with:  $\$0.60 + \$6.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$13.20 - \$6.50 = \$6.70$  (profit of \$6.70)

I think \$7.00 is appropriate given the very high trust shown (88% vs the 70% I've been sending).

usage:

```
{
 "input_tokens": 1827,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.4, it became \$13.200000000000001, and you returned \$7.0. Your visible payoff was \$6.200000000000001.

Your total visible earnings across all rounds are \$39.46000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.91, so you received \$11.73. How much do you return? (0-11.73)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

content:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left

This is high trust, similar to previous high-trust rounds. Comparing to my history:

- Round 1: 66.8% sent ? returned \$5.00
- Round 6: 88% sent ? returned \$7.00

At 78.2%, this falls between those two. I'll return proportionally.

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

```
{'return': 6.50}
```

reasoning:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left (\$5 - \$3.91)

This is high trust, though not quite as high as the 88% I saw in Round 6. Let me compare to my previous responses:

Round 1: Sender sent 66.8%, I received \$10.02, I returned \$5.00 (about 50%)

Round 3: Sender sent 50.4%, I received \$7.56, I returned \$4.00 (about 53%)

Round 6: Sender sent 88%, I received \$13.20, I returned \$7.00 (about 53%)

This sender sent 78.2%, which is between Round 1 (66.8%) and Round 6 (88%).

If I return \$6.00:

- Sender ends with:  $\$1.09 + \$6.00 = \$7.09$  (profit of \$2.09)

- I end with:  $\$11.73 - \$6.00 = \$5.73$  (profit of \$5.73)

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)

- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

Given the high trust level (78.2%), I think \$6.50 is appropriate, falling between my Round 1 return (\$5.00 for 66.8%) and Round 6 return (\$7.00 for 88%).

usage:

```
{
 "input_tokens": 2156,
 "output_tokens": 639,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.91, it became \$11.73, and you returned \$6.5. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$44.69000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my consistent success as sender:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)

- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)

- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)

The \$3.50 strategy has proven reliable and profitable. I'll continue with it.

```
{'send': 3.50}
```

**Reasoning:**

Let me review my history as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)

- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

- Round 5: Sent \$3.50, received \$6.50 back (payoff \$8.00)

My sending strategy of \$3.50 has been consistently successful, with returns ranging from \$6.50 to \$6.95.

As receiver, I've been fair and reciprocal:

- Round 1: Received \$10.02, returned \$5.00

- Round 3: Received \$7.56, returned \$4.00

- Round 6: Received \$13.20, returned \$7.00

- Round 7: Received \$11.73, returned \$6.50

The cooperative strategy has worked well. I should continue with \$3.50.

Usage: {

"input\_tokens": 2234,

"output\_tokens": 359,

"reasoning\_tokens": 0

}

**GAME - Agent\_4 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:40:33  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8  
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_4"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.49, it became \$7.470000000000001, and you returned \$3.5. Your visible payoff was \$3.970000000000006.

Your total visible earnings across all rounds are \$3.970000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

In Round 1, when I was receiver, the sender showed trust by sending about 50% (\$2.49), and I reciprocated by returning a fair amount that gave us both profits.

Now as sender, I want to:

1. Show similar trust and cooperation
2. Signal I'm willing to engage in mutually beneficial exchanges
3. Send a moderate amount that balances risk and potential reward

I'll send \$2.50 (50% of my endowment), which mirrors the trust level shown to me in Round 1.

{"send": 2.50}

**reasoning:**

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was a receiver and:

- The sender sent \$2.49 (about 50%)
- I received \$7.47
- I returned \$3.50
- I kept \$3.97

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. This is a multi-round game with different opponents each round
2. My current co-player is different from Round 1
3. There may be communication noise
4. I want to build a reputation as a cooperative player
5. I demonstrated willingness to return a reasonable amount when I was receiver

Strategic thinking:

- If I send nothing, I guarantee \$5 but miss potential gains and hurt my reputation
- If I send a lot, I show trust but risk getting little back
- The previous sender sent about 50% (\$2.49), which seemed like a reasonable trust signal
- I returned a decent amount, showing I'm cooperative

I think I should send a similar amount to what I received in Round 1 - around 50% or so. This shows:

1. Consistency with my cooperative behavior
2. Trust in the system
3. Willingness to earn through mutual benefit

Let me send \$2.50 (50% of my endowment). This is almost exactly what the previous sender sent to me, and it's a clear signal of moderate trust.

If the receiver returns about 50% of what they get (like I did), they'd receive \$7.50 and return \$3.75, giving me \$2.50 remaining + \$3.75 = \$6.25 total, which is a profit of \$1.25.

That seems reasonable.

**usage:**

```
{
 "input_tokens": 897,
 "output_tokens": 538,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.

Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

**reasoning:**

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff: \$2.50 + \$3.93 = \$6.43

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

**usage:**

```
{
 "input_tokens": 1153,
 "output_tokens": 447,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with: \$0.52 + \$7.00 = \$7.52 (profit of ~\$2.52)
- I keep: \$13.44 - \$7.00 = \$6.44 (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

{"return": 7.00}

**reasoning:**

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with: \$0.52 + \$6.50 = \$7.02 (profit of ~\$2)
- I would keep: \$13.44 - \$6.50 = \$6.94 (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with: \$0.52 + \$7 = \$7.52 (profit of \$2.52)
- I keep: \$13.44 - \$7 = \$6.44 (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be: \$13.44 \* 0.5 = \$6.72

...

**usage:**

```
{
 "input_tokens": 1412,
 "output_tokens": 682,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5



In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001. Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of  $\sim\$2.22$ )
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of  $\sim\$6.34$ )

This maintains a similar return rate ( $\sim 51\%$ ) as Round 4 ( $\sim 52\%$ ) and both of us profit well.

{"return": 6.50}

**reasoning:**

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocation pattern for high trust.

The sender showed  $\sim 86\%$  trust level, which is slightly less than the  $\sim 90\%$  in Round 4, but still very high.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of  $\sim\$2.22$ )
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of  $\sim\$6.34$ )

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

**usage:**

```
{
 "input_tokens": 1745,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.28, it became \$12.84, and you returned \$6.5. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$28.540000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my history:

- As sender (Rounds 2-3): I've consistently sent \$2.50 (50%), with varying returns
- As receiver (Rounds 1, 4-5): I've reciprocated well when others showed trust

My 50% sending strategy has been reasonably successful, giving both parties opportunities to profit. While some agents have sent higher amounts (85-90%) when they're senders, I'll maintain my consistent approach that balances trust and risk.

```
{"send": 2.50}
```

**reasoning:**

Let me review my history:

Round 1 (RECEIVER): Received \$7.47 from sender who sent \$2.49 (50%), returned \$3.50 (~47% of what I received). Payoff: \$3.97

Round 2 (SENDER): Sent \$2.50 (50%), received \$3.93 back. Payoff: \$6.43

Round 3 (SENDER): Sent \$2.50 (50%), received \$2.86 back. Payoff: \$5.36

Round 4 (RECEIVER): Received \$13.44 from sender who sent \$4.48 (90%), returned \$7.00 (~52% of what I received). Payoff: \$6.44

Round 5 (RECEIVER): Received \$12.84 from sender who sent \$4.28 (86%), returned \$6.50 (~51% of what I received). Payoff: \$6.34

Now Round 6 (SENDER): I need to decide how much to send.

Looking at the pattern:

- When I've been sender, I've sent \$2.50 (50%) consistently
- When I'm receiver, I've gotten good returns when others send high amounts (50-90%)
- I've been rewarded for reciprocating high trust with good returns

However, I've noticed that when I sent 50% in rounds 2 and 3, I got varying returns:

- Round 2: \$3.93 (very good return, about 52% of what they received)
- Round 3: \$2.86 (lower return, about 38% of what they received)

Other agents have been sending higher amounts (85-90%) to me when they're senders, and I've reciprocated well. This has worked out for both parties.

Should I increase my sending? Or stick with my 50% strategy?

...

**usage:**

```
{
 "input_tokens": 2073,
 "output_tokens": 696,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.76 to you. Your visible payoff was \$7.26.

Your total visible earnings across all rounds are \$35.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my sending history:

- Round 2: Sent \$2.50, received \$3.93 back
- Round 3: Sent \$2.50, received \$2.86 back
- Round 6: Sent \$2.50, received \$4.76 back

The 50% strategy continues to work well on average. Round 6 showed particularly good reciprocity with a \$4.76 return. I'll maintain this consistent, moderate trust level.

```
{"send": 2.50}
```

**reasoning:**

Looking at my history:

Round 2 (SENDER): Sent \$2.50, received \$3.93 back, payoff \$6.43  
Round 3 (SENDER): Sent \$2.50, received \$2.86 back, payoff \$5.36  
Round 6 (SENDER): Sent \$2.50, received \$4.76 back, payoff \$7.26

As sender, my 50% strategy has been working reasonably well. In Round 6, I received an excellent return of \$4.76.

As receiver:

Round 1: Received \$7.47, returned \$3.50  
Round 4: Received \$13.44, returned \$7.00  
Round 5: Received \$12.84, returned \$6.50

I've been consistent and cooperative. My 50% sending strategy has been reasonable and I should continue with it.

**usage:**

```
{
 "input_tokens": 2318,
 "output_tokens": 335,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.1 to you. Your visible payoff was \$4.6.

Your total visible earnings across all rounds are \$40.400000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender showed good trust by sending \$3.41 (about 68% of their endowment)
- I received \$10.23 after the 3x multiplication
- The sender currently has \$1.59 remaining

In previous rounds as receiver, I've returned roughly 47-52% of what I received, depending on the trust shown.

If I return \$5.00:

- Sender ends with: \$1.59 + \$5.00 = \$6.59 (profit of ~\$1.59)
- I keep: \$10.23 - \$5.00 = \$5.23 (profit of ~\$5.23)

This is a fair reciprocation that rewards their trust while both of us profit.

```
{"return": 5.00}
```

**Reasoning:**

Let me analyze this situation:

- The sender sent \$3.41 (about 68% of their endowment) - showing good trust
- I received \$10.23 after the 3x multiplication
- The sender currently has \$5 - \$3.41 = \$1.59 remaining

Looking at my history as receiver:

- Round 1: Received \$7.47, returned \$3.50 (47% return rate)
- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- Round 5: Received \$12.84, returned \$6.50 (51% return rate)

The sender showed good but not exceptional trust (68% vs the 85-90% I saw in rounds 4-5).

If I return \$5.00:  
- Sender ends with:  $\$1.59 + \$5.00 = \$6.59$  (profit of \$1.59)  
- I keep:  $\$10.23 - \$5.00 = \$5.23$  (profit of \$5.23)

This gives about a 49% return rate, which is consistent with my previous behavior. Both parties profit reasonably.

Let me consider \$5.00 as a fair reciprocation.

```
Usage: {
 "input_tokens": 2283,
 "output_tokens": 486,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_3 - Sender - dyad\_4

Timestamp: 2026-08-12T09:40:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.89, it became \$11.67, and you returned \$5.28. Your visible payoff was \$6.39.

Your total visible earnings across all rounds are \$6.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the previous round, my co-player showed considerable trust by sending 77.8% of their endowment, which led to a fair outcome for both of us.

I'll reciprocate with a similar level of trust.

{'send': 4.00}

**reasoning:**

In the previous round, I was the receiver and someone sent me \$3.89 (77.8% of their endowment). I received \$11.67 and returned \$5.28, ending with \$6.39.

Now I'm the sender with \$5. I need to decide how much to send.

The previous sender showed high trust by sending 77.8% of their endowment. That worked well - they ended up with  $\$1.11 + \$5.28 = \$6.39$ , and I also got \$6.39. This was a fair outcome.

Since the previous round demonstrated that high trust can lead to good outcomes for both parties, I should consider sending a similar proportion. If I send around 75-80% of my endowment, that would be \$3.75-\$4.00.

Let me think about what's reasonable:

- If I send \$4.00, the receiver gets \$12.00
- If they return proportionally (like I did, which was about 45% of what I received), they'd return about \$5.40
- I'd end up with  $\$1.00 + \$5.40 = \$6.40$
- They'd end up with  $\$12.00 - \$5.40 = \$6.60$

That seems fair and similar to the previous round's outcome.

Alternatively, if I send \$3.75:

- Receiver gets \$11.25
- If they return ~45%, that's about \$5.06
- I'd end up with  $\$1.25 + \$5.06 = \$6.31$
- They'd end up with  $\$11.25 - \$5.06 = \$6.19$

I'll go with \$4.00 to show strong trust, similar to what was shown to me in the previous round.

**usage:**

```
{
 "input_tokens": 787,
 "output_tokens": 478,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

```
{'send': 4.00}
```

**reasoning:**

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was  $\$5 - \$4 + \$5.20 = \$6.20$

The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

**usage:**

```
{
 "input_tokens": 970,
 "output_tokens": 251,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created:  $\$0.63$  (with sender) +  $\$13.11$  (with me) =  $\$13.74$
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Return

ning \$6.24 would give us both around \$6.87.

```
{'return': 6.24}
```

**reasoning:**

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has \$5 - \$4.37 = \$0.63

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created: \$0.63 (sender has) + \$13.11 (I have) = \$13.74

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have: \$0.63 + return = \$6.87, so return = \$6.24
- I would have: \$13.11 - \$6.24 = \$6.87

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)
- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

```
{'return': 5.90}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender:  $\$0.80 + \$6.30 = \$7.10$
- Me:  $\$12.60 - \$6.30 = \$6.30$

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return:  $\$13.40 - \$6.70 = \$6.70$

- Sender would get:  $\$0.80 + \$6.70 = \$7.50$
- I would get:  $\$12.60 - \$6.70 = \$5.90$

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets:  $\$0.80 + X$
- I get:  $\$12.60 - X$

For equal outcomes:

$$\$0.80 + X = \$12.60 - X$$

$$2X = \$12.60 - \$0.80$$

$$2X = \$11.80$$

$$X = \$5.90$$

Check:

- Sender:  $\$0.80 + \$5.90 = \$6.70$
- Me:  $\$12.60 - \$5.90 = \$6.70$

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

usage:

```
{
 "input_tokens": 1423,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.2, it became \$12.600000000000001, and you returned \$5.9. Your visible payoff was \$6.700000000000001.

Your total visible earnings across all rounds are \$31.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience across rounds, sending around \$4.00 has consistently led to cooperative outcomes. I'll continue with this trusting strategy.

```
{'send': 4.00}
```

reasoning:

In this round, I'm the sender. Let me review my recent experiences:

Round 1: I was receiver, sender sent \$3.89, I returned \$5.28, both got ~\$6.39

Round 2: I sent \$4.00, received back \$5.20, got \$6.20

Round 3: I sent \$4.00, received back \$4.73, got \$5.73

Round 4: I was receiver, sender sent \$4.37, I returned \$6.24, both got ~\$6.87

Round 5: I was receiver, sender sent \$4.20, I returned \$5.90, both got \$6.70

When I've been a receiver, I've been fairly generous in my returns, aiming for equal splits. When I've been a sender, I've sent \$4.00 and received reasonable returns.



My strategy has been cooperative and trusting. I should continue with this approach, sending a high amount to show trust and hopefully receive a fair return.

Sending \$4.00 has worked reasonably well in the past. I'll continue with this level.

**usage:**

```
{
 "input_tokens": 1662,
 "output_tokens": 301,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.91 to you. Your visible payoff was \$8.91.

Your total visible earnings across all rounds are \$40.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment), showing moderate trust
- Total pool created: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- This is less trust than most previous rounds

I'll still reciprocate fairly. Returning \$2.00 would give us both approximately \$4.75.

```
{'return': 2.00}
```

**reasoning:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment)
- This is notably less than previous senders I've encountered who sent around 77-87%
- Total pool: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- They showed moderate trust compared to others

Since they sent less than previous players, I should still reciprocate fairly but proportionally to their trust level.

For an equal split: each gets \$4.74

That means I'd return  $\$2.76 + (\text{total} - \text{sender's remaining}) / 2 = \dots$  let me recalculate

Sender has:  $\$5 - \$2.24 = \$2.76$

I received: \$6.72

Total created:  $\$2.76 + \$6.72 = \$9.48$

For equal split of \$4.74 each:

Sender needs:  $\$4.74 - \$2.76 = \$1.98$  returned

I keep:  $\$6.72 - \$1.98 = \$4.74$

But given they showed less trust than previous players, maybe I should be slightly less generous. However, I should still be fair and cooperative.

I'll return around \$2.00-\$2.50 to maintain fairness.

Let me go with \$2.30, which gives:

- Sender:  $\$2.76 + \$2.30 = \$5.06$
- Me:  $\$6.72 - \$2.30 = \$4.42$

This is slightly favoring them which encourages cooperation but not as generous as when they send more.

Actually, let me reconsider. I've been fairly cooperative throughout. Even though they sent less, being consistent in fairness might be better. Let me aim closer to equal:

...

```
usage:
{
 "input_tokens": 1846,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.24, it became \$6.720000000000001, and you returned \$2.0. Your visible payoff was \$4.720000000000001.

Your total visible earnings across all rounds are \$45.519999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience, sending \$4.00 has generally led to good cooperative outcomes. I'll continue with this strategy.

```
{'send': 4.00}
```

**Reasoning:**

Looking at my history:

- Round 1 (as RECEIVER): They sent \$3.89, I returned \$5.28
- Round 2 (as SENDER): I sent \$4.00, they returned \$5.20
- Round 3 (as SENDER): I sent \$4.00, they returned \$4.73
- Round 4 (as RECEIVER): They sent \$4.37, I returned \$6.24
- Round 5 (as RECEIVER): They sent \$4.20, I returned \$5.90
- Round 6 (as SENDER): I sent \$4.00, they returned \$7.91
- Round 7 (as RECEIVER): They sent \$2.24, I returned \$2.00

I've been consistently cooperative, and most co-players have shown high trust levels (sending \$4+). The exception was Round 7 where someone only sent \$2.24, and I responded proportionally.

I should continue with my trusting strategy. Sending \$4.00 has worked well overall.

Usage: {

```
 "input_tokens": 1885,
 "output_tokens": 286,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_5 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:40:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
 Message 2 role=user
 content:
 Round 2

 In your most recent previous game round, you were the SENDER against another agent. You se
 nt $3.0, it became $9.0, and you saw them return $2.93 to you. Your visible payoff was $
 4.93.
 Your total visible earnings across all rounds are $4.93.

 This round, you are the RECEIVER against your current co-player. They sent you $2.82, so y
 ou received $8.459999999999999. How much do you return? (0-8.459999999999999)
 Respond exactly as JSON: {'return': <amount>}
 Message 3 role=assistant
 content:
 Round 2 - RECEIVER Decision

 Looking at this situation:
 - My current co-player sent me $2.82 (56% of their endowment) - a substantial show of trus
 t
 - I received $8.46 after the 3x multiplier
 - They currently have $2.18 remaining

```

My co-player demonstrated significant trust by sending more than half their endowment. To encourage cooperation and ensure mutual benefit:

- If I return \$4.00:
  - They end up with:  $\$2.18 + \$4.00 = \$6.18$
  - I end up with:  $\$8.46 - \$4.00 = \$4.46$
  - Both of us benefit compared to keeping the initial \$5

This seems like a fair reciprocation of their trust.

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 1 recap:\*\***

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.93 to me
- My payoff:  $\$5 - \$3 + \$2.93 = \$4.93$
- Their payoff:  $\$9.0 - \$2.93 = \$6.07$

So the receiver kept about 67% of what they received and returned about 33%.

**\*\*Round 2:\*\***

- I'm the RECEIVER
- The sender sent me \$2.82
- I received \$8.46 (after 3x multiplication)
- I need to decide how much to return

The note says there may be communication noise, so the \$2.82 might not be exactly what was sent.

Now, how should I think about this?

From a reciprocity perspective:

- In round 1, my co-player returned about 33% of what they received (\$2.93 out of \$9.0)
- But that was a different co-player

For this round with my current co-player:

- They sent me \$2.82 out of their \$5, which is 56.4% - a fairly generous amount
- This created \$8.46 for me
- A fair return might be to split the gains

Let me think about fairness:

- The sender kept:  $\$5 - \$2.82 = \$2.18$
- I received: \$8.46
- Total pie:  $\$2.18 + \$8.46 = \$10.64$

If we split evenly: each gets \$5.32

- For that, I'd need to return:  $\$8.46 - \$5.32 = \$3.14$

Alternatively, I could match the return rate from my previous round experience (33%):

- Return:  $\$8.46 \times 0.33 = \$2.79$

Or I could be more generous given they sent a majority of their endowment. Since they demonstrated substantial trust by sending 56.4% of their stake, returning around 47% of what I received-roughly \$4-seems appropriate. This rewards their cooperation while still letting me benefit from the gains, giving them a final payoff of about \$6.18....

**usage:**

```
{
 "input_tokens": 775,
 "output_tokens": 861,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Round 3 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

**usage:**

```
{
 "input_tokens": 1105,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 4 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

**\*\*Round 4 - RECEIVER Decision\*\***

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have  $\$5 - \$2.84 = \$2.16$  remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately.

**usage:**

```
{
 "input_tokens": 1398,
 "output_tokens": 518,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 5 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything  
. I should reciprocate proportionally:

- If I return \$6.50:
  - They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
  - I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)
  - Both benefit substantially, rewarding their exceptional trust

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust

I received \$12.96 after the 3x multiplier

They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount.  
Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
  - They end up with:  $\$0.68 + \$6.00 = \$6.68$
  - I end up with:  $\$12.96 - \$6.00 = \$6.96$
  - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
  - They end up with:  $\$0.68 + \$7.00 = \$7.68$
  - I end up with:  $\$12.96 - \$7.00 = \$5.96$
  - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
- I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

**usage:**

```
{
 "input_tokens": 1715,
 "output_tokens": 645,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.32, it became \$12.96, and you returned \$6.5. Your visible payoff was \$6.460000000000001.

Your total visible earnings across all rounds are \$26.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 6 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My experience as sender has been positive:

- Round 1: Sent \$3, got \$2.93 back
- Round 3: Sent \$3, got \$3.75 back

Both times sending \$3 led to reasonable cooperation. This creates \$9 for the receiver, allowing room for mutual benefit. I'll continue this consistent approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Round 6 - I'm the SENDER with \$5.

Let me review my experience so far:

As SENDER:

- Round 1: Sent \$3, received ~\$2.93 back
- Round 3: Sent \$3, received \$3.75 back

As RECEIVER:

- Round 2: Received \$2.82? \$8.46, returned \$4.00
- Round 4: Received \$2.84? \$8.52, returned \$4.00
- Round 5: Received \$4.32? \$12.96, returned \$6.50

The game has been showing good cooperation overall. As a sender, I've gotten decent returns (around \$3-\$3.75 on \$3 sent). As a receiver, I've been fair and reciprocal.

Now I'm facing a new co-player. The cooperation has been working well. I could:

1. Continue with \$3 (my previous sender strategy)
2. Send more to signal stronger trust (like the \$4.32 I saw in Round 5)
3. Send less to be more cautious

Given the positive cooperation trends and that this is a new co-player, I'll stick with my consistent sender strategy of \$3. This has worked well and creates good mutual benefit opportunities.

**usage:**

```
{
 "input_tokens": 2048,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.38 to you. Your visible payoff was \$5.38.

Your total visible earnings across all rounds are \$31.499999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 7 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My track record as sender:

- Round 1: Sent \$3 ? received \$2.93 back (profit of \$0.93)
- Round 3: Sent \$3 ? received \$3.75 back (profit of \$0.75)
- Round 6: Sent \$3 ? received \$3.38 back (profit of \$0.38)

Sending \$3 has consistently worked well, showing decent cooperation from different co-players. I'll stick with this strategy that balances trust and risk management.

```
'''json
{"send": 3}
'''
```

**reasoning:**



I'm the sender again with \$5.

Looking at my history as sender:

- Round 1: Sent \$3, got \$2.93 back (visible payoff \$4.93)
- Round 3: Sent \$3, got \$3.75 back (visible payoff \$5.75)
- Round 6: Sent \$3, got \$3.38 back (visible payoff \$5.38)

All three times I sent \$3, and I got decent returns ranging from about \$2.93 to \$3.75. This strategy has been working well - I'm consistently making a small profit.

I'm facing a new co-player now. Sending \$3 has been my consistent strategy and it's been working. I should continue with this.

**usage:**

```
{
 "input_tokens": 2294,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$7.03 to you. Your visible payoff was \$9.030000000000001.

Your total visible earnings across all rounds are \$40.53.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Round 8 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$3.72 (74% of their endowment) - high trust
- I received \$11.16 after the 3x multiplier
- They currently have \$1.28 remaining

This is substantial trust, between my Round 2/4 experiences (~57%) and Round 5 (86%).

Looking at my pattern as receiver:

- Rounds 2 & 4: Returned ~47% of received amount
- Round 5: Returned 50% of received amount

For consistency, returning around 50% makes sense:

- If I return \$5.50:
  - They end up with:  $\$1.28 + \$5.50 = \$6.78$  (profit of \$1.78)
  - I end up with:  $\$11.16 - \$5.50 = \$5.66$  (profit of \$0.66)
  - Both benefit, rewarding their trust appropriately

```
'''json
{"return": 5.5}
'''
```

**Reasoning:**

This is Round 8, and I'm the receiver.

My co-player sent me \$3.72 (74% of their endowment) - that's a high level of trust.

I received \$11.16 after the 3x multiplier.

They currently have \$1.28 remaining.

Looking at my history as receiver:

- Round 2: Received \$8.46, returned \$4.00 (kept \$4.46)
- Round 4: Received \$8.52, returned \$4.00 (kept \$4.52)
- Round 5: Received \$12.96, returned \$6.50 (kept \$6.46)

In Round 5, when I received \$12.96 (from a send of \$4.32), I returned \$6.50, which was about 50% of what I received.

In this round, I received \$11.16 (from a send of \$3.72).

The sender has \$1.28 left. If I return \$5.50:  
- They end up with:  $\$1.28 + \$5.50 = \$6.78$  (gain of \$1.78)  
- I end up with:  $\$11.16 - \$5.50 = \$5.66$  (gain of \$0.66)

That seems fair - they sent 74% of their endowment, showing high trust. I should reciprocate proportionally.

Actually, let me think about this more carefully. They sent \$3.72, which is 74% of \$5. In Round 5, they sent \$4.32 which is 86% and I returned \$6.50 (50% of what I received).

For \$3.72 sent (74% trust level), returning around \$5.00-\$5.50 seems appropriate.

If I return \$5.50:  
- Sender ends with \$6.78 (profit of \$1.78 from initial \$5)  
- I end with \$5.66 (profit of \$0.66 from initial \$5)

This rewards their trust reasonably while ensuring both benefit. Looking at my return patterns across rounds, I've consistently returned roughly 47-50% of what I received, with higher returns when the sender showed greater trust. A \$5.50 return maintains that consistency...

```
Usage: {
 "input_tokens": 2433,
 "output_tokens": 935,
 "reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_8=Receiver  
dyad\_2: Agent\_5=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_2=Sender, Agent\_7=Receiver  
dyad\_4: Agent\_6=Sender, Agent\_4=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3.25; received \$9.75; Agent\_8 returned \$4.5; payoffs Agent\_1=\$6.25, Agent\_8=\$5.25; noise/visible: sent visible=\$2.96, received visible=\$8.88, returned visible=\$4.05  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.75, "min": 0, "raw": 4.5}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}

dyad\_2: Agent\_5 sent \$3.7; received \$11.1; Agent\_3 returned \$3.34; payoffs Agent\_5=\$4.64, Agent\_3=\$7.76; noise/visible: sent visible=\$2.92, received visible=\$8.76, returned visible=\$3.66  
Action validation: {"returned": {"amount": 3.34, "clamped": false, "max": 11.1, "min": 0, "raw": 3.34}, "sent": {"amount": 3.7, "clamped": false, "max": 5, "min": 0, "raw": 3.7}}

dyad\_3: Agent\_2 sent \$3.5; received \$10.5; Agent\_7 returned \$7.5; payoffs Agent\_2=\$9, Agent\_7=\$3; noise/visible: sent visible=\$4.27, received visible=\$12.81, returned visible=\$6.69  
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 10.5, "min": 0, "raw": 7.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad\_4: Agent\_6 sent \$3.5; received \$10.5; Agent\_4 returned \$5; payoffs Agent\_6=\$6.5, Agent\_4=\$5.5; noise/visible: sent visible=\$3.39, received visible=\$10.17, returned visible=\$5.74  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

```
"Agent_1": 59.4,
"Agent_2": 55.74,
"Agent_3": 52.59,
"Agent_4": 49.1,
"Agent_5": 52.89,
"Agent_6": 50.199999999999996,
"Agent_7": 46.230000000000004,
"Agent_8": 54.75
}
```

Visible balances after round: {

```
"Agent_1": 58.66,
"Agent_2": 59.700000000000002,
"Agent_3": 57.22,
"Agent_4": 50.800000000000001,
"Agent_5": 51.15,
"Agent_6": 50.800000000000001,
"Agent_7": 46.230000000000004,
"Agent_8": 54.75
}
```

```

"Agent_6": 50.94,
"Agent_7": 47.600000000000001,
"Agent_8": 48.930000000000001
}

```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:41:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.82 to you. Your visible payoff was \$6.57.

Your total visible earnings across all rounds are \$12.35.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

{'return': 3.50}

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.78 (55.6% of their endowment)
- I received \$8.34 after the 3x multiplier
- The sender currently has  $\$5 - \$2.78 = \$2.22$  remaining

In Round 1, when I was receiver:

- Sender sent \$3.26 (65.2%)
- I returned \$4.00
- This gave them \$5.74 total and me \$5.78

In Round 2, when I was sender:

- I sent \$3.25 (65%)
- They returned \$4.82
- This gave me \$6.57 total and them \$4.93

The sender in this round sent \$2.78, which is less generous than what I've seen before (55.6% vs 65%). However, they still showed some trust.

If I return \$3.50:

- I keep:  $\$8.34 - \$3.50 = \$4.84$
- Sender ends with:  $\$2.22 + \$3.50 = \$5.72$

If I return \$3.00:

- I keep:  $\$8.34 - \$3.00 = \$5.34$
- Sender ends with:  $\$2.22 + \$3.00 = \$5.22$

If I return \$4.00:

- I keep:  $\$8.34 - \$4.00 = \$4.34$
- Sender ends with:  $\$2.22 + \$4.00 = \$6.22$

Given they sent about 55.6%, I should reciprocate proportionally. In Round 1, for a 65% send, I returned about 41% of what I received. For a 55.6% send, I might return a bit less.

Let me return \$3.50, which is about 42% of what I received. ...

**usage:**

```
{
 "input_tokens": 960,
 "output_tokens": 465,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.  
Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze the pattern from previous rounds:

Round 1 (RECEIVER):

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with:  $\$1.74 + \$4.00 = \$5.74$

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff:  $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with:  $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

**usage:**

```
{
 "input_tokens": 1093,
 "output_tokens": 468,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.  
Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78

Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57

Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84

Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)

- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).

Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

**usage:**

```
{
 "input_tokens": 1229,
 "output_tokens": 282,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.55 to you. Your visible payoff was \$8.3.

Your total visible earnings across all rounds are \$34.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.04, so you received \$6.12. How much do you return? (0-6.12)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

{'return': 2.75}

**reasoning:**

Let me analyze this situation:

The sender sent \$2.04 out of \$5, which is about 40.8% of their endowment.

I received \$6.12 after the 3x multiplier.

The sender currently has: \$5 - \$2.04 = \$2.96

This is a more cautious send compared to what I've been seeing and sending (around \$3.25 or 65%).

Looking at my history:

- Round 1: Received \$9.78, returned \$4.00

- Round 2: Sent \$3.25, received back \$4.82

- Round 3: Received \$8.34, returned \$3.50

- Round 4: Sent \$3.25, received back \$7.34

- Round 5: Sent \$3.25, received back \$6.55

- Round 6: Received \$6.12, need to decide return

The sender showed moderate trust (40.8%). I should reciprocate proportionally.

If I return \$2.50:

- I keep: \$6.12 - \$2.50 = \$3.62

- Sender ends with: \$2.96 + \$2.50 = \$5.46

If I return \$3.00:  
- I keep:  $\$6.12 - \$3.00 = \$3.12$   
- Sender ends with:  $\$2.96 + \$3.00 = \$5.96$

If I return \$2.00:  
- I keep:  $\$6.12 - \$2.00 = \$4.12$   
- Sender ends with:  $\$2.96 + \$2.00 = \$4.96$

Given that they sent less than what I typically send, but still showed some trust, I'll re turn around \$2.50 to \$3.00. Let me go with \$2.75 as a middle ground.

...

**usage:**

```
{
 "input_tokens": 1374,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.04, it became \$6.12, and you returned \$2.75. Your visible payoff was \$3.37.

Your total visible earnings across all rounds are \$37.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

```
{'send': 3.25}
```

**reasoning:**

Let me analyze the previous round with this co-player:

- They sent me \$2.04 (about 41% of their endowment)
- I received \$6.12
- I returned \$2.75
- They ended with:  $\$5 - \$2.04 + \$2.75 = \$5.71$
- I ended with:  $\$6.12 - \$2.75 = \$3.37$

So they sent a moderate amount (\$2.04), and I returned \$2.75, which gave them a slight gain overall.

Now I'm the sender. Given that:

1. They sent me a moderate amount last round (\$2.04, which is about 41%)
2. I've been generally cooperative, sending around \$3.25 in other rounds
3. This is a repeated interaction with this specific co-player

I should probably send a similar or slightly higher amount than they sent to maintain cooperation. My typical send has been \$3.25, which has worked well with other players who returned good amounts.

I'll send \$3.25 to continue building trust with this co-player.

**usage:**

```
{
 "input_tokens": 1510,
 "output_tokens": 268,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.52 to you. Your visible payoff was \$8.27.

Your total visible earnings across all rounds are \$46.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so y

ou received \$11.64. How much do you return? (0-11.64)  
Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. I'm now in Round 8 as a RECEIVER
2. My co-player sent me \$3.88, which became \$11.64
3. I need to decide how much to return

Looking at my history:

- Round 2: I was RECEIVER, got \$9.78, returned \$4.0 (kept \$5.78)
- Round 3: I was SENDER, sent \$3.25, got back \$4.82 (kept \$6.57)
- Round 3 (second part): I was RECEIVER, got \$8.34, returned \$3.5 (kept \$4.84)
- Round 4: I was SENDER, sent \$3.25, got back \$7.34 (kept \$9.09)
- Round 5: I was SENDER, sent \$3.25, got back \$6.55 (kept \$8.3)
- Round 6: I was RECEIVER, got \$6.12, returned \$2.75 (kept \$3.37)
- Round 7: I was SENDER, sent \$3.25, got back \$6.52 (kept \$8.27)
- Round 8: I'm RECEIVER, got \$11.64, need to decide

My pattern as SENDER has been consistent: I send \$3.25 each time.

As RECEIVER:

- Round 2: received \$9.78, returned \$4.0 (return ratio =  $4.0/9.78 = 40.9\%$ )
- Round 3: received \$8.34, returned \$3.5 (return ratio =  $3.5/8.34 = 42.0\%$ )
- Round 6: received \$6.12, returned \$2.75 (return ratio =  $2.75/6.12 = 44.9\%$ )

So I've been returning around 40-45% of what I receive.

My co-player sent \$3.88, which is notably higher than the typical \$2-3 range I've observed. With \$11.64 received, applying my established return ratio suggests returning between \$4.66-\$5.24. Since they demonstrated stronger trust with a larger initial send, I'll return approximately \$5.00 to acknowledge that while staying consistent with my historical pattern of around 43%.

Looking back at my previous rounds, I notice I've consistently ensured my co-player ends up with roughly \$5.70-\$5.74 total—slightly above their initial \$5 investment. When I received \$9.78 and returned \$4.00, they netted \$5.74; at \$8.34 with a \$3.50 return, they got \$5.72; and with \$6.12 returned at \$2.75, they ended at \$5.71. This suggests I should aim for a return that keeps them in that same range. If I return \$5.63 here, they'd receive \$6.75 total, which is higher than my usual pattern. I need to consider what happens when I send larger amounts myself to calibrate appropriately....

**usage:**

```
{
 "input_tokens": 1437,
 "output_tokens": 1113,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.88, it became \$11.64, and you returned \$5.0. Your visible payoff was \$6.6400000000000001.

Your total visible earnings across all rounds are \$52.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

{'send': 3.25}

**Reasoning:**

Let me analyze the situation:

1. In Round 8, my co-player sent me \$3.88 (which became \$11.64), and I returned \$5.00.
2. Now in Round 9, I'm the sender against the same co-player.

Let me look at the pattern:



- When I've been sender, I've consistently sent \$3.25
- When I receive amounts, I try to return a fair portion

In Round 8, my co-player sent me \$3.88, which is higher than my usual \$3.25. They showed trust

Looking at my returns as receiver:

- Round 3: Received \$8.34 (from \$2.78 sent), returned \$3.50
- Round 6: Received \$6.12 (from \$2.04 sent), returned \$2.75
- Round 8: Received \$11.64 (from \$3.88 sent), returned \$5.00

My return pattern seems to aim for roughly giving back a bit more than what was originally sent but not the full multiplied amount.

Now as sender against the same co-player who sent me \$3.88 in Round 8:

- They were generous with \$3.88
- I should probably match or reciprocate their level of trust
- I've been sending \$3.25 consistently, but maybe I should send more to match their \$3.88

Actually, let me stick with my consistent strategy of \$3.25. This has been working well overall and my total earnings are good at \$52.86.

```
Usage: {
 "input_tokens": 1436,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:41:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.33, it became \$12.99, and you returned \$6.5. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$11.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.29 (about 66% of their endowment) - showing good trust
- I received \$9.87
- They have \$1.71 remaining before my return

I'll continue the pattern of fair reciprocation from last round. Returning approximately 50% of what I received:

```
'''json
{"return": 5.00}
'''
```

This gives them a total of \$6.71 and me \$4.87, which fairly splits the gains from our cooperation.

**reasoning:**

**\*\*Round 3 Analysis:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.29 (about 66% of their endowment)
- I received \$9.87
- They have \$1.71 remaining before my return

This sender showed moderate-high trust (66% of endowment). I should reciprocate fairly.

Looking at last round, I returned 50% of what I received when the sender sent 87%. This sender sent a bit less (66%), but still showed good trust.

Let me think about fair reciprocation:

- They sent \$3.29
- I received \$9.87
- If I return around 50% of what I received:  $\$9.87 * 0.5 = \$4.935$

That would give them:

- Sender total:  $\$1.71 + \$4.935 = \$6.645$
- My total:  $\$9.87 - \$4.935 = \$4.935$

This seems fair - we both benefit from the cooperation, with the sender getting a bit more since they took the initial risk.

Alternatively, I could return a bit less since they sent less than the previous sender. But maintaining consistency at around 50% return rate seems like a good strategy to build trust.

I'll return \$5.00 (just over 50% of received) to be fair and keep things simple.

Actually, let me reconsider. They sent \$3.29 out of \$5. If I return \$5, they get:

- Total:  $\$1.71 + \$5 = \$6.71$
- I keep:  $\$9.87 - \$5 = \$4.87$

That's a reasonable split. The sender ends up slightly better, which rewards their trust.

I'll go with \$5.00.

**usage:**

```
{
 "input_tokens": 1053,
 "output_tokens": 544,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Round 4 Analysis:\*\***

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send **\*\*\$4\*\*** to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 4 Analysis:\*\***

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)

Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)

Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

usage:

```
{
 "input_tokens": 1305,
 "output_tokens": 469,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

**Round 5 Analysis:**

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send **\$4** again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

reasoning:

**Round 5 Analysis:**

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)

- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

usage:

```
{
 "input_tokens": 1557,
 "output_tokens": 384,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$30.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.3, so you received \$6.899999999999995. How much do you return? (0-6.899999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

I'll maintain my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 3.50}
'''
```

This gives them a total of \$6.20 and me \$3.40, fairly splitting the gains.

reasoning:

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

This is a lower send amount than I've been receiving. I should still reciprocate fairly but proportionally to the trust shown.

Following my pattern of returning about 50% of what I received:

- 50% of \$6.90 = \$3.45

This would give them:

- Sender: \$2.70 + \$3.45 = \$6.15 total
- Me: \$6.90 - \$3.45 = \$3.45

This seems fair and proportional to their investment.

usage:

```
{
 "input_tokens": 1803,
 "output_tokens": 293,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.5. Your visible payoff was \$3.399999999999995.

Your total visible earnings across all rounds are \$33.77.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing strong trust
- I received \$11.04
- They have \$1.32 remaining before my return

I'll continue my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 5.50}
'''
```

This gives them a total of \$6.82 and me \$5.54, fairly splitting the gains from cooperation

**reasoning:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing good trust
- I received \$11.04
- They have \$1.32 remaining before my return

I should continue my pattern of fair reciprocation. Returning about 50% of what I received  
:

50% of \$11.04 = \$5.52

This would give them \$1.32 + \$5.52 = \$6.84 total  
And I would keep \$11.04 - \$5.52 = \$5.52

That seems fair and consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 274,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001. Your total visible earnings across all rounds are \$39.31.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER with a new co-player.

My track record:

- As sender: sending \$4 has yielded good returns (\$6.14 and \$5.84)
- As receiver: I've consistently returned ~50% of what I received
- Total earnings: \$39.31

I'll continue with **\*\*\$4\*\*** - this amount has proven successful in building cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER now with a new co-player.

Let me review my history:

- Round 1 (sender): sent \$3, got back \$3.03 - fair
- Round 2 (receiver): received \$12.99, returned \$6.50
- Round 3 (receiver): received \$9.87, returned \$5.00
- Round 4 (sender): sent \$4, got back \$6.14 - good
- Round 5 (sender): sent \$4, got back \$5.84 - good
- Round 6 (receiver): received \$6.90, returned \$3.50
- Round 7 (receiver): received \$11.04, returned \$5.50

As a sender, I've sent \$3 and \$4 twice. Both \$4 sends worked well (got back \$6.14 and \$5.84).

As a receiver, I've consistently returned about 50% of what I received, which seems fair.

I'm doing well overall with \$39.31 in earnings. The strategy of sending \$4 has been working well for me.

I'll continue with sending \$4, which signals strong cooperation while not being too risky.

usage:

```
{
 "input_tokens": 2150,
 "output_tokens": 400,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$4.24 to you. Your visible payoff was \$5.24.

Your total visible earnings across all rounds are \$44.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.96, so you received \$8.879999999999999. How much do you return? (0-8.879999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Round 9 Analysis:\*\***

- Same co-player as last round! Last round I sent \$4, they returned \$4.24
- This round they sent \$2.96 (59% of their endowment) - moderate trust
- I received \$8.88
- They have \$2.04 remaining before my return

I'll maintain my cooperative strategy of returning about 50% of what I received:

```
'''json
{"return": 4.50}
'''
```

This gives them a total of \$6.54 and me \$4.38, maintaining fair reciprocation.

**Reasoning:**

Round 9 Analysis:

- Last round I was sender with this same co-player: sent \$4, received \$4.24 back - they returned about 35% of the \$12 they received
- This round they sent \$2.96 (59% of their endowment) - moderate trust
- I received \$8.88
- They have \$2.04 remaining before my return

This is interesting - I'm playing against the same co-player from last round. They returned \$4.24 when I sent \$4 (which gave them \$12). That's about 35% return rate.

Now they sent me \$2.96, giving me \$8.88.

I should be fair but also consider their previous behavior. They were somewhat conservative in their return last round. However, I generally try to maintain a cooperative 50% return strategy.

Let me return around 50% of what I received: about \$4.50

This would give them \$2.04 + \$4.50 = \$6.54 total

And I'd keep \$8.88 - \$4.50 = \$4.38

That seems fair given the circumstances.

```
Usage: {
 "input_tokens": 2126,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_5 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:41:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_3"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_3": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.



The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.82, it became \$8.459999999999999, and you returned \$4.0. Your visible payoff was \$4.459999999999999.

Your total visible earnings across all rounds are \$9.389999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

**\*\*Round 3 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player. Looking at my experience:

- Round 1: Sent \$3, received ~\$2.93 back (reasonable cooperation)
- Round 2: As receiver, got \$2.82 and returned \$4.00 (fair exchange)

Both previous interactions showed decent cooperation. Since I'm with a new co-player and cooperation has been working, I'll continue with a similar strategy.

Sending \$3 creates \$9 for them to work with, which allows for mutual benefit if they reciprocate reasonably.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my situation:

Round 1: I was SENDER, sent \$3, received \$2.93 back, payoff was \$4.93

Round 2: I was RECEIVER, received \$8.46 (from \$2.82 sent), returned \$4.00, payoff was \$4.46

Total visible earnings: \$9.39

Now I'm SENDER again in Round 3 with a new co-player.

From my experience:

- Round 1: I sent \$3, got \$2.93 back - this was roughly fair (they kept \$6.07 from \$9, I ended with \$4.93)
- Round 2: My co-player sent \$2.82 (good trust), I returned \$4.00 which gave them \$6.18 total and me \$4.46

Both rounds showed reasonable cooperation. Since I'm playing with a new co-player this round and don't know their behavior, I should continue with a cooperative strategy.

Sending \$3 worked reasonably well in Round 1. I could:

- Send \$3 again as a consistent strategy
- Send more to show high cooperation
- Send less to be more cautious

Given that cooperation has generally worked so far, I'll stick with sending \$3 as a baseline cooperative move.

**usage:**

```
{
 "input_tokens": 1105,
```

```
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Round 4 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit substantially

```
'''json
{
 "return": 4.0
}
'''
```

**reasoning:**

**\*\*Round 4 - RECEIVER Decision\*\***

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have  $\$5 - \$2.84 = \$2.16$  remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately

**usage:**

```
{
 "input_tokens": 1398,
 "output_tokens": 518,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 5 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything. I should reciprocate proportionally:

- If I return \$6.50:
  - They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
  - I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)
  - Both benefit substantially, rewarding their exceptional trust

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust

I received \$12.96 after the 3x multiplier

They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount. Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
  - They end up with:  $\$0.68 + \$6.00 = \$6.68$
  - I end up with:  $\$12.96 - \$6.00 = \$6.96$
  - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
  - They end up with:  $\$0.68 + \$7.00 = \$7.68$
  - I end up with:  $\$12.96 - \$7.00 = \$5.96$
  - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
- I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

**usage:**

```
{
 "input_tokens": 1715,
 "output_tokens": 645,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.32, it became \$12.96, and you returned \$6.5. Your visible payoff was \$6.460000000000001.

Your total visible earnings across all rounds are \$26.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 6 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My experience as sender has been positive:

- Round 1: Sent \$3, got \$2.93 back
- Round 3: Sent \$3, got \$3.75 back

Both times sending \$3 led to reasonable cooperation. This creates \$9 for the receiver, allowing room for mutual benefit. I'll continue this consistent approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Round 6 - I'm the SENDER with \$5.

Let me review my experience so far:

As SENDER:

- Round 1: Sent \$3, received ~\$2.93 back
- Round 3: Sent \$3, received \$3.75 back

As RECEIVER:

- Round 2: Received \$2.82? \$8.46, returned \$4.00
- Round 4: Received \$2.84? \$8.52, returned \$4.00
- Round 5: Received \$4.32? \$12.96, returned \$6.50

The game has been showing good cooperation overall. As a sender, I've gotten decent returns (around \$3-\$3.75 on \$3 sent). As a receiver, I've been fair and reciprocal.

Now I'm facing a new co-player. The cooperation has been working well. I could:

1. Continue with \$3 (my previous sender strategy)
2. Send more to signal stronger trust (like the \$4.32 I saw in Round 5)
3. Send less to be more cautious

Given the positive cooperation trends and that this is a new co-player, I'll stick with my consistent sender strategy of \$3. This has worked well and creates good mutual benefit opportunities.

**usage:**

```
{
 "input_tokens": 2048,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.38 to you. Your visible payoff was \$5.38.

Your total visible earnings across all rounds are \$31.499999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 7 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My track record as sender:

- Round 1: Sent \$3 ? received \$2.93 back (profit of \$0.93)
- Round 3: Sent \$3 ? received \$3.75 back (profit of \$0.75)
- Round 6: Sent \$3 ? received \$3.38 back (profit of \$0.38)

Sending \$3 has consistently worked well, showing decent cooperation from different co-players. I'll stick with this strategy that balances trust and risk management.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm the sender again with \$5.

Looking at my history as sender:

- Round 1: Sent \$3, got \$2.93 back (visible payoff \$4.93)
- Round 3: Sent \$3, got \$3.75 back (visible payoff \$5.75)
- Round 6: Sent \$3, got \$3.38 back (visible payoff \$5.38)

All three times I sent \$3, and I got decent returns ranging from about \$2.93 to \$3.75. This strategy has been working well - I'm consistently making a small profit.

I'm facing a new co-player now. Sending \$3 has been my consistent strategy and it's been working. I should continue with this.

**usage:**

```
{
 "input_tokens": 2294,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$7.03 to you. Your visible payoff was \$9.030000000000001.

Your total visible earnings across all rounds are \$40.53.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 8 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$3.72 (74% of their endowment) - high trust
- I received \$11.16 after the 3x multiplier
- They currently have \$1.28 remaining

This is substantial trust, between my Round 2/4 experiences (~57%) and Round 5 (86%).

Looking at my pattern as receiver:

- Rounds 2 & 4: Returned ~47% of received amount
- Round 5: Returned 50% of received amount

For consistency, returning around 50% makes sense:

- If I return \$5.50:
  - They end up with: \$1.28 + \$5.50 = \$6.78 (profit of \$1.78)
  - I end up with: \$11.16 - \$5.50 = \$5.66 (profit of \$0.66)
  - Both benefit, rewarding their trust appropriately

```
'''json
{"return": 5.5}
'''
```

**reasoning:**

This is Round 8, and I'm the receiver.

My co-player sent me \$3.72 (74% of their endowment) - that's a high level of trust.  
I received \$11.16 after the 3x multiplier.  
They currently have \$1.28 remaining.

Looking at my history as receiver:

- Round 2: Received \$8.46, returned \$4.00 (kept \$4.46)
- Round 4: Received \$8.52, returned \$4.00 (kept \$4.52)
- Round 5: Received \$12.96, returned \$6.50 (kept \$6.46)

In Round 5, when I received \$12.96 (from a send of \$4.32), I returned \$6.50, which was about 50% of what I received.

In this round, I received \$11.16 (from a send of \$3.72).

The sender has \$1.28 left. If I return \$5.50:

- They end up with:  $\$1.28 + \$5.50 = \$6.78$  (gain of \$1.78)
- I end up with:  $\$11.16 - \$5.50 = \$5.66$  (gain of \$0.66)

That seems fair - they sent 74% of their endowment, showing high trust. I should reciprocate proportionally.

Actually, let me think about this more carefully. They sent \$3.72, which is 74% of \$5. In Round 5, they sent \$4.32 which is 86% and I returned \$6.50 (50% of what I received).

For \$3.72 sent (74% trust level), returning around \$5.00-\$5.50 seems appropriate.

If I return \$5.50:

- Sender ends with \$6.78 (profit of \$1.78 from initial \$5)
- I end with \$5.66 (profit of \$0.66 from initial \$5)

This rewards their trust reasonably while ensuring both benefit. Looking at my return patterns across rounds, I've consistently returned roughly 47-50% of what I received, with higher returns when the sender showed greater trust. A \$5.50 return maintains that consistency....

**usage:**

```
{
 "input_tokens": 2433,
 "output_tokens": 935,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.72, it became \$11.16, and you returned \$5.5. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$46.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Round 9 - SENDER Decision\*\***

I'm playing against the same co-player from Round 8! They showed strong trust by sending \$3.72 (74% of their endowment), and I returned \$5.50, giving them a profit of \$1.78.

Since they demonstrated high trust with me, I should reciprocate. Matching their trust level of 74%, I'll send approximately \$3.70.

This creates \$11.10 for them and shows I value the cooperative relationship we've built.

```
'''json
{"send": 3.7}
'''
```

**Reasoning:**

This is interesting - I'm playing against the same co-player as Round 8. They sent me \$3.72 (74% of their endowment) and I returned \$5.50.

Let me think about this:

- They showed high trust by sending \$3.72 (74%)
- I returned \$5.50, giving them a profit of \$1.78
- They ended with \$6.78

Now I'm the sender. Given that they showed strong trust last round, I should reciprocate with similarly strong trust.

My typical strategy has been to send \$3, but this co-player demonstrated extra trust with me. I should consider sending more to match their level of trust.

They sent 74% of their endowment. If I match that, I would send  $74\% \times \$5 = \$3.70$ .

Actually, let me think about this more carefully. They sent \$3.72 and after my return of \$5.50, they made a profit of \$1.78. That's a good outcome for them.

If I send \$3.70:

- It becomes \$11.10
- If they return proportionally like I did (about 50%), they'd return ~\$5.55
- I'd end up with:  $\$5 - \$3.70 + \$5.55 = \$6.85$
- They'd end up with:  $\$11.10 - \$5.55 = \$5.55$

This seems fair and reciprocates their trust level.

Alternatively, I could send \$4 to show even more trust:

- It becomes \$12
- If they return 50%, they return \$6
- I end up with:  $\$5 - \$4 + \$6 = \$7$
- They end up with:  $\$12 - \$6 = \$6$

...

```
Usage: {
 "input_tokens": 2465,
 "output_tokens": 571,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_3 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:41:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_3"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_3": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.2 to you. Your visible payoff was \$6.2.

Your total visible earnings across all rounds are \$12.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my recent experience, sending \$4.00 led to a reasonable return of \$5.20, showing my co-player was cooperative.

I'll continue with a trusting strategy.

{'send': 4.00}

**reasoning:**

In the previous round:

- I sent \$4.00
- It became \$12.00 for the receiver
- They returned \$5.20
- My payoff was \$5 - \$4 + \$5.20 = \$6.20



The receiver was fairly generous, returning \$5.20 out of \$12.00 (43.3%), which gave us both around \$6.20 and \$6.80 respectively.

This is a new round with potentially a new co-player (since pairings are randomized each round). I should continue with a cooperative strategy since:

1. My previous experiences have been positive
2. Sending a substantial amount has led to good returns
3. Building trust tends to yield better outcomes

I'll send a similar amount to what I sent before, around \$4.00, to maintain a cooperative approach.

**usage:**

```
{
 "input_tokens": 970,
 "output_tokens": 251,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was \$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created: \$0.63 (with sender) + \$13.11 (with me) = \$13.74
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

{'return': 6.24}

**reasoning:**

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has \$5 - \$4.37 = \$0.63

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created: \$0.63 (sender has) + \$13.11 (I have) = \$13.74

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have: \$0.63 + return = \$6.87, so return = \$6.24
- I would have: \$13.11 - \$6.24 = \$6.87

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)
- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

{'return': 5.90}

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender: \$0.80 + \$6.30 = \$7.10
- Me: \$12.60 - \$6.30 = \$6.30

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return: \$13.40 - \$6.70 = \$6.70

- Sender would get: \$0.80 + \$6.70 = \$7.50
- I would get: \$12.60 - \$6.70 = \$5.90

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets: \$0.80 + X
- I get: \$12.60 - X

For equal outcomes:

$\$0.80 + X = \$12.60 - X$

$2X = \$12.60 - \$0.80$

$2X = \$11.80$

$X = \$5.90$

Check:

- Sender: \$0.80 + \$5.90 = \$6.70
- Me: \$12.60 - \$5.90 = \$6.70

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.2, it became \$12.600000000000001, and you returned \$5.9. Your visible payoff was \$6.700000000000001.

Your total visible earnings across all rounds are \$31.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience across rounds, sending around \$4.00 has consistently led to cooperative outcomes. I'll continue with this trusting strategy.

```
{'send': 4.00}
```

**reasoning:**

In this round, I'm the sender. Let me review my recent experiences:

Round 1: I was receiver, sender sent \$3.89, I returned \$5.28, both got ~\$6.39

Round 2: I sent \$4.00, received back \$5.20, got \$6.20

Round 3: I sent \$4.00, received back \$4.73, got \$5.73

Round 4: I was receiver, sender sent \$4.37, I returned \$6.24, both got ~\$6.87

Round 5: I was receiver, sender sent \$4.20, I returned \$5.90, both got \$6.70

When I've been a receiver, I've been fairly generous in my returns, aiming for equal splits. When I've been a sender, I've sent \$4.00 and received reasonable returns.

My strategy has been cooperative and trusting. I should continue with this approach, sending a high amount to show trust and hopefully receive a fair return.

Sending \$4.00 has worked reasonably well in the past. I'll continue with this level.

**usage:**

```
{
 "input_tokens": 1662,
 "output_tokens": 301,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.91 to you. Your visible payoff was \$8.91.

Your total visible earnings across all rounds are \$40.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment), showing moderate trust
- Total pool created: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- This is less trust than most previous rounds

I'll still reciprocate fairly. Returning \$2.00 would give us both approximately \$4.75.

```
{'return': 2.00}
```

**reasoning:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment)
- This is notably less than previous senders I've encountered who sent around 77-87%
- Total pool: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- They showed moderate trust compared to others

Since they sent less than previous players, I should still reciprocate fairly but proportionally to their trust level.

For an equal split: each gets \$4.74

That means I'd return  $\$2.76 + (\text{total} - \text{sender's remaining}) / 2 = \dots$  let me recalculate

Sender has:  $\$5 - \$2.24 = \$2.76$

I received: \$6.72

Total created:  $\$2.76 + \$6.72 = \$9.48$

For equal split of \$4.74 each:

Sender needs:  $\$4.74 - \$2.76 = \$1.98$  returned

I keep:  $\$6.72 - \$1.98 = \$4.74$

But given they showed less trust than previous players, maybe I should be slightly less generous. However, I should still be fair and cooperative.

I'll return around \$2.00-\$2.50 to maintain fairness.

Let me go with \$2.30, which gives:

- Sender:  $\$2.76 + \$2.30 = \$5.06$
- Me:  $\$6.72 - \$2.30 = \$4.42$

This is slightly favoring them which encourages cooperation but not as generous as when they send more.

Actually, let me reconsider. I've been fairly cooperative throughout. Even though they sent less, being consistent in fairness might be better. Let me aim closer to equal:

...

**usage:**

```
{
 "input_tokens": 1846,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.24, it became \$6.720000000000001, and you returned \$2.0. Your visible payoff was \$4.720000000000001.

Your total visible earnings across all rounds are \$45.519999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my experience, sending \$4.00 has generally led to good cooperative outcomes. I'll continue with this strategy.

```
{'send': 4.00}
```

**reasoning:**

Looking at my history:

- Round 1 (as RECEIVER): They sent \$3.89, I returned \$5.28
- Round 2 (as SENDER): I sent \$4.00, they returned \$5.20
- Round 3 (as SENDER): I sent \$4.00, they returned \$4.73
- Round 4 (as RECEIVER): They sent \$4.37, I returned \$6.24
- Round 5 (as RECEIVER): They sent \$4.20, I returned \$5.90
- Round 6 (as SENDER): I sent \$4.00, they returned \$7.91

- Round 7 (as RECEIVER): They sent \$2.24, I returned \$2.00

I've been consistently cooperative, and most co-players have shown high trust levels (sending \$4+). The exception was Round 7 where someone only sent \$2.24, and I responded proportionally.

I should continue with my trusting strategy. Sending \$4.00 has worked well overall.

**usage:**

```
{
 "input_tokens": 1885,
 "output_tokens": 286,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$5.28 to you. Your visible payoff was \$6.28.

Your total visible earnings across all rounds are \$51.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.92 (58.4% of their endowment), showing moderate trust
- Total pool created: \$2.08 (with sender) + \$8.76 (with me) = \$10.84
- This is the same co-player from last round who returned \$5.28 to me, showing fair reciprocation

I'll reciprocate fairly. Returning \$3.34 would give us both approximately \$5.42 each.

{'return': 3.34}

**Reasoning:**

Looking at this situation:

- The sender sent \$2.92 (58.4% of their endowment)
- Total pool created: \$2.08 (with sender) + \$8.76 (with me) = \$10.84
- They showed moderate trust

In the previous round with this same co-player:

- I sent \$4.00, which became \$12.00
- They returned \$5.28
- I ended up with \$6.28, they ended up with \$6.72

So in round 8 (the previous round with this co-player), they showed reasonable reciprocity by returning \$5.28 on the \$12.00 they received.

Now they're sending me \$2.92. This is less than what I sent them last time (\$4.00), but still a reasonable amount.

For fairness, aiming for an equal split:

- They have: \$2.08
- I have: \$8.76
- Total: \$10.84
- Equal split: ~\$5.42 each

To give them \$5.42, I need to return: \$5.42 - \$2.08 = \$3.34

Let me double check:

- Sender keeps: \$5 - \$2.92 + \$3.34 = \$5.42
- I keep: \$8.76 - \$3.34 = \$5.42

Yes, that works for an equal split.

Given they were cooperative in the previous round (returning \$5.28 on \$12.00), I'll reciprocate fairly here.

```
Usage: {
 "input_tokens": 1880,
 "output_tokens": 494,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:41:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_7"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.73 to you. Your visible payoff was \$8.23.

Your total visible earnings across all rounds are \$13.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment), showing moderate trust
- I received \$7.56 after the 3x multiplication
- The sender currently has \$2.48 left

In Round 1, when a sender showed high trust (66.8%), I returned about 50% of what I received. This sender showed moderate trust, so I'll return a fair amount that ensures we both profit.

If I return \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

This rewards their trust and maintains cooperation.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

Round 3 - I'm the RECEIVER

- The sender sent \$2.52 (which is 50.4% of their \$5 endowment)
- I received \$7.56 after the 3x multiplication
- The sender currently has  $\$5 - \$2.52 = \$2.48$  left

Last round I sent \$3.50 and got back \$6.73, which was quite generous (about 64% return rate).

In Round 1, when I received \$10.02, I returned \$5.00 (about 50% return rate).

Now I received \$7.56. The sender showed moderate trust (50.4% of endowment). I should return a fair amount.

If I return around 50% again:  $\$7.56 * 0.5 = \$3.78$

Let me check the outcomes:

- If I return \$3.78:
  - Sender ends with:  $\$2.48 + \$3.78 = \$6.26$  (profit of \$1.26)
  - I end with:  $\$7.56 - \$3.78 = \$3.78$  (loss of \$1.22 compared to keeping everything)

Actually, let me think about this more carefully. The sender sent 50.4% of their endowment, which is moderate trust. In Round 1, when someone sent 66.8%, I returned 50%.

For 50.4% trust level, I could return a proportional amount. Let me return about \$4.00:

- Sender ends with:  $\$2.48 + \$4.00 = \$6.48$  (profit of \$1.48)
- I end with:  $\$7.56 - \$4.00 = \$3.56$  (profit of \$3.56)

```
...
usage:
{
 "input_tokens": 1072,
 "output_tokens": 643,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005. Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

```
{'send': 3.50}
```

**reasoning:**

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

**usage:**

```
{
 "input_tokens": 1400,
 "output_tokens": 305,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was \$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it



t.

```
{'send': 3.50}
```

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well.  
I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

**usage:**

```
{
 "input_tokens": 1587,
 "output_tokens": 211,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.5 to you. Your visible payoff was \$8.0.

Your total visible earnings across all rounds are \$33.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.40 (88% of their endowment)
- I received \$13.20 after the 3x multiplication
- The sender currently has only \$0.60 left

This is the highest trust I've seen so far, exceeding even Round 1 (66.8%). I should reciprocate generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust while ensuring we both profit well.

```
{'return': 7.00}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.40 (88% of their endowment) - this is very high trust
- I received \$13.20 after the 3x multiplication
- The sender currently has \$0.60 left (\$5 - \$4.40)

This sender showed exceptional trust by sending 88% of their endowment. I should reciprocate generously.

Looking at my past behavior as receiver:

- Round 1: Received \$10.02, returned \$5.00 (about 50%)
- Round 3: Received \$7.56, returned \$4.00 (about 53%)

For this higher trust situation, I should return more generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust significantly while still maintaining good profit for both.

Alternatively, if I return \$6.50:

- Sender ends with:  $\$0.60 + \$6.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$13.20 - \$6.50 = \$6.70$  (profit of \$6.70)

I think \$7.00 is appropriate given the very high trust shown (88% vs the 70% I've been sending).

usage:

```
{
 "input_tokens": 1827,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.4, it became \$13.200000000000001, and you returned \$7.0. Your visible payoff was \$6.200000000000001.

Your total visible earnings across all rounds are \$39.46000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.91, so you received \$11.73. How much do you return? (0-11.73)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left

This is high trust, similar to previous high-trust rounds. Comparing to my history:

- Round 1: 66.8% sent ? returned \$5.00
- Round 6: 88% sent ? returned \$7.00

At 78.2%, this falls between those two. I'll return proportionally.

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

{'return': 6.50}

reasoning:

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left (\$5 - \$3.91)

This is high trust, though not quite as high as the 88% I saw in Round 6. Let me compare to my previous responses:

Round 1: Sender sent 66.8%, I received \$10.02, I returned \$5.00 (about 50%)

Round 3: Sender sent 50.4%, I received \$7.56, I returned \$4.00 (about 53%)

Round 6: Sender sent 88%, I received \$13.20, I returned \$7.00 (about 53%)

This sender sent 78.2%, which is between Round 1 (66.8%) and Round 6 (88%).

If I return \$6.00:

- Sender ends with:  $\$1.09 + \$6.00 = \$7.09$  (profit of \$2.09)
- I end with:  $\$11.73 - \$6.00 = \$5.73$  (profit of \$5.73)

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

Given the high trust level (78.2%), I think \$6.50 is appropriate, falling between my Round 1 return (\$5.00 for 66.8%) and Round 6 return (\$7.00 for 88%).

**usage:**

```
{
 "input_tokens": 2156,
 "output_tokens": 639,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.91, it became \$11.73, and you returned \$6.5. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$44.69000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my consistent success as sender:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)

The \$3.50 strategy has proven reliable and profitable. I'll continue with it.

```
{'send': 3.50}
```

**reasoning:**

Let me review my history as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50, received \$6.50 back (payoff \$8.00)

My sending strategy of \$3.50 has been consistently successful, with returns ranging from \$6.50 to \$6.95.

As receiver, I've been fair and reciprocal:

- Round 1: Received \$10.02, returned \$5.00
- Round 3: Received \$7.56, returned \$4.00
- Round 6: Received \$13.20, returned \$7.00
- Round 7: Received \$11.73, returned \$6.50

The cooperative strategy has worked well. I should continue with \$3.50.

**usage:**

```
{
 "input_tokens": 2234,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$5.32 to you. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$51.51000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my sender history:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)
- Round 8: Sent \$3.50 ? received \$5.32 back (payoff \$6.82)

Even with some variation in returns, the high-trust strategy of sending \$3.50 has been consistent

ently profitable. I'll maintain this approach.

```
{'send': 3.50}
```

**Reasoning:**

Looking at my track record as sender:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)
- Round 8: Sent \$3.50 ? received \$5.32 back (payoff \$6.82)

The last round had a lower return (\$5.32) but still profitable. The overall strategy of sending \$3.50 has been working well on average. I've been maintaining trust and cooperation, and while returns vary, I'm still profiting overall.

I should continue with the \$3.50 strategy as it has proven reliable over multiple rounds.

```
Usage: {
 "input_tokens": 2286,
 "output_tokens": 372,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:41:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_7"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.08, it became \$6.24, and you returned \$3.5. Your visible payoff was \$2.74.

Your total visible earnings across all rounds are \$9.83.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

#### Message 3 role=assistant

##### content:

**\*\*Analysis:\*\***

- My co-player sent \$4.18 (84% of their endowment) - showing very high trust!
- I received \$12.54
- This created a substantial gain of \$8.36

**\*\*Considerations:\*\***

1. This is a strong signal of trust - sending over 80% of their endowment
2. I should reciprocate generously to reward and maintain this cooperative relationship
3. Need to balance fairness with mutual benefit

If I return \$7:

- They end with: \$5 - \$4.18 + \$7 = \$7.82 (profit of \$2.82)
- I end with: \$12.54 - \$7 = \$5.54 (profit of \$5.54)
- Both of us profit significantly

This rewards their high trust while maintaining a mutually beneficial outcome.

{"return": 7}

##### reasoning:

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$4.18 (about 84% of their \$5 endowment) - this is very high trust!
- I received \$12.54
- This created a gain of \$12.54 - \$4.18 = \$8.36

**\*\*Previous rounds context:\*\***

- Round 1: I sent \$3, received back \$5.09 (generous return)
- Round 2: I received \$6.24, returned \$3.50

**\*\*Considerations:\*\***

1. My current co-player showed HIGH trust by sending \$4.18 (84% of endowment)
2. I should reciprocate generously to reward this trust
3. Fair reciprocation would give them a good profit while I also benefit

**\*\*Options:\*\***

If I return \$6:

- They end with:  $\$5 - \$4.18 + \$6 = \$6.82$  (profit of \$1.82)
- I end with:  $\$12.54 - \$6 = \$6.54$  (profit of \$6.54)

If I return \$7:

- They end with:  $\$5 - \$4.18 + \$7 = \$7.82$  (profit of \$2.82)
- I end with:  $\$12.54 - \$7 = \$5.54$  (profit of \$5.54)

If I return \$8:

- They end with:  $\$5 - \$4.18 + \$8 = \$8.82$  (profit of \$3.82)
- I end with:  $\$12.54 - \$8 = \$4.54$  (profit of \$4.54)

Given their high trust (sending 84%), I should return generously. Returning about 55-60% of what I received seems fair and rewards their trust.

\$6-\$7 seems reasonable. I'll go with \$7 to reward their high trust while still benefiting from the transaction.

**usage:**

```
{
 "input_tokens": 1143,
 "output_tokens": 671,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous sending experience:\*\***

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

**\*\*Strategy:\*\***

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

```
{"send": 3}
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

**\*\*Strategic considerations:\*\***

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

**\*\*Options:\*\***

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 529,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)
- Both parties gain significantly

This rewards their trust while creating mutual benefit.

{"return": 5.50}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*Previous round:\*\***

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received

- This was fair but not overly generous

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with:  $\$5 - \$3.40 + \$5.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$10.20 - \$5.50 = \$4.70$  (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

**usage:**

```
{
 "input_tokens": 1755,
 "output_tokens": 512,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.4, it became \$10.2, and you returned \$5.5. Your visible payoff was \$4.6999999999999999.

Your total visible earnings across all rounds are \$25.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

Both instances were profitable, showing that \$3 is a reasonable amount that encourages reciprocity.

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their behavior
- My consistent moderate strategy has been working

**\*\*Strategy:\*\***

Sending \$3:

- Creates \$9 for the receiver
- Demonstrates cooperative intent
- Provides room for mutually beneficial returns
- Has proven effective in previous rounds

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3, received \$5.09 back - good return



- Round 4: Sent \$3, received \$3.66 back - fair return

Both times I sent \$3, I received positive returns that made the exchange worthwhile.

**\*\*Current situation:\*\***

- New co-player this round
- No specific information about their reciprocity
- My moderate sending strategy has been working well

**\*\*Strategy:\*\***

- Sending \$3 has proven to be a good balanced approach
- Creates \$9 for receiver, creating room for mutual benefit
- Shows cooperation without risking everything
- Consistent with my established pattern

I'll continue with \$3 as it's been effective.

**usage:**

```
{
 "input_tokens": 2072,
 "output_tokens": 350,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.32 to you. Your visible payoff was \$4.32.

Your total visible earnings across all rounds are \$30.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 6: I sent \$3, they returned \$2.32 (about 26% of what they received)

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They sent \$3.70 (74% of their endowment) - high trust!
- I received \$11.10
- This created \$7.40 in surplus

**\*\*Considerations:\*\***

1. They're showing high trust now by sending 74%
2. Despite lower return last round, they're signaling cooperation
3. Should reciprocate fairly to maintain the relationship

If I return \$6:

- They end with: \$5 - \$3.70 + \$6 = \$7.30 (profit: \$2.30)
- I end with: \$11.10 - \$6 = \$5.10 (profit: \$5.10)
- Both parties benefit significantly

This rewards their trust and encourages continued cooperation.

{"return": 6}

**reasoning:**

**\*\*Analysis of the situation:\*\***

Previous round (Round 6):

- I sent \$3 to this co-player
- They received \$9
- They returned \$2.32 to me
- This was a somewhat low return (about 26% of what they received)

Current round (Round 7):

- Same co-player (continuing interaction)
- They sent me \$3.70 (74% of their endowment)
- I received \$11.10
- This is a strong trust signal

**\*\*Considerations:\*\***

1. They showed high trust by sending \$3.70 (74%)
2. In the previous round, they returned less than I might have hoped (\$2.32 from \$9)
3. This is a continued interaction, so my response matters for future cooperation
4. I should be fair and reciprocate their trust appropriately

**\*\*Strategic thinking:\*\***

Even though they returned less in Round 6, they're now showing high trust by sending \$3.70

. I should:

- Reciprocate fairly to maintain cooperation
- Not punish them excessively for the previous lower return
- Create mutual benefit

If I return \$6:

- They end with:  $\$5 - \$3.70 + \$6 = \$7.30$  (profit of \$2.30)
- I end with:  $\$11.10 - \$6 = \$5.10$  (profit of \$5.10)
- Both profit significantly

This rewards their current trust while maintaining fair reciprocity.

**usage:**

```
{
 "input_tokens": 2393,
 "output_tokens": 586,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.7, it became \$11.100000000000001, and you returned \$6.0. Your visible payoff was \$5.100000000000001.

Your total visible earnings across all rounds are \$35.150000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3 ? received \$5.09 (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 (profit: \$0.66)
- Round 6: Sent \$3 ? received \$2.32 (loss: -\$0.68)

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No information about their reciprocity patterns
- My moderate strategy has mostly been effective

**\*\*Strategy:\*\***

Continue with \$3:

- Creates \$9 for the receiver
- Shows cooperative intent
- Balanced risk approach
- Has generally yielded positive outcomes (2 out of 3 profitable)

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

- Round 6: Sent \$3 ? received \$2.32 back (profit: -\$0.68, actually a loss)

**\*\*Current situation:\*\***

- This is Round 8, so a new co-player (randomized pairings)
- No specific information about their behavior
- My \$3 strategy has been generally effective, though with varying results

**\*\*Strategy considerations:\*\***

The \$3 sending amount has been my consistent approach:

- Creates \$9 for receiver
- Shows cooperative intent but not excessive risk
- Has yielded mixed but generally positive results

I'll continue with \$3 as it's a balanced approach that:

1. Demonstrates willingness to cooperate
2. Creates significant surplus
3. Doesn't overcommit resources
4. Has worked reasonably well overall

```
{"send": 3}
```

**usage:**

```
{
 "input_tokens": 2571,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$42.290000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$4.27 (85.4% of their endowment) - very high trust!
- I received \$12.81
- This created a surplus of \$8.54

**\*\*Considerations:\*\***

1. They showed exceptional trust by sending over 85% of their endowment
2. Should reciprocate generously to reward this high trust
3. Both parties should benefit substantially from the exchange

If I return \$7.50:

- They end with: \$5 - \$4.27 + \$7.50 = \$8.23 (profit: \$3.23)
- I end with: \$12.81 - \$7.50 = \$5.31 (profit: \$5.31)
- Both parties gain significantly

This rewards their exceptional trust while maintaining mutual benefit.

```
{"return": 7.50}
```

**Reasoning:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$4.27 (85.4% of their endowment) - very high trust!
- I received \$12.81
- This created a gain of \$8.54

**\*\*My receiving history:\*\***

- Round 2: Received \$6.24, returned \$3.50 (kept \$2.74)
- Round 3: Received \$12.54, returned \$7.00 (kept \$5.54)

- Round 5: Received \$10.20, returned \$5.50 (kept \$4.70)
- Round 7: Received \$11.10, returned \$6.00 (kept \$5.10)

Looking at my pattern:

- When I receive ~\$6.24: returned ~56%
- When I receive ~\$12.54: returned ~56%
- When I receive ~\$10.20: returned ~54%
- When I receive ~\$11.10: returned ~54%

I generally return about 54-56% of what I receive, which seems fair and encourages cooperation

**\*\*For this round:\*\***

- They sent \$4.27 (very high trust - 85.4%)
- I received \$12.81
- Returning about 55% would be \$7.05

If I return \$7:

- They end with:  $\$5 - \$4.27 + \$7 = \$7.73$  (profit: \$2.73)
- I end with:  $\$12.81 - \$7 = \$5.81$  (profit: \$5.81)

Both parties profit substantially. This rewards their very high trust level.

Let me calculate if I should be a bit more generous given the very high trust (85.4%):

If I return \$7.50:

- They end with:  $\$5 - \$4.27 + \$7.50 = \$8.23$  (profit: \$3.23)

...

```
Usage: {
 "input_tokens": 2551,
 "output_tokens": 752,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:42:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.25, it became \$9.75, and you returned \$6.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$11.49.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my experience so far:

**\*\*Analysis:\*\***

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 as receiver: Got \$9.75, returned \$6 - both parties benefited
- This is a new co-player, but the pattern suggests cooperation is working

**\*\*My decision:\*\***

Based on the positive reciprocity I've seen, I'll send **\*\*\$3.50\*\*** - slightly more than Round 1 to show continued trust and willingness to cooperate. This creates a \$10.50 pot for the receiver.

If they reciprocate at similar rates (~60-65%), we should both benefit nicely.

{"send": 3.5}

**reasoning:**

Let me analyze my history:

Round 1: I was SENDER  
- Sent \$3.0  
- Received back \$5.74  
- Payoff:  $\$5 - \$3 + \$5.74 = \$7.74$

Round 2: I was RECEIVER  
- Received \$9.75 (from \$3.25 sent)  
- Returned \$6.0  
- Payoff:  $\$9.75 - \$6.0 = \$3.75$

Total visible earnings: \$11.49

Now Round 3: I am SENDER again with a new co-player.

Things to consider:

1. In Round 1, when I sent \$3, I got back \$5.74 - this worked well
2. In Round 2, when someone sent me \$3.25, I returned \$6 (about 61.5% of what I received)
3. I have no information about this particular co-player for Round 3
4. There's communication noise mentioned in the instructions

The game is working reasonably well when people send moderate amounts. My experience suggests:

- Sending around \$3 has been successful
- Receivers have been returning reasonable amounts
- We're building some positive reciprocity

I should probably send a similar amount to Round 1, maybe slightly more to show continued trust. I'll send \$3 again or perhaps \$3.5.

Actually, let me think about the optimal strategy. If receivers are returning about 60-65% of what they receive:

- If I send \$3, they get \$9, might return ~\$5.5-6
- If I send \$4, they get \$12, might return ~\$7-8
- If I send \$5, they get \$15, might return ~\$9-10

Sending \$4 seems like a good balance - shows strong trust without going all-in on an unknown co-player.

Let me go with \$3.5 as a middle ground - slightly more than last time but not too aggressive.

...

usage:

```
{
 "input_tokens": 1132,
 "output_tokens": 711,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

**\*\*My decision:\*\***

Given their high trust, I want to reciprocate generously. Returning about 64% would be \*\*\$8\*\*.

If I return \$8:

- Sender ends with:  $\$0.86 + \$8 = \$8.86$  (net gain of \$3.86)
- I end with:  $\$12.42 - \$8 = \$4.42$  (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

```
{"return": 8}
```

**reasoning:**

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets  $\$0.86 + \$7.50 = \$8.36$  (gain of \$3.36), I get  $\$12.42 - \$7.50 = \$4.92$  (gain of \$4.92)
- If I return \$8: Sender gets  $\$0.86 + \$8 = \$8.86$  (gain of \$3.86), I get  $\$12.42 - \$8 = \$4.42$  (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 699,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my track record:

**\*\*Analysis:\*\***

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

**\*\*My decision:\*\***

Given the strong pattern of reciprocity, I'll send \*\*\$4\*\* to show trust and create a \$12 pot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

```
{"send": 4}
```

**reasoning:**

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)
- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

**usage:**

```
{
 "input_tokens": 1763,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.31 to you. Your visible payoff was \$7.31.

Your total visible earnings across all rounds are \$31.179999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.47, so you received \$7.41. How much do you return? (0-7.41)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.47 (keeping \$2.53) - more cautious than previous senders
- I received \$7.41
- I've consistently returned 60-65% as a receiver
- Need to maintain reciprocal behavior

**\*\*My decision:\*\***

Maintaining my pattern, I'll return about 62% which is \*\*\$4.60\*\*.

If I return \$4.60:

- Sender ends with: \$2.53 + \$4.60 = \$7.13 (net gain of \$2.13)
- I end with: \$7.41 - \$4.60 = \$2.81 (net gain of \$2.81)

Both players benefit, and I'm being fair and consistent with my established pattern.

```
{"return": 4.6}
```

**reasoning:**



Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.47 (keeping \$2.53)
- I received \$7.41
- This is a more cautious send compared to previous rounds I've seen

**\*\*My previous behavior as receiver:\*\***

- Round 2: Received \$9.75, returned \$6 (about 61.5%)
- Round 4: Received \$12.42, returned \$8 (about 64.4%)

**\*\*My pattern:\*\***

I've been returning around 60-65% of what I receive, which has been working well.

**\*\*Considerations:\*\***

- This sender was more cautious (\$2.47 vs the \$3-4+ I've seen)
- I should maintain consistency in my return rate
- 60-65% of \$7.41 would be about \$4.45 to \$4.82

**\*\*My decision:\*\***

I'll return around 62-63% to maintain my reciprocal pattern. That would be about \$4.60.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of \$2.13)
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of \$2.81)

Both benefit, and I'm being consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 526,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.47, it became \$7.41, and you returned \$4.6. Your visible payoff was \$2.8100000000000005.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Consistent returns around \$6-6.50 when sending \$3-4
- As receiver: I've been fair and reciprocal
- Total earnings are healthy at \$34
- Cooperation has generally paid off

**\*\*My decision:\*\***

I'll send **\*\*\$4\*\*** to maintain my cooperative strategy. This creates a \$12 pot for the receiver.

Based on patterns, I expect a reasonable return that benefits both of us.

```
{"return": 4}
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3 ? got back \$5.74
- Round 3: Sent \$3.5 ? got back \$6.46
- Round 5: Sent \$4 ? got back \$6.31

As RECEIVER:

- Round 2: Received \$9.75, returned \$6
- Round 4: Received \$12.42, returned \$8
- Round 6: Received \$7.41, returned \$4.6

I've been experiencing good cooperation overall. When I send, I generally get back decent returns (though Round 5 was slightly lower relative to what I sent). When I receive, I've been consistently fair.

For Round 7 as sender, I should continue the cooperative strategy. My last send was \$4 and got \$6.31 back. I could:

- Send \$4 again (proven to work)
- Send \$3.5 (slightly more conservative)
- Send \$4.5 (slightly more trusting)

Given the overall positive trend and my healthy total earnings (\$34), I'll continue with a trusting approach. I'll send \$4 again since that worked reasonably well last time.

```
{"send": 4}
```

**usage:**

```
{
 "input_tokens": 2389,
 "output_tokens": 410,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.4 to you. Your visible payoff was \$6.4.

Your total visible earnings across all rounds are \$40.38999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.92 (keeping \$2.08) - moderate trust
- I received \$8.76
- I've consistently returned around 60-65% as a receiver (Round 2: 61.5%, Round 4: 64.4%, Round 6: 62%)
- Need to maintain fair and consistent behavior

**\*\*My decision:\*\***

Staying with my pattern, I'll return about 62% which is **\*\*\$5.45\*\***.

If I return \$5.45:

- Sender ends with:  $\$2.08 + \$5.45 = \$7.53$  (net gain of \$2.53)
- I end with:  $\$8.76 - \$5.45 = \$3.31$  (net gain of \$3.31)

Both players benefit and I maintain my reciprocal reputation.

```
{"return": 5.45}
```

**reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.92 (keeping \$2.08) - moderate trust
- I received \$8.76
- I've been consistently returning around 60-65% as a receiver
- This has been my pattern: Round 2: returned  $6/9.75 = 61.5\%$ , Round 4: returned  $8/12.42 = 64.4\%$ , Round 6: returned  $4.6/7.41 = 62\%$

**\*\*My decision:\*\***

Maintaining my consistent pattern, I'll return about 62% which is approximately \$5.45.

If I return \$5.45:  
- Sender ends with:  $\$2.08 + \$5.45 = \$7.53$  (net gain of \$2.53)  
- I end with:  $\$8.76 - \$5.45 = \$3.31$  (net gain of \$3.31)

Both players benefit, and I'm being fair and consistent.

Let me calculate:  $8.76 * 0.62 = 5.43$

I'll go with \$5.45 to be slightly generous while maintaining my pattern.

**usage:**

```
{
 "input_tokens": 2455,
 "output_tokens": 494,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$5.45. Your visible payoff was \$3.3099999999999996.

Your total visible earnings across all rounds are \$43.699999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Returns have varied (\$5.40-\$6.46) but cooperation has paid off
- As receiver: I've been consistently fair (60-65% returns)
- Total earnings are strong at \$43.70
- This is a new co-player

**\*\*My decision:\*\***

I'll send **\*\*\$3.5\*\*** to show cooperation while being moderately cautious with a new partner. This creates a \$10.50 pot for the receiver.

Based on typical reciprocity patterns, I expect a reasonable return that benefits both of us.

```
{"send": 3.5}
```

**Reasoning:**

Let me analyze my track record:

As SENDER:

- Round 1: Sent \$3 ? got \$5.74 back (payoff: \$7.74)
- Round 3: Sent \$3.5 ? got \$6.46 back (payoff: \$7.96)
- Round 5: Sent \$4 ? got \$6.31 back (payoff: \$7.31)
- Round 7: Sent \$4 ? got \$5.4 back (payoff: \$6.4)

As RECEIVER:

- Round 2: Received \$9.75 ? returned \$6 (payoff: \$3.75)
- Round 4: Received \$12.42 ? returned \$8 (payoff: \$4.42)
- Round 6: Received \$7.41 ? returned \$4.6 (payoff: \$2.81)
- Round 8: Received \$8.76 ? returned \$5.45 (payoff: \$3.31)

Total earnings: \$43.70

Pattern analysis:

- When I send more, returns have generally been good but not always proportionally better
- I've been consistent as a receiver, returning 60-65%
- Cooperation has been working overall
- Last round as sender (Round 7) with \$4 got back \$5.4, which is 45% of the tripled amount (\$12)

I'm in Round 9 now, as a sender again with a new co-player. Given:

1. I have good total earnings

2. Cooperation has been working
3. I want to maintain trust but also be strategic

I'll send around \$3.5-\$4 to show cooperation while not being overly risky.

```
Usage: {
 "input_tokens": 2443,
 "output_tokens": 558,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_4 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:42:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### **GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.93 to you. Your visible payoff was \$6.43.  
Your total visible earnings across all rounds are \$10.4.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my recent experience:

In Round 2, I sent \$2.50 (50%) and received a good return of \$3.93, which was cooperative and gave me a profit of \$1.43.

Since I'm playing against a new co-player this round and the cooperative strategy has been working well, I'll continue with a similar approach. Sending around 50% signals trust and willingness to cooperate, which has been reciprocated so far.

```
{"send": 2.50}
```

**reasoning:**

Let me analyze what happened:

- Round 1: I was RECEIVER. Sender sent \$2.49, I received \$7.47, returned \$3.50. My payoff: \$3.97
- Round 2: I was SENDER. I sent \$2.50, they received \$7.50, they returned \$3.93. My payoff: \$2.50 + \$3.93 = \$6.43

In Round 2, I showed trust by sending \$2.50 (50%), and my co-player reciprocated well by returning \$3.93 out of \$7.50 (about 52%), which gave me a profit of \$3.93 (I invested \$2.50, got back \$3.93, net gain \$1.43).

Now I'm SENDER again in Round 3 with a new co-player.

The pattern so far shows:

- When senders show moderate-high trust (~50%), receivers tend to reciprocate
- Both interactions yielded positive outcomes
- I'm building a cooperative reputation

Since I'm facing a new co-player each round (randomized pairings from the pool of 8 agents), I should:

1. Continue to signal cooperation
2. Send a reasonable amount to invite reciprocity
3. Possibly send slightly more since the previous round went well and I'm building goodwill

I'll send \$2.50 again, maintaining consistency in my strategy.

**usage:**

```
{
 "input_tokens": 1153,
 "output_tokens": 447,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with:  $\$0.52 + \$7.00 = \$7.52$  (profit of  $\sim\$2.52$ )
- I keep:  $\$13.44 - \$7.00 = \$6.44$  (profit of  $\sim\$6.44$ )

Both of us profit well, and the sender is rewarded for their exceptional trust.

{"return": 7.00}

**reasoning:**

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with:  $\$0.52 + \$6.50 = \$7.02$  (profit of  $\sim\$2$ )
- I would keep:  $\$13.44 - \$6.50 = \$6.94$  (profit of  $\sim\$7$ )

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with:  $\$0.52 + \$7 = \$7.52$  (profit of  $\$2.52$ )
- I keep:  $\$13.44 - \$7 = \$6.44$  (profit of  $\$6.44$ )

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be:  $\$13.44 * 0.5 = \$6.72$

...

**usage:**

```
{
 "input_tokens": 1412,
 "output_tokens": 682,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001. Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of  $\sim\$2.22$ )
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of  $\sim\$6.34$ )

This maintains a similar return rate ( $\sim 51\%$ ) as Round 4 ( $\sim 52\%$ ) and both of us profit well.

{"return": 6.50}

**reasoning:**

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocation pattern for high trust.

The sender showed  $\sim 86\%$  trust level, which is slightly less than the  $\sim 90\%$  in Round 4, but still very high.

If I return \$6.50:

- Sender ends with:  $\$0.72 + \$6.50 = \$7.22$  (profit of  $\sim\$2.22$ )
- I keep:  $\$12.84 - \$6.50 = \$6.34$  (profit of  $\sim\$6.34$ )

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

**usage:**

```
{
 "input_tokens": 1745,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.28, it became \$12.84, and you returned \$6.5. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$28.540000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my history:

- As sender (Rounds 2-3): I've consistently sent \$2.50 (50%), with varying returns
- As receiver (Rounds 1, 4-5): I've reciprocated well when others showed trust

My 50% sending strategy has been reasonably successful, giving both parties opportunities to profit. While some agents have sent higher amounts (85-90%) when they're senders, I'll maintain my consistent approach that balances trust and risk.

{"send": 2.50}

**reasoning:**

Let me review my history:

Round 1 (RECEIVER): Received \$7.47 from sender who sent \$2.49 (50%), returned \$3.50 (~47% of what I received). Payoff: \$3.97

Round 2 (SENDER): Sent \$2.50 (50%), received \$3.93 back. Payoff: \$6.43

Round 3 (SENDER): Sent \$2.50 (50%), received \$2.86 back. Payoff: \$5.36

Round 4 (RECEIVER): Received \$13.44 from sender who sent \$4.48 (90%), returned \$7.00 (~52% of what I received). Payoff: \$6.44

Round 5 (RECEIVER): Received \$12.84 from sender who sent \$4.28 (86%), returned \$6.50 (~51% of what I received). Payoff: \$6.34

Now Round 6 (SENDER): I need to decide how much to send.

Looking at the pattern:

- When I've been sender, I've sent \$2.50 (50%) consistently
- When I'm receiver, I've gotten good returns when others send high amounts (50-90%)
- I've been rewarded for reciprocating high trust with good returns

However, I've noticed that when I sent 50% in rounds 2 and 3, I got varying returns:

- Round 2: \$3.93 (very good return, about 52% of what they received)
- Round 3: \$2.86 (lower return, about 38% of what they received)

Other agents have been sending higher amounts (85-90%) to me when they're senders, and I've reciprocated well. This has worked out for both parties.

Should I increase my sending? Or stick with my 50% strategy?

...

**usage:**

```
{
 "input_tokens": 2073,
 "output_tokens": 696,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.76 to you. Your visible payoff was \$7.26.

Your total visible earnings across all rounds are \$35.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)



Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my sending history:

- Round 2: Sent \$2.50, received \$3.93 back
- Round 3: Sent \$2.50, received \$2.86 back
- Round 6: Sent \$2.50, received \$4.76 back

The 50% strategy continues to work well on average. Round 6 showed particularly good reciprocity with a \$4.76 return. I'll maintain this consistent, moderate trust level.

{"send": 2.50}

**reasoning:**

Looking at my history:

Round 2 (SENDER): Sent \$2.50, received \$3.93 back, payoff \$6.43  
Round 3 (SENDER): Sent \$2.50, received \$2.86 back, payoff \$5.36  
Round 6 (SENDER): Sent \$2.50, received \$4.76 back, payoff \$7.26

As sender, my 50% strategy has been working reasonably well. In Round 6, I received an excellent return of \$4.76.

As receiver:

Round 1: Received \$7.47, returned \$3.50  
Round 4: Received \$13.44, returned \$7.00  
Round 5: Received \$12.84, returned \$6.50

I've been consistent and cooperative. My 50% sending strategy has been reasonable and I should continue with it.

**usage:**

```
{
 "input_tokens": 2318,
 "output_tokens": 335,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.1 to you. Your visible payoff was \$4.6.

Your total visible earnings across all rounds are \$40.400000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender showed good trust by sending \$3.41 (about 68% of their endowment)
- I received \$10.23 after the 3x multiplication
- The sender currently has \$1.59 remaining

In previous rounds as receiver, I've returned roughly 47-52% of what I received, depending on the trust shown.

If I return \$5.00:

- Sender ends with: \$1.59 + \$5.00 = \$6.59 (profit of ~\$1.59)
- I keep: \$10.23 - \$5.00 = \$5.23 (profit of ~\$5.23)

This is a fair reciprocation that rewards their trust while both of us profit.

{"return": 5.00}

**reasoning:**

Let me analyze this situation:

- The sender sent \$3.41 (about 68% of their endowment) - showing good trust

- I received \$10.23 after the 3x multiplication
- The sender currently has  $\$5 - \$3.41 = \$1.59$  remaining

Looking at my history as receiver:

- Round 1: Received \$7.47, returned \$3.50 (47% return rate)
- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- Round 5: Received \$12.84, returned \$6.50 (51% return rate)

The sender showed good but not exceptional trust (68% vs the 85-90% I saw in rounds 4-5).

If I return \$5.00:

- Sender ends with:  $\$1.59 + \$5.00 = \$6.59$  (profit of \$1.59)
- I keep:  $\$10.23 - \$5.00 = \$5.23$  (profit of \$5.23)

This gives about a 49% return rate, which is consistent with my previous behavior. Both parties profit reasonably.

Let me consider \$5.00 as a fair reciprocation.

**usage:**

```
{
 "input_tokens": 2283,
 "output_tokens": 486,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.41, it became \$10.23, and you returned \$5.0. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$45.63000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.39, so you received \$10.17. How much do you return? (0-10.17)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$3.39 (about 68% of their endowment) - very similar trust to Round 8
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 remaining

This is nearly identical to Round 8 where someone sent \$3.41 and I returned \$5.00. I should be consistent in my reciprocation.

If I return \$5.00:

- Sender ends with:  $\$1.61 + \$5.00 = \$6.61$  (profit of ~\$1.61)
- I keep:  $\$10.17 - \$5.00 = \$5.17$  (profit of ~\$5.17)

```
{"return": 5.00}
```

**Reasoning:**

Looking at this situation:

- The sender sent \$3.39 (about 67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 remaining

This is very similar to Round 8 where someone sent \$3.41 and I returned \$5.00. I should be consistent.

If I return \$5.00:

- Sender ends with:  $\$1.61 + \$5.00 = \$6.61$  (profit of ~\$1.61)
- I keep:  $\$10.17 - \$5.00 = \$5.17$  (profit of ~\$5.17)

This maintains consistency with my Round 8 response and fairly reciprocates their trust.

**Usage:**

```
{
 "input_tokens": 2338,
 "output_tokens": 368,
 "reasoning_tokens": 0
}
```

=====

## ROUND 10

=====

Roles: Agent\_1=Receiver, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Receiver,  
Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_8=Sender, Agent\_5=Receiver  
dyad\_2: Agent\_3=Sender, Agent\_6=Receiver  
dyad\_3: Agent\_4=Sender, Agent\_1=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_8 sent \$4; received \$12; Agent\_5 returned \$4.5; payoffs Agent\_8=\$5.5, Agent\_5=\$7.5;  
noise/visible: sent visible=\$3.19, received visible=\$9.57, returned visible=\$4.86  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 12.0, "min": 0, "raw": 4.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_2: Agent\_3 sent \$4; received \$12; Agent\_6 returned \$9.2; payoffs Agent\_3=\$10.2, Agent\_6=\$2.8;  
noise/visible: sent visible=\$4.75, received visible=\$14.25, returned visible=\$9.16  
Action validation: {"returned": {"amount": 9.2, "clamped": false, "max": 12.0, "min": 0, "raw": 9.2}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_4 sent \$2.5; received \$7.5; Agent\_1 returned \$3.25; payoffs Agent\_4=\$5.75, Agent\_1=\$4.25; noise/visible: sent visible=\$2.53, received visible=\$7.59, returned visible=\$3.45  
Action validation: {"returned": {"amount": 3.25, "clamped": false, "max": 7.5, "min": 0, "raw": 3.25}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}  
dyad\_4: Agent\_7 sent \$4; received \$12; Agent\_2 returned \$6.5; payoffs Agent\_7=\$7.5, Agent\_2=\$5.5;  
noise/visible: sent visible=\$3.86, received visible=\$11.58, returned visible=\$6.58  
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent\_1": 63.65,  
"Agent\_2": 61.24,  
"Agent\_3": 62.790000000000006,  
"Agent\_4": 54.85,  
"Agent\_5": 60.39,  
"Agent\_6": 53.0,  
"Agent\_7": 53.730000000000004,  
"Agent\_8": 60.25

}

Visible balances after round: {

"Agent\_1": 63.0,  
"Agent\_2": 64.780000000000002,  
"Agent\_3": 67.38,  
"Agent\_4": 56.750000000000014,  
"Agent\_5": 56.22,  
"Agent\_6": 55.989999999999995,  
"Agent\_7": 55.180000000000001,  
"Agent\_8": 54.790000000000006

}

## LLM CALLS

### GAME - Agent\_8 - Sender - dyad\_1

Timestamp: 2026-08-12T09:42:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_5",  
"pairing": {  
  "agents": [  
    "Agent\_8",  
    "Agent\_5"  
  ],  
  "investor": "Agent\_8",  
  "roles": {  
    "Agent\_5": "trustee",  
    "Agent\_8": "investor"  
  }  
},

```

 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$5.0. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$16.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

**\*\*Round 4 Analysis:\*\***

I'm the SENDER now with a new co-player. Based on my experience:

- Round 1 as sender: sent \$3, got back \$3.03 - fair reciprocation
- Rounds 2-3 as receiver: I returned about 50% of what I received

I'll send \*\*\$4\*\* to signal strong willingness to cooperate and create value through the 3x multiplier, while managing some risk since I don't know this co-player's strategy.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 4 Analysis:\*\***

I'm now the SENDER again. Let me think about what I've learned:

Round 1: I sent \$3, received \$3.03 back (roughly equal return)  
Round 2: As receiver, I got \$12.99, returned \$6.50 (50% return rate)  
Round 3: As receiver, I got \$9.87, returned \$5.00 (50% return rate)

I've been playing fairly as a receiver, returning about 50% of what I receive. My experience as a sender in Round 1 was positive - I got back roughly what I sent.

Now I'm playing with a new co-player (since pairings are randomized each round). I don't know their history or strategy.

Given that:

1. The 3x multiplier creates significant value from cooperation
2. I've seen moderate cooperation work well (Round 1)
3. I should signal willingness to cooperate but manage risk
4. Sending around 60-80% seems optimal to balance value creation and risk

I'll send \$4 this time - a bit more than Round 1, showing I'm willing to trust and create value while still keeping some safety margin.

If they return fairly (around 50% of what they receive = \$6), I'll end up with \$5 - \$4 + \$6 = \$7, they'll get \$6. If they return less, I still have \$1 kept.

**usage:**

```
{
 "input_tokens": 1305,
 "output_tokens": 469,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$23.53.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Round 5 Analysis:\*\***

Previous round went well - sent \$4, received \$6.14 back, netting \$2.14 profit. That's good reciprocation.

Now with a new co-player as sender. My experience suggests players tend to reciprocate fairly when I send substantial amounts.

I'll send \*\*\$4\*\* again - this has worked well and signals strong cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 5 Analysis:\*\***

Previous round outcome:

- I sent \$4, it became \$12
- They returned \$6.14 (about 51% of what they received)
- My payoff: \$7.14 - excellent! I gained \$2.14 from cooperating

This is very positive reciprocation. Now I'm facing a new co-player as sender again.

My track record as sender:

- Round 1: sent \$3, got back \$3.03
- Round 4: sent \$4, got back \$6.14

Both times I received fair to good returns. This suggests either:

1. Players in this pool tend to reciprocate fairly
2. I've been lucky with cooperative partners

As receiver (Rounds 2-3), I returned about 50% consistently.

For Round 5 with a new co-player:

- I could send \$5 (maximum trust and value creation)
- I could send \$4 (high cooperation, what worked well last round)
- I could send \$3 (moderate)

Given the positive experiences and that fair reciprocation seems to be the norm, I'll send \$4 again. This shows strong cooperation without maximum risk exposure.

**usage:**

```
{
 "input_tokens": 1557,
 "output_tokens": 384,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$30.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.3, so you received \$6.8999999999999995. How much do you return? (0-6.8999999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

I'll maintain my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 3.50}
'''
```

This gives them a total of \$6.20 and me \$3.40, fairly splitting the gains.

**reasoning:**

**\*\*Round 6 Analysis:\*\***

- My co-player sent \$2.30 (46% of their endowment) - moderate trust
- I received \$6.90
- They have \$2.70 remaining before my return

This is a lower send amount than I've been receiving. I should still reciprocate fairly but proportionally to the trust shown.

Following my pattern of returning about 50% of what I received:

- 50% of \$6.90 = \$3.45

This would give them:  
- Sender:  $\$2.70 + \$3.45 = \$6.15$  total  
- Me:  $\$6.90 - \$3.45 = \$3.45$

This seems fair and proportional to their investment.

**usage:**

```
{
 "input_tokens": 1803,
 "output_tokens": 293,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.5. Your visible payoff was \$3.399999999999995.

Your total visible earnings across all rounds are \$33.77.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing strong trust
- I received \$11.04
- They have \$1.32 remaining before my return

I'll continue my fair reciprocation pattern of returning about 50% of what I received:

```
'''json
{"return": 5.50}
'''
```

This gives them a total of \$6.82 and me \$5.54, fairly splitting the gains from cooperation

**reasoning:**

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.68 (about 74% of their endowment) - showing good trust
- I received \$11.04
- They have \$1.32 remaining before my return

I should continue my pattern of fair reciprocation. Returning about 50% of what I received  
:

50% of \$11.04 = \$5.52

This would give them  $\$1.32 + \$5.52 = \$6.84$  total  
And I would keep  $\$11.04 - \$5.52 = \$5.52$

That seems fair and consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 274,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001.

Your total visible earnings across all rounds are \$39.31.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER with a new co-player.

My track record:

- As sender: sending \$4 has yielded good returns (\$6.14 and \$5.84)
- As receiver: I've consistently returned ~50% of what I received
- Total earnings: \$39.31

I'll continue with **\*\*\$4\*\*** - this amount has proven successful in building cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

**\*\*Round 8 Analysis:\*\***

I'm the SENDER now with a new co-player.

Let me review my history:

- Round 1 (sender): sent \$3, got back \$3.03 - fair
- Round 2 (receiver): received \$12.99, returned \$6.50
- Round 3 (receiver): received \$9.87, returned \$5.00
- Round 4 (sender): sent \$4, got back \$6.14 - good
- Round 5 (sender): sent \$4, got back \$5.84 - good
- Round 6 (receiver): received \$6.90, returned \$3.50
- Round 7 (receiver): received \$11.04, returned \$5.50

As a sender, I've sent \$3 and \$4 twice. Both \$4 sends worked well (got back \$6.14 and \$5.84).

As a receiver, I've consistently returned about 50% of what I received, which seems fair.

I'm doing well overall with \$39.31 in earnings. The strategy of sending \$4 has been working well for me.

I'll continue with sending \$4, which signals strong cooperation while not being too risky.

**usage:**

```
{
 "input_tokens": 2150,
 "output_tokens": 400,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$4.24 to you. Your visible payoff was \$5.24.

Your total visible earnings across all rounds are \$44.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.96, so you received \$8.879999999999999. How much do you return? (0-8.879999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 9 Analysis:\*\***

- Same co-player as last round! Last round I sent \$4, they returned \$4.24
- This round they sent \$2.96 (59% of their endowment) - moderate trust
- I received \$8.88
- They have \$2.04 remaining before my return

I'll maintain my cooperative strategy of returning about 50% of what I received:

```
'''json
{"return": 4.50}
'''
```



'''

This gives them a total of \$6.54 and me \$4.38, maintaining fair reciprocation.

**reasoning:**

Round 9 Analysis:

- Last round I was sender with this same co-player: sent \$4, received \$4.24 back - they returned about 35% of the \$12 they received
- This round they sent \$2.96 (59% of their endowment) - moderate trust
- I received \$8.88
- They have \$2.04 remaining before my return

This is interesting - I'm playing against the same co-player from last round. They returned \$4.24 when I sent \$4 (which gave them \$12). That's about 35% return rate.

Now they sent me \$2.96, giving me \$8.88.

I should be fair but also consider their previous behavior. They were somewhat conservative in their return last round. However, I generally try to maintain a cooperative 50% return strategy.

Let me return around 50% of what I received: about \$4.50

This would give them \$2.04 + \$4.50 = \$6.54 total

And I'd keep \$8.88 - \$4.50 = \$4.38

That seems fair given the circumstances.

**usage:**

```
{
 "input_tokens": 2126,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.96, it became \$8.879999999999999, and you returned \$4.5. Your visible payoff was \$4.379999999999999.

Your total visible earnings across all rounds are \$48.930000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Round 10 Analysis:\*\***

I'm the SENDER with a new co-player.

My sending strategy has been consistent at \$4, with generally good returns:

- Returns ranging from \$4.24 to \$6.14
- All profitable outcomes
- Total earnings: \$48.93

I'll continue with **\*\*\$4\*\*** to maintain strong cooperation signals.

```
'''json
{"send": 4}
'''
```

**Reasoning:**

Round 10 Analysis:

Looking at my history:

- Round 4: SENDER, sent \$4, received \$6.14 back, payoff \$7.14
- Round 5: SENDER, sent \$4, received \$5.84 back, payoff \$6.84
- Round 6: RECEIVER, received \$6.90, returned \$3.50, payoff \$3.40
- Round 7: RECEIVER, received \$11.04, returned \$5.50, payoff \$5.54
- Round 8: SENDER, sent \$4, received \$4.24 back, payoff \$5.24
- Round 9: RECEIVER (same co-player as Round 8), received \$8.88, returned \$4.50, payoff \$4.38

Now Round 10: I'm the SENDER with a new co-player.

My strategy as sender has been to send \$4 consistently, which has worked well. The returns have been:

- \$6.14 (51% of \$12)
- \$5.84 (49% of \$12)
- \$4.24 (35% of \$12)

The third one was a bit lower, but still net positive. Overall, sending \$4 has been profitable and demonstrates strong cooperation.

I'll continue with \$4 as it's been my successful strategy.

```
Usage: {
 "input_tokens": 2125,
 "output_tokens": 414,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_5 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:42:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_5"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_5": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.75 to you. Your visible payoff was \$5.75.

Your total visible earnings across all rounds are \$15.139999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

**\*\*Round 4 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$2.84 (57% of their endowment) - strong trust
- I received \$8.52 after the 3x multiplier
- They currently have \$2.16 remaining

This is very similar to Round 2 where I received \$8.46 and returned \$4.00. To maintain consistency and reward their trust:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit substantially

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

**\*\*Round 4 - RECEIVER Decision\*\***

Let me analyze the situation:

- My current co-player sent me \$2.84 (56.8% of their endowment) - a strong show of trust
- I received \$8.52 after the 3x multiplier
- They currently have  $\$5 - \$2.84 = \$2.16$  remaining

Looking at my recent history:

- Round 1 (Sender): Sent \$3, got \$2.93 back ? ended with \$4.93
- Round 2 (Receiver): Got \$2.82 (became \$8.46), returned \$4.00 ? ended with \$4.46
- Round 3 (Sender): Sent \$3, got \$3.75 back ? ended with \$5.75

My co-player has shown similar trust to Round 2, sending about \$2.84. In Round 2, I returned \$4.00 when I received \$8.46. This time I received \$8.52, which is very similar.

To be consistent and fair:

- If I return \$4.00:
  - They end up with:  $\$2.16 + \$4.00 = \$6.16$
  - I end up with:  $\$8.52 - \$4.00 = \$4.52$
  - Both benefit from cooperation

This maintains consistency with my previous behavior and rewards their trust appropriately

usage:

```
{
 "input_tokens": 1398,
 "output_tokens": 518,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.84, it became \$8.52, and you returned \$4.0. Your visible payoff was \$4.52.

Your total visible earnings across all rounds are \$19.659999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

**\*\*Round 5 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$4.32 (86% of their endowment!) - exceptional trust
- I received \$12.96 after the 3x multiplier
- They currently have only \$0.68 remaining

This is much higher trust than previous rounds (~\$2.82-\$2.84). They sent almost everything. I should reciprocate proportionally:

- If I return \$6.50:
  - They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
  - I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)
  - Both benefit substantially, rewarding their exceptional trust

```
'''json
{"return": 6.5}
'''
```

reasoning:

Round 5 - I'm the RECEIVER again

My current co-player sent \$4.32 (86% of their endowment!) - this is a very high level of trust

I received \$12.96 after the 3x multiplier

They currently have \$0.68 remaining

This is significantly more trust than previous rounds where I received ~\$8.46-\$8.52. They sent almost everything they had.

To reciprocate this high level of trust appropriately, I should return a generous amount.

Let me think about what's fair:

- They sent \$4.32 and kept only \$0.68
- If I return \$6.00:
  - They end up with:  $\$0.68 + \$6.00 = \$6.68$
  - I end up with:  $\$12.96 - \$6.00 = \$6.96$
  - They get more than their initial \$5, I get significantly more
- If I return \$7.00:
  - They end up with:  $\$0.68 + \$7.00 = \$7.68$
  - I end up with:  $\$12.96 - \$7.00 = \$5.96$
  - They gain \$2.68, I gain \$0.96

Given they showed exceptional trust (sending 86% vs the ~56% in previous rounds), I should return proportionally more. Returning around \$6-7 would be appropriate.

I'll go with \$6.50 as a middle ground that rewards their exceptional trust:

- They end up with:  $\$0.68 + \$6.50 = \$7.18$  (gain of \$2.18)
- I end up with:  $\$12.96 - \$6.50 = \$6.46$  (gain of \$1.46)

Both of us benefit well, and it appropriately rewards their very high trust.

**usage:**

```
{
 "input_tokens": 1715,
 "output_tokens": 645,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.32, it became \$12.96, and you returned \$6.5. Your visible payoff was \$6.460000000000001.

Your total visible earnings across all rounds are \$26.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Round 6 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My experience as sender has been positive:

- Round 1: Sent \$3, got \$2.93 back
- Round 3: Sent \$3, got \$3.75 back

Both times sending \$3 led to reasonable cooperation. This creates \$9 for the receiver, allowing room for mutual benefit. I'll continue this consistent approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Round 6 - I'm the SENDER with \$5.

Let me review my experience so far:

As SENDER:

- Round 1: Sent \$3, received ~\$2.93 back
- Round 3: Sent \$3, received \$3.75 back

As RECEIVER:

- Round 2: Received \$2.82? \$8.46, returned \$4.00
- Round 4: Received \$2.84? \$8.52, returned \$4.00
- Round 5: Received \$4.32? \$12.96, returned \$6.50

The game has been showing good cooperation overall. As a sender, I've gotten decent returns (around \$3-\$3.75 on \$3 sent). As a receiver, I've been fair and reciprocal.

Now I'm facing a new co-player. The cooperation has been working well. I could:

1. Continue with \$3 (my previous sender strategy)
2. Send more to signal stronger trust (like the \$4.32 I saw in Round 5)
3. Send less to be more cautious

Given the positive cooperation trends and that this is a new co-player, I'll stick with my consistent sender strategy of \$3. This has worked well and creates good mutual benefit opportunities.

**usage:**

```
{
```

```
"input_tokens": 2048,
"output_tokens": 424,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.38 to you. Your visible payoff was \$5.38.

Your total visible earnings across all rounds are \$31.499999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 7 - SENDER Decision\*\***

I'm the sender with \$5 facing a new co-player.

My track record as sender:

- Round 1: Sent \$3 ? received \$2.93 back (profit of \$0.93)
- Round 3: Sent \$3 ? received \$3.75 back (profit of \$0.75)
- Round 6: Sent \$3 ? received \$3.38 back (profit of \$0.38)

Sending \$3 has consistently worked well, showing decent cooperation from different co-players. I'll stick with this strategy that balances trust and risk management.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm the sender again with \$5.

Looking at my history as sender:

- Round 1: Sent \$3, got \$2.93 back (visible payoff \$4.93)
- Round 3: Sent \$3, got \$3.75 back (visible payoff \$5.75)
- Round 6: Sent \$3, got \$3.38 back (visible payoff \$5.38)

All three times I sent \$3, and I got decent returns ranging from about \$2.93 to \$3.75. This strategy has been working well - I'm consistently making a small profit.

I'm facing a new co-player now. Sending \$3 has been my consistent strategy and it's been working. I should continue with this.

**usage:**

```
{
 "input_tokens": 2294,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$7.03 to you. Your visible payoff was \$9.030000000000001.

Your total visible earnings across all rounds are \$40.53.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Round 8 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$3.72 (74% of their endowment) - high trust

- I received \$11.16 after the 3x multiplier
- They currently have \$1.28 remaining

This is substantial trust, between my Round 2/4 experiences (~57%) and Round 5 (86%).

Looking at my pattern as receiver:

- Rounds 2 & 4: Returned ~47% of received amount
- Round 5: Returned 50% of received amount

For consistency, returning around 50% makes sense:

- If I return \$5.50:
  - They end up with:  $\$1.28 + \$5.50 = \$6.78$  (profit of \$1.78)
  - I end up with:  $\$11.16 - \$5.50 = \$5.66$  (profit of \$0.66)
  - Both benefit, rewarding their trust appropriately

```
'''json
{"return": 5.5}
'''
```

#### reasoning:

This is Round 8, and I'm the receiver.

My co-player sent me \$3.72 (74% of their endowment) - that's a high level of trust.  
I received \$11.16 after the 3x multiplier.  
They currently have \$1.28 remaining.

Looking at my history as receiver:

- Round 2: Received \$8.46, returned \$4.00 (kept \$4.46)
- Round 4: Received \$8.52, returned \$4.00 (kept \$4.52)
- Round 5: Received \$12.96, returned \$6.50 (kept \$6.46)

In Round 5, when I received \$12.96 (from a send of \$4.32), I returned \$6.50, which was about 50% of what I received.

In this round, I received \$11.16 (from a send of \$3.72).

The sender has \$1.28 left. If I return \$5.50:

- They end up with:  $\$1.28 + \$5.50 = \$6.78$  (gain of \$1.78)
- I end up with:  $\$11.16 - \$5.50 = \$5.66$  (gain of \$0.66)

That seems fair - they sent 74% of their endowment, showing high trust. I should reciprocate proportionally.

Actually, let me think about this more carefully. They sent \$3.72, which is 74% of \$5. In Round 5, they sent \$4.32 which is 86% and I returned \$6.50 (50% of what I received).

For \$3.72 sent (74% trust level), returning around \$5.00-\$5.50 seems appropriate.

If I return \$5.50:

- Sender ends with \$6.78 (profit of \$1.78 from initial \$5)
- I end with \$5.66 (profit of \$0.66 from initial \$5)

This rewards their trust reasonably while ensuring both benefit. Looking at my return patterns across rounds, I've consistently returned roughly 47-50% of what I received, with higher returns when the sender showed greater trust. A \$5.50 return maintains that consistency....

#### usage:

```
{
 "input_tokens": 2433,
 "output_tokens": 935,
 "reasoning_tokens": 0
}
```

#### Message 12 role=user

##### content:

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.72, it became \$11.16, and you returned \$5.5. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$46.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Round 9 - SENDER Decision\*\***

I'm playing against the same co-player from Round 8! They showed strong trust by sending \$3.72 (74% of their endowment), and I returned \$5.50, giving them a profit of \$1.78.

Since they demonstrated high trust with me, I should reciprocate. Matching their trust level of 74%, I'll send approximately \$3.70.

This creates \$11.10 for them and shows I value the cooperative relationship we've built.

```
'''json
{"send": 3.7}
'''
```

**reasoning:**

This is interesting - I'm playing against the same co-player as Round 8. They sent me \$3.72 (74% of their endowment) and I returned \$5.50.

Let me think about this:

- They showed high trust by sending \$3.72 (74%)
- I returned \$5.50, giving them a profit of \$1.78
- They ended with \$6.78

Now I'm the sender. Given that they showed strong trust last round, I should reciprocate with similarly strong trust.

My typical strategy has been to send \$3, but this co-player demonstrated extra trust with me. I should consider sending more to match their level of trust.

They sent 74% of their endowment. If I match that, I would send  $74\% \times \$5 = \$3.70$ .

Actually, let me think about this more carefully. They sent \$3.72 and after my return of \$5.50, they made a profit of \$1.78. That's a good outcome for them.

If I send \$3.70:

- It becomes \$11.10
- If they return proportionally like I did (about 50%), they'd return ~\$5.55
- I'd end up with:  $\$5 - \$3.70 + \$5.55 = \$6.85$
- They'd end up with:  $\$11.10 - \$5.55 = \$5.55$

This seems fair and reciprocates their trust level.

Alternatively, I could send \$4 to show even more trust:

- It becomes \$12
- If they return 50%, they return \$6
- I end up with:  $\$5 - \$4 + \$6 = \$7$
- They end up with:  $\$12 - \$6 = \$6$

...

**usage:**

```
{
 "input_tokens": 2465,
 "output_tokens": 571,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.7, it became \$11.100000000000001, and you saw them return \$3.66 to you. Your visible payoff was \$4.96.

Your total visible earnings across all rounds are \$51.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)



Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Round 10 - RECEIVER Decision\*\***

Analyzing the situation:

- My current co-player sent \$3.19 (64% of their endowment) - good trust
- I received \$9.57 after the 3x multiplier
- They currently have \$1.81 remaining

Looking at my receiver pattern:

- I've consistently returned around 47-50% of what I receive
- This amount (\$9.57) is similar to Rounds 2 & 4 where I returned \$4.00

For consistency:

- If I return \$4.50:
  - They end up with:  $\$1.81 + \$4.50 = \$6.31$  (profit of \$1.31)
  - I end up with:  $\$9.57 - \$4.50 = \$5.07$  (profit of \$0.07)
  - Both benefit from cooperation

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Round 10 - RECEIVER Decision

My current co-player sent \$3.19 (64% of their endowment) - moderate-high trust  
I received \$9.57 after the 3x multiplier  
They currently have \$1.81 remaining

Looking at my receiver history:

- Round 2: Received \$8.46, returned \$4.00 (47%)
- Round 4: Received \$8.52, returned \$4.00 (47%)
- Round 5: Received \$12.96, returned \$6.50 (50%)
- Round 8: Received \$11.16, returned \$5.50 (49%)

I've been returning around 47-50% of what I receive.

For \$9.57, returning around 47-50% would be:

- 47% = \$4.50
- 50% = \$4.79

If I return \$4.50:

- They end up with:  $\$1.81 + \$4.50 = \$6.31$  (profit of \$1.31)
- I end up with:  $\$9.57 - \$4.50 = \$5.07$  (profit of \$0.07)
- Both benefit

If I return \$5.00:

- They end up with:  $\$1.81 + \$5.00 = \$6.81$  (profit of \$1.81)
- I end up with:  $\$9.57 - \$5.00 = \$4.57$  (loss of \$0.43)
- They benefit more

I'll return \$4.50 to maintain my pattern of returning about 47% while ensuring both parties benefit.

```
Usage: {
 "input_tokens": 2452,
 "output_tokens": 618,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:42:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_6",
 "pairing": {
```

```

"agents": [
 "Agent_3",
 "Agent_6"
],
"investor": "Agent_3",
"roles": {
 "Agent_3": "investor",
 "Agent_6": "trustee"
},
"trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.73 to you. Your visible payoff was

\$5.73.

Your total visible earnings across all rounds are \$18.32.

This round, you are the RECEIVER against your current co-player. They sent you \$4.37, so you received \$13.11. How much do you return? (0-13.11)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.37 (87.4% of their endowment)
- Total pool created: \$0.63 (with sender) + \$13.11 (with me) = \$13.74
- They took significant risk to create mutual benefit

To reciprocate their strong trust fairly, I'll aim for an approximately equal split. Returning \$6.24 would give us both around \$6.87.

{'return': 6.24}

**reasoning:**

Let me analyze this situation:

In Round 4, I'm the RECEIVER.

- The sender sent \$4.37 (87.4% of their endowment)
- I received \$13.11 (after 3x multiplication)
- The sender now has \$5 - \$4.37 = \$0.63

The sender showed very high trust by sending 87.4% of their endowment. This is quite generous.

Total pool created: \$0.63 (sender has) + \$13.11 (I have) = \$13.74

To be fair and reciprocate their trust, I should aim for a roughly equal split. If we each get around \$6.87, that would mean:

- Sender would have: \$0.63 + return = \$6.87, so return = \$6.24
- I would have: \$13.11 - \$6.24 = \$6.87

This seems like a fair reciprocation of their high trust.

Looking at my past behavior:

- Round 1: As receiver, got \$11.67, returned \$5.28 (kept \$6.39, sender got \$6.39)
- Round 2: As sender, sent \$4.00
- Round 3: As sender, sent \$4.00

I've been fairly cooperative and trusting. The sender in this round showed even more trust than the sender in Round 1 (87.4% vs 77.8%), so I should reciprocate appropriately.

Returning \$6.24 would give us both about \$6.87, which is fair.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.37, it became \$13.11, and you returned \$6.24. Your visible payoff was \$6.869999999999999.

Your total visible earnings across all rounds are \$25.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$4.20 (84% of their endowment)

- Total pool created: \$0.80 (with sender) + \$12.60 (with me) = \$13.40
- They took significant risk to create mutual benefit

To reciprocate their trust fairly, I'll aim for an equal split. Returning \$5.90 would give us both \$6.70.

```
{'return': 5.90}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.20 (84% of their endowment)
- I received \$12.60 (after 3x multiplication)
- The sender has \$0.80 left
- Total pool: \$0.80 + \$12.60 = \$13.40

The sender showed high trust. To be fair and reciprocate, I should aim for roughly equal payoffs.

If I return \$6.30, we'd both have:

- Sender: \$0.80 + \$6.30 = \$7.10
- Me: \$12.60 - \$6.30 = \$6.30

That's not quite equal. Let me recalculate.

Total pool = \$13.40

For equal split: each gets \$6.70

So I should return: \$13.40 - \$6.70 = \$6.70

- Sender would get: \$0.80 + \$6.70 = \$7.50
- I would get: \$12.60 - \$6.70 = \$5.90

Hmm, that's not right either. Let me think again.

If I return X:

- Sender gets: \$0.80 + X
- I get: \$12.60 - X

For equal outcomes:

$$\$0.80 + X = \$12.60 - X$$

$$2X = \$12.60 - \$0.80$$

$$2X = \$11.80$$

$$X = \$5.90$$

Check:

- Sender: \$0.80 + \$5.90 = \$6.70
- Me: \$12.60 - \$5.90 = \$6.70

Perfect! That's an equal split.

Given the sender's high trust (84%), I should reciprocate fairly. ...

**usage:**

```
{
 "input_tokens": 1423,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.2, it became \$12.600000000000001, and you returned \$5.9. Your visible payoff was \$6.700000000000001. Your total visible earnings across all rounds are \$31.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience across rounds, sending around \$4.00 has consistently led to cooperative outcomes. I'll continue with this trusting strategy.

```
{'send': 4.00}
```

**reasoning:**

In this round, I'm the sender. Let me review my recent experiences:

Round 1: I was receiver, sender sent \$3.89, I returned \$5.28, both got ~\$6.39

Round 2: I sent \$4.00, received back \$5.20, got \$6.20

Round 3: I sent \$4.00, received back \$4.73, got \$5.73

Round 4: I was receiver, sender sent \$4.37, I returned \$6.24, both got ~\$6.87

Round 5: I was receiver, sender sent \$4.20, I returned \$5.90, both got \$6.70

When I've been a receiver, I've been fairly generous in my returns, aiming for equal splits. When I've been a sender, I've sent \$4.00 and received reasonable returns.

My strategy has been cooperative and trusting. I should continue with this approach, sending a high amount to show trust and hopefully receive a fair return.

Sending \$4.00 has worked reasonably well in the past. I'll continue with this level.

**usage:**

```
{
 "input_tokens": 1662,
 "output_tokens": 301,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.91 to you. Your visible payoff was \$8.91.

Your total visible earnings across all rounds are \$40.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.24, so you received \$6.720000000000001. How much do you return? (0-6.720000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment), showing moderate trust
- Total pool created: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- This is less trust than most previous rounds

I'll still reciprocate fairly. Returning \$2.00 would give us both approximately \$4.75.

```
{'return': 2.00}
```

**reasoning:**

Looking at this situation:

- The sender sent \$2.24 (44.8% of their endowment)
- This is notably less than previous senders I've encountered who sent around 77-87%
- Total pool: \$2.76 (with sender) + \$6.72 (with me) = \$9.48
- They showed moderate trust compared to others

Since they sent less than previous players, I should still reciprocate fairly but proportionally to their trust level.

For an equal split: each gets \$4.74

That means I'd return  $\$2.76 + (\text{total} - \text{sender's remaining}) / 2 = \dots$  let me recalculate

Sender has:  $\$5 - \$2.24 = \$2.76$

I received: \$6.72

Total created:  $\$2.76 + \$6.72 = \$9.48$

For equal split of \$4.74 each:

Sender needs:  $\$4.74 - \$2.76 = \$1.98$  returned

I keep:  $\$6.72 - \$1.98 = \$4.74$

But given they showed less trust than previous players, maybe I should be slightly less generous. However, I should still be fair and cooperative.

I'll return around \$2.00-\$2.50 to maintain fairness.

Let me go with \$2.30, which gives:

- Sender:  $\$2.76 + \$2.30 = \$5.06$
- Me:  $\$6.72 - \$2.30 = \$4.42$

This is slightly favoring them which encourages cooperation but not as generous as when they send more.

Actually, let me reconsider. I've been fairly cooperative throughout. Even though they send less, being consistent in fairness might be better. Let me aim closer to equal:

...

**usage:**

```
{
 "input_tokens": 1846,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.24, it became \$6.720000000000001, and you returned \$2.0. Your visible payoff was \$4.720000000000001.

Your total visible earnings across all rounds are \$45.519999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience, sending \$4.00 has generally led to good cooperative outcomes. I'll continue with this strategy.

{'send': 4.00}

**reasoning:**

Looking at my history:

- Round 1 (as RECEIVER): They sent \$3.89, I returned \$5.28
- Round 2 (as SENDER): I sent \$4.00, they returned \$5.20
- Round 3 (as SENDER): I sent \$4.00, they returned \$4.73
- Round 4 (as RECEIVER): They sent \$4.37, I returned \$6.24
- Round 5 (as RECEIVER): They sent \$4.20, I returned \$5.90
- Round 6 (as SENDER): I sent \$4.00, they returned \$7.91
- Round 7 (as RECEIVER): They sent \$2.24, I returned \$2.00

I've been consistently cooperative, and most co-players have shown high trust levels (sending \$4+). The exception was Round 7 where someone only sent \$2.24, and I responded proportionally.

I should continue with my trusting strategy. Sending \$4.00 has worked well overall.

**usage:**

```
{
 "input_tokens": 1885,
 "output_tokens": 286,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$5.28 to you. Your visible payoff was \$6.28.

Your total visible earnings across all rounds are \$51.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so y

ou received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.92 (58.4% of their endowment), showing moderate trust
- Total pool created: \$2.08 (with sender) + \$8.76 (with me) = \$10.84
- This is the same co-player from last round who returned \$5.28 to me, showing fair reciprocation

I'll reciprocate fairly. Returning \$3.34 would give us both approximately \$5.42 each.

{'return': 3.34}

**reasoning:**

Looking at this situation:

- The sender sent \$2.92 (58.4% of their endowment)
- Total pool created: \$2.08 (with sender) + \$8.76 (with me) = \$10.84
- They showed moderate trust

In the previous round with this same co-player:

- I sent \$4.00, which became \$12.00
- They returned \$5.28
- I ended up with \$6.28, they ended up with \$6.72

So in round 8 (the previous round with this co-player), they showed reasonable reciprocity by returning \$5.28 on the \$12.00 they received.

Now they're sending me \$2.92. This is less than what I sent them last time (\$4.00), but still a reasonable amount.

For fairness, aiming for an equal split:

- They have: \$2.08
- I have: \$8.76
- Total: \$10.84
- Equal split: ~\$5.42 each

To give them \$5.42, I need to return:  $\$5.42 - \$2.08 = \$3.34$

Let me double check:

- Sender keeps:  $\$5 - \$2.92 + \$3.34 = \$5.42$
- I keep:  $\$8.76 - \$3.34 = \$5.42$

Yes, that works for an equal split.

Given they were cooperative in the previous round (returning \$5.28 on \$12.00), I'll reciprocate fairly here.

**usage:**

```
{
 "input_tokens": 1880,
 "output_tokens": 494,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$3.34. Your visible payoff was \$5.42.

Your total visible earnings across all rounds are \$57.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my consistent strategy of sending \$4.00, which has led to positive outcomes overall, I'll continue with this approach.

```
{'return': 4.00}
```

**Reasoning:**

Looking at my game history:

- Round 4: I was sender, sent \$4.0, got back \$4.73, payoff \$5.73
- Round 5: I was receiver, they sent \$4.37, I returned \$6.24, payoff \$6.87
- Round 6: I was receiver, they sent \$4.2, I returned \$5.9, payoff \$6.70
- Round 7: I was sender, sent \$4.0, got back \$7.91, payoff \$8.91
- Round 8: I was receiver, they sent \$2.24, I returned \$2.0, payoff \$4.72
- Round 9: I was sender, sent \$4.0, got back \$5.28, payoff \$6.28
- Round 10: I was receiver, they sent \$2.92, I returned \$3.34, payoff \$5.42

My strategy has been to send \$4.00 when I'm the sender, which has been working well overall. I've gotten back varying amounts but generally reasonable reciprocation. I'll continue with this strategy.

Usage: {

```
"input_tokens": 1949,
"output_tokens": 305,
"reasoning_tokens": 0
```

}

**GAME - Agent\_6 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:42:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_3",
"pairing": {
 "agents": [
 "Agent_3",
 "Agent_6"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
```

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN



- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.46 to you. Your visible payoff was \$7.96.

Your total visible earnings across all rounds are \$19.45.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

#### Message 3 role=assistant

##### content:

Looking at this situation:

##### \*\*Analysis:\*\*

- My co-player sent \$4.14 (keeping only \$0.86) - very high trust!
- I received \$12.42
- In previous rounds, return rates have been around 60-65% of what receivers got
- This sender showed exceptional generosity

##### \*\*My decision:\*\*

Given their high trust, I want to reciprocate generously. Returning about 64% would be \*\*\$8\*\*.

If I return \$8:

- Sender ends with: \$0.86 + \$8 = \$8.86 (net gain of \$3.86)
- I end with: \$12.42 - \$8 = \$4.42 (net gain of \$4.42)

Both of us benefit significantly, and this rewards their trust appropriately.

{"return": 8}

##### reasoning:

Looking at this situation:

- My co-player sent \$4.14 (keeping \$0.86)
- I received \$12.42 (approximately)
- This is a very generous send - they sent most of their endowment!

Looking at my history:

- Round 1: Sent \$3, got back \$5.74 (they returned ~64% of \$9)
- Round 2: Received \$9.75, returned \$6 (~62%)
- Round 3: Sent \$3.5, got back \$6.46 (~62% of \$10.5)

The pattern shows returns around 60-65% of what the receiver gets.

My co-player sent \$4.14, showing significant trust. They kept only \$0.86.

If I return around 60-65%:

- 60% of \$12.42 = \$7.45
- 65% of \$12.42 = \$8.07

Let me think about fairness:

- If I return \$7.50: Sender gets  $\$0.86 + \$7.50 = \$8.36$  (gain of \$3.36), I get  $\$12.42 - \$7.50 = \$4.92$  (gain of \$4.92)
- If I return \$8: Sender gets  $\$0.86 + \$8 = \$8.86$  (gain of \$3.86), I get  $\$12.42 - \$8 = \$4.42$  (gain of \$4.42)

Given they sent such a high amount (\$4.14 out of \$5), I should be generous back. Let me return \$8.

Actually, let me reconsider. They sent \$4.14 and kept \$0.86. If I return \$8:

- They end with \$8.86 total (started with \$5, so net gain of \$3.86)

...

**usage:**

```
{
 "input_tokens": 1438,
 "output_tokens": 699,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$8.0. Your visible payoff was \$4.419999999999998.

Your total visible earnings across all rounds are \$23.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my track record:

**\*\*Analysis:\*\***

- As sender: Sent \$3 ? got \$5.74 back; Sent \$3.5 ? got \$6.46 back
- As receiver: I've returned fairly (60-65% of what I received)
- Cooperation has been consistently rewarded
- Total earnings are healthy at \$23.87

**\*\*My decision:\*\***

Given the strong pattern of reciprocity, I'll send **\*\*\$4\*\*** to show trust and create a \$12 pot for the receiver.

If they reciprocate at the rates I've been seeing (~60-65%), I should get around \$7-8 back, resulting in a good mutual benefit.

{"send": 4}

**reasoning:**

Looking at my track record:

- Round 1: Sent \$3, received back \$5.74 - good outcome
- Round 2 (receiver): Got \$9.75, returned \$6 - both benefited
- Round 3: Sent \$3.5, received back \$6.46 - good outcome
- Round 4 (receiver): Got \$12.42, returned \$8 - both benefited

The pattern shows cooperation is working well. Each time I've been the sender, I've gotten good returns. Each time I've been the receiver, I've returned fairly and both parties benefited.

This is Round 5, and I'm the sender again with a new co-player.

Given that cooperation has been consistently rewarded, and that agents in this experiment seem to be reciprocating fairly, I could:

- Send \$3 (conservative, similar to Round 1)

- Send \$3.5 (similar to Round 3)
- Send \$4+ (more trusting)

Since cooperation has been working and I want to maintain the positive momentum, I'll send \$4. This shows strong trust and creates a \$12 pot. If they return around 60-65%, I'd get around \$7.20-7.80 back, giving me a net gain of ~\$2.20-2.80.

Let me send \$4.

**usage:**

```
{
 "input_tokens": 1763,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.31 to you. Your visible payoff was \$7.31.

Your total visible earnings across all rounds are \$31.179999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.47, so you received \$7.41. How much do you return? (0-7.41)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.47 (keeping \$2.53) - more cautious than previous senders
- I received \$7.41
- I've consistently returned 60-65% as a receiver
- Need to maintain reciprocal behavior

**\*\*My decision:\*\***

Maintaining my pattern, I'll return about 62% which is **\*\*\$4.60\*\***.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of \$2.13)
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of \$2.81)

Both players benefit, and I'm being fair and consistent with my established pattern.

**{"return": 4.6}**

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.47 (keeping \$2.53)
- I received \$7.41
- This is a more cautious send compared to previous rounds I've seen

**\*\*My previous behavior as receiver:\*\***

- Round 2: Received \$9.75, returned \$6 (about 61.5%)
- Round 4: Received \$12.42, returned \$8 (about 64.4%)

**\*\*My pattern:\*\***

I've been returning around 60-65% of what I receive, which has been working well.

**\*\*Considerations:\*\***

- This sender was more cautious (\$2.47 vs the \$3-4+ I've seen)
- I should maintain consistency in my return rate
- 60-65% of \$7.41 would be about \$4.45 to \$4.82

**\*\*My decision:\*\***

I'll return around 62-63% to maintain my reciprocal pattern. That would be about \$4.60.

If I return \$4.60:

- Sender ends with:  $\$2.53 + \$4.60 = \$7.13$  (net gain of  $\$2.13$ )
- I end with:  $\$7.41 - \$4.60 = \$2.81$  (net gain of  $\$2.81$ )

Both benefit, and I'm being consistent with my previous behavior.

**usage:**

```
{
 "input_tokens": 2066,
 "output_tokens": 526,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.47, it became \$7.41, and you returned \$4.6. Your visible payoff was \$2.8100000000000005.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Consistent returns around \$6-6.50 when sending \$3-4
- As receiver: I've been fair and reciprocal
- Total earnings are healthy at \$34
- Cooperation has generally paid off

**\*\*My decision:\*\***

I'll send **\*\*\$4\*\*** to maintain my cooperative strategy. This creates a \$12 pot for the receiver.

Based on patterns, I expect a reasonable return that benefits both of us.

```
{"return": 4}
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3 ? got back \$5.74
- Round 3: Sent \$3.5 ? got back \$6.46
- Round 5: Sent \$4 ? got back \$6.31

As RECEIVER:

- Round 2: Received \$9.75, returned \$6
- Round 4: Received \$12.42, returned \$8
- Round 6: Received \$7.41, returned \$4.6

I've been experiencing good cooperation overall. When I send, I generally get back decent returns (though Round 5 was slightly lower relative to what I sent). When I receive, I've been consistently fair.

For Round 7 as sender, I should continue the cooperative strategy. My last send was \$4 and got \$6.31 back. I could:

- Send \$4 again (proven to work)
- Send \$3.5 (slightly more conservative)
- Send \$4.5 (slightly more trusting)

Given the overall positive trend and my healthy total earnings (\$34), I'll continue with a trusting approach. I'll send \$4 again since that worked reasonably well last time.

```
{"send": 4}
```

**usage:**

```
{
 "input_tokens": 2389,
 "output_tokens": 410,
}
```

```

 "reasoning_tokens": 0
 }
Message 10 role=user
content:
 Round 8

 In your most recent previous game round, you were the SENDER against another agent. You sent $4.0, it became $12.0, and you saw them return $5.4 to you. Your visible payoff was $6.4.
 Your total visible earnings across all rounds are $40.38999999999999.

 This round, you are the RECEIVER against your current co-player. They sent you $2.92, so you received $8.76. How much do you return? (0-8.76)
 Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
 Looking at this situation:

 Analysis:
 - My co-player sent $2.92 (keeping $2.08) - moderate trust
 - I received $8.76
 - I've consistently returned around 60-65% as a receiver (Round 2: 61.5%, Round 4: 64.4%, Round 6: 62%)
 - Need to maintain fair and consistent behavior

 My decision:
 Staying with my pattern, I'll return about 62% which is **$5.45**.

 If I return $5.45:
 - Sender ends with: $2.08 + $5.45 = $7.53 (net gain of $2.53)
 - I end with: $8.76 - $5.45 = $3.31 (net gain of $3.31)

 Both players benefit and I maintain my reciprocal reputation.

 {"return": 5.45}
reasoning:
 Looking at this situation:

 Analysis:
 - My co-player sent $2.92 (keeping $2.08) - moderate trust
 - I received $8.76
 - I've been consistently returning around 60-65% as a receiver
 - This has been my pattern: Round 2: returned 6/9.75 = 61.5%, Round 4: returned 8/12.42 = 64.4%, Round 6: returned 4.6/7.41 = 62%

 My decision:
 Maintaining my consistent pattern, I'll return about 62% which is approximately $5.45.

 If I return $5.45:
 - Sender ends with: $2.08 + $5.45 = $7.53 (net gain of $2.53)
 - I end with: $8.76 - $5.45 = $3.31 (net gain of $3.31)

 Both players benefit, and I'm being fair and consistent.

 Let me calculate: $8.76 * 0.62 = 5.43$

 I'll go with $5.45 to be slightly generous while maintaining my pattern.
usage:
 {
 "input_tokens": 2455,
 "output_tokens": 494,
 "reasoning_tokens": 0
 }
Message 12 role=user
content:
 Round 9

 In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $2.92, it became $8.76, and you returned $5.45. Your visible payoff was $3.3099999999999996.

```

Your total visible earnings across all rounds are \$43.699999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my experience:

**\*\*Analysis:\*\***

- As sender: Returns have varied (\$5.40-\$6.46) but cooperation has paid off
- As receiver: I've been consistently fair (60-65% returns)
- Total earnings are strong at \$43.70
- This is a new co-player

**\*\*My decision:\*\***

I'll send **\*\*\$3.5\*\*** to show cooperation while being moderately cautious with a new partner. This creates a \$10.50 pot for the receiver.

Based on typical reciprocity patterns, I expect a reasonable return that benefits both of us.

{"send": 3.5}

**reasoning:**

Let me analyze my track record:

As SENDER:

- Round 1: Sent \$3 ? got \$5.74 back (payoff: \$7.74)
- Round 3: Sent \$3.5 ? got \$6.46 back (payoff: \$7.96)
- Round 5: Sent \$4 ? got \$6.31 back (payoff: \$7.31)
- Round 7: Sent \$4 ? got \$5.4 back (payoff: \$6.4)

As RECEIVER:

- Round 2: Received \$9.75 ? returned \$6 (payoff: \$3.75)
- Round 4: Received \$12.42 ? returned \$8 (payoff: \$4.42)
- Round 6: Received \$7.41 ? returned \$4.6 (payoff: \$2.81)
- Round 8: Received \$8.76 ? returned \$5.45 (payoff: \$3.31)

Total earnings: \$43.70

Pattern analysis:

- When I send more, returns have generally been good but not always proportionally better
- I've been consistent as a receiver, returning 60-65%
- Cooperation has been working overall
- Last round as sender (Round 7) with \$4 got back \$5.4, which is 45% of the tripled amount (\$12)

I'm in Round 9 now, as a sender again with a new co-player. Given:

1. I have good total earnings
2. Cooperation has been working
3. I want to maintain trust but also be strategic

I'll send around \$3.5-\$4 to show cooperation while not being overly risky.

**usage:**

```
{
 "input_tokens": 2443,
 "output_tokens": 558,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$5.74 to you. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$50.94.

This round, you are the RECEIVER against your current co-player. They sent you \$4.75, so you received \$14.25. How much do you return? (0-14.25)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.75 (keeping only \$0.25) - extremely high trust! That's 95% of their endowment
- I received \$14.25
- I've consistently returned 60-65% as a receiver
- This is even more generous than the \$4.14 sender in Round 4 (where I returned \$8)

**\*\*My decision:\*\***

Given their exceptional generosity, I'll return **\*\*\$9.20\*\*** (about 64.6%).

If I return \$9.20:

- Sender ends with:  $\$0.25 + \$9.20 = \$9.45$  (net gain of \$4.45)
- I end with:  $\$14.25 - \$9.20 = \$5.05$  (net gain of \$5.05)

Both of us benefit significantly, and this appropriately rewards their extraordinary trust.

{"return": 9.2}

**Reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.75 (keeping only \$0.25) - this is VERY high trust! They sent 95% of their endowment
- I received \$14.25
- I've been consistently returning 60-65% as a receiver
- This sender showed exceptional generosity, even more than the \$4.14 sender in Round 4

**\*\*My decision:\*\***

Given their extremely high trust (sending \$4.75 out of \$5), I should reciprocate generously. Let me calculate:

If I return around 63% (my typical rate):  $\$14.25 \times 0.63 = \$8.98$

If I return \$9:

- Sender ends with:  $\$0.25 + \$9 = \$9.25$  (net gain of \$4.25)
- I end with:  $\$14.25 - \$9 = \$5.25$  (net gain of \$5.25)

Both of us benefit significantly. The sender showed exceptional trust and should be rewarded proportionally.

Actually, let me think about this more carefully. They sent \$4.75, which is even more generous than the \$4.14 in Round 4 (when I returned \$8).

In Round 4: received \$12.42, returned \$8 (64.4%)

Here: received \$14.25

If I use a similar percentage:  $\$14.25 \times 0.644 = \$9.18$

I could return \$9 or \$9.20 to be consistent and reward their exceptional trust.

Let me go with \$9.20 to be slightly more generous given they sent nearly everything.

If I return \$9.20:

- Sender ends with:  $\$0.25 + \$9.20 = \$9.45$  (net gain of \$4.45)

...

Usage: {

"input\_tokens": 2438,  
"output\_tokens": 697,  
"reasoning\_tokens": 0

}

**GAME - Agent\_4 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:42:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_1": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**



**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.86 to you. Your visible payoff was \$5.3599999999999999.  
Your total visible earnings across all rounds are \$15.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.48, so you received \$13.440000000000001. How much do you return? (0-13.440000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.48 (nearly 90% of their endowment!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This is much higher trust than I've been showing (I've been sending 50%). Such strong trust deserves strong reciprocation.

If I return \$7.00:

- Sender ends with:  $\$0.52 + \$7.00 = \$7.52$  (profit of ~\$2.52)
- I keep:  $\$13.44 - \$7.00 = \$6.44$  (profit of ~\$6.44)

Both of us profit well, and the sender is rewarded for their exceptional trust.

{"return": 7.00}

**reasoning:**

Let me analyze this situation:

Round 4:

- I'm the RECEIVER
- The sender sent \$4.48 (which is 89.6% of their \$5 endowment - very high trust!)
- I received \$13.44 after the 3x multiplication
- The sender currently has only \$0.52 remaining

This sender showed enormous trust by sending almost 90% of their endowment. This is much higher than what I've been sending (50%).

I need to reciprocate this high level of trust appropriately:

- If I return too little, I'm exploiting their trust
- If I return a fair amount, both of us profit well

Let me think about what would be fair:

- The sender took a big risk sending \$4.48
- They currently have \$0.52
- I have \$13.44

If I return around \$6-7:

- Sender would end with:  $\$0.52 + \$6.50 = \$7.02$  (profit of ~\$2)
- I would keep:  $\$13.44 - \$6.50 = \$6.94$  (profit of ~\$7)

Actually, given they sent 90%, I should be more generous. Let me recalculate:

If I return \$7:

- Sender ends with:  $\$0.52 + \$7 = \$7.52$  (profit of \$2.52)
- I keep:  $\$13.44 - \$7 = \$6.44$  (profit of \$6.44)

This seems more equitable - we both profit significantly, with me keeping slightly less than the sender's final amount, but I still profit more in absolute terms.

Actually, let me think about this differently. They sent \$4.48 and are left with \$0.52. If I want to give them a good return on their trust:

A 50% return rate would be:  $\$13.44 * 0.5 = \$6.72$

...

**usage:**

{

```
"input_tokens": 1412,
"output_tokens": 682,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.48, it became \$13.440000000000001, and you returned \$7.0. Your visible payoff was \$6.440000000000001.

Your total visible earnings across all rounds are \$22.200000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.28, so you received \$12.84. How much do you return? (0-12.84)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$4.28 (about 85.6% of their endowment)
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is similar to Round 4 where someone sent \$4.48 and I returned \$7.00. I want to be consistent in rewarding high trust.

If I return \$6.50:

- Sender ends with: \$0.72 + \$6.50 = \$7.22 (profit of ~\$2.22)
- I keep: \$12.84 - \$6.50 = \$6.34 (profit of ~\$6.34)

This maintains a similar return rate (~51%) as Round 4 (~52%) and both of us profit well.

```
{"return": 6.50}
```

**reasoning:**

Looking at this situation:

- The sender sent \$4.28 (around 85.6% of their endowment) - very high trust
- I received \$12.84 after the 3x multiplication
- The sender currently has \$0.72 remaining

This is again very high trust, similar to the previous round where someone sent \$4.48.

In Round 4, I returned \$7.00 when I received \$13.44. Let me be consistent with my reciprocation pattern for high trust.

The sender showed ~86% trust level, which is slightly less than the ~90% in Round 4, but still very high.

If I return \$6.50:

- Sender ends with: \$0.72 + \$6.50 = \$7.22 (profit of ~\$2.22)
- I keep: \$12.84 - \$6.50 = \$6.34 (profit of ~\$6.34)

This is proportionally similar to what I did in Round 4. Both of us profit well.

Let me think about the ratio:

- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- This round: Received \$12.84, returning \$6.50 (51% return rate)

That seems consistent and fair.

**usage:**

```
{
 "input_tokens": 1745,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.28, it became \$12.84, and you returned \$6.5. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$28.540000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history:

- As sender (Rounds 2-3): I've consistently sent \$2.50 (50%), with varying returns
- As receiver (Rounds 1, 4-5): I've reciprocated well when others showed trust

My 50% sending strategy has been reasonably successful, giving both parties opportunities to profit. While some agents have sent higher amounts (85-90%) when they're senders, I'll maintain my consistent approach that balances trust and risk.

```
{"send": 2.50}
```

**reasoning:**

Let me review my history:

Round 1 (RECEIVER): Received \$7.47 from sender who sent \$2.49 (50%), returned \$3.50 (~47% of what I received). Payoff: \$3.97

Round 2 (SENDER): Sent \$2.50 (50%), received \$3.93 back. Payoff: \$6.43

Round 3 (SENDER): Sent \$2.50 (50%), received \$2.86 back. Payoff: \$5.36

Round 4 (RECEIVER): Received \$13.44 from sender who sent \$4.48 (90%), returned \$7.00 (~52% of what I received). Payoff: \$6.44

Round 5 (RECEIVER): Received \$12.84 from sender who sent \$4.28 (86%), returned \$6.50 (~51% of what I received). Payoff: \$6.34

Now Round 6 (SENDER): I need to decide how much to send.

Looking at the pattern:

- When I've been sender, I've sent \$2.50 (50%) consistently
- When I'm receiver, I've gotten good returns when others send high amounts (50-90%)
- I've been rewarded for reciprocating high trust with good returns

However, I've noticed that when I sent 50% in rounds 2 and 3, I got varying returns:

- Round 2: \$3.93 (very good return, about 52% of what they received)
- Round 3: \$2.86 (lower return, about 38% of what they received)

Other agents have been sending higher amounts (85-90%) to me when they're senders, and I've reciprocated well. This has worked out for both parties.

Should I increase my sending? Or stick with my 50% strategy?

...

**usage:**

```
{
 "input_tokens": 2073,
 "output_tokens": 696,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.76 to you. Your visible payoff was \$7.26.

Your total visible earnings across all rounds are \$35.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my sending history:

- Round 2: Sent \$2.50, received \$3.93 back
- Round 3: Sent \$2.50, received \$2.86 back
- Round 6: Sent \$2.50, received \$4.76 back

The 50% strategy continues to work well on average. Round 6 showed particularly good reciprocity with a \$4.76 return. I'll maintain this consistent, moderate trust level.

{"send": 2.50}

**reasoning:**

Looking at my history:

Round 2 (SENDER): Sent \$2.50, received \$3.93 back, payoff \$6.43  
Round 3 (SENDER): Sent \$2.50, received \$2.86 back, payoff \$5.36  
Round 6 (SENDER): Sent \$2.50, received \$4.76 back, payoff \$7.26

As sender, my 50% strategy has been working reasonably well. In Round 6, I received an excellent return of \$4.76.

As receiver:

Round 1: Received \$7.47, returned \$3.50  
Round 4: Received \$13.44, returned \$7.00  
Round 5: Received \$12.84, returned \$6.50

I've been consistent and cooperative. My 50% sending strategy has been reasonable and I should continue with it.

**usage:**

```
{
 "input_tokens": 2318,
 "output_tokens": 335,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.1 to you. Your visible payoff was \$4.6.

Your total visible earnings across all rounds are \$40.400000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender showed good trust by sending \$3.41 (about 68% of their endowment)
- I received \$10.23 after the 3x multiplication
- The sender currently has \$1.59 remaining

In previous rounds as receiver, I've returned roughly 47-52% of what I received, depending on the trust shown.

If I return \$5.00:

- Sender ends with:  $\$1.59 + \$5.00 = \$6.59$  (profit of ~\$1.59)
- I keep:  $\$10.23 - \$5.00 = \$5.23$  (profit of ~\$5.23)

This is a fair reciprocation that rewards their trust while both of us profit.

{"return": 5.00}

**reasoning:**

Let me analyze this situation:

- The sender sent \$3.41 (about 68% of their endowment) - showing good trust
- I received \$10.23 after the 3x multiplication

- The sender currently has  $\$5 - \$3.41 = \$1.59$  remaining

Looking at my history as receiver:

- Round 1: Received \$7.47, returned \$3.50 (47% return rate)
- Round 4: Received \$13.44, returned \$7.00 (52% return rate)
- Round 5: Received \$12.84, returned \$6.50 (51% return rate)

The sender showed good but not exceptional trust (68% vs the 85-90% I saw in rounds 4-5).

If I return \$5.00:

- Sender ends with:  $\$1.59 + \$5.00 = \$6.59$  (profit of \$1.59)
- I keep:  $\$10.23 - \$5.00 = \$5.23$  (profit of \$5.23)

This gives about a 49% return rate, which is consistent with my previous behavior. Both parties profit reasonably.

Let me consider \$5.00 as a fair reciprocation.

**usage:**

```
{
 "input_tokens": 2283,
 "output_tokens": 486,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.41, it became \$10.23, and you returned \$5.0. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$45.63000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.39, so you received \$10.17. How much do you return? (0-10.17)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.39 (about 68% of their endowment) - very similar trust to Round 8
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 remaining

This is nearly identical to Round 8 where someone sent \$3.41 and I returned \$5.00. I should be consistent in my reciprocation.

If I return \$5.00:

- Sender ends with:  $\$1.61 + \$5.00 = \$6.61$  (profit of ~\$1.61)
- I keep:  $\$10.17 - \$5.00 = \$5.17$  (profit of ~\$5.17)

```
{"return": 5.00}
```

**reasoning:**

Looking at this situation:

- The sender sent \$3.39 (about 67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 remaining

This is very similar to Round 8 where someone sent \$3.41 and I returned \$5.00. I should be consistent.

If I return \$5.00:

- Sender ends with:  $\$1.61 + \$5.00 = \$6.61$  (profit of ~\$1.61)
- I keep:  $\$10.17 - \$5.00 = \$5.17$  (profit of ~\$5.17)

This maintains consistency with my Round 8 response and fairly reciprocates their trust.

**usage:**

```
{
 "input_tokens": 2338,
 "output_tokens": 368,
 "reasoning_tokens": 0
}
```

```

 }
Message 14 role=user
content:
 Round 10

 In your most recent previous game round, you were the RECEIVER against another agent. You
 saw them send $3.39, it became $10.17, and you returned $5.0. Your visible payoff was $5
 .17.
 Your total visible earnings across all rounds are $50.80000000000001.

 This round, you are the SENDER against your current co-player. You have $5. How much do yo
 u send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Assistant output:
 Looking at my sending history:

 - Round 2: Sent $2.50, received $3.93 back
 - Round 3: Sent $2.50, received $2.86 back
 - Round 6: Sent $2.50, received $4.76 back
 - Round 7: Sent $2.50, received $2.10 back

 My consistent 50% strategy has yielded varying returns but remains a balanced approach. I'll c
 ontinue with this moderate trust level.

 {"send": 2.50}
Reasoning:
 Looking at my history as sender:
 - Round 2: Sent $2.50, received $3.93 back (payoff: $6.43)
 - Round 3: Sent $2.50, received $2.86 back (payoff: $5.36)
 - Round 6: Sent $2.50, received $4.76 back (payoff: $7.26)
 - Round 7: Sent $2.50, received $2.10 back (payoff: $4.60)

 Average return when I send $2.50: about $3.41 on average

 I've been consistently sending $2.50 (50%) which has been working reasonably well. This is my
 5th time as sender. I'll maintain consistency.
 Usage: {
 "input_tokens": 2408,
 "output_tokens": 312,
 "reasoning_tokens": 0
 }

GAME - Agent_1 - Receiver - dyad_3
 Timestamp: 2026-08-12T09:43:08
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_1": "trustee",
 "Agent_4": "investor"
 }
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
 }
Exact input messages sent to model:

```

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$3.5. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$17.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze the pattern from previous rounds:

**Round 1 (RECEIVER):**

- Sender sent \$3.26 (65.2%)
- I received \$9.78
- I returned \$4.00
- My payoff: \$5.78
- Sender ended with: \$1.74 + \$4.00 = \$5.74

Round 2 (SENDER):

- I sent \$3.25 (65%)
- It became \$9.75
- Receiver returned \$4.82
- My payoff:  $\$1.75 + \$4.82 = \$6.57$

Round 3 (RECEIVER):

- Sender sent \$2.78 (55.6%)
- I received \$8.34
- I returned \$3.50
- My payoff: \$4.84
- Sender ended with:  $\$2.22 + \$3.50 = \$5.72$

Now in Round 4, I'm the SENDER again.

Looking at the pattern:

- In Round 2, I sent \$3.25 and got back \$4.82 (good return)
- In Round 3, the sender sent \$2.78 (less than I sent in R2) and I returned \$3.50

The returns I've seen have been reasonable. In Round 2, I sent \$3.25 and got \$4.82 back, which was a good return ratio.

I should continue with a cooperative strategy. Since I sent \$3.25 last time as sender and got a good return, I could either:

1. Send the same amount to show consistency
2. Send slightly more to show increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working well, I'll send around the same amount as before, maybe slightly adjusting. I'll send \$3.00 to \$3.50.

Let me send \$3.25 again to maintain consistency with my previous sender behavior.

**usage:**

```
{
 "input_tokens": 1093,
 "output_tokens": 468,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$7.34 to you. Your visible payoff was \$9.09.

Your total visible earnings across all rounds are \$26.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze my history:

Round 1: RECEIVER - Received \$9.78, returned \$4.00, kept \$5.78

Round 2: SENDER - Sent \$3.25, received back \$4.82, kept \$6.57

Round 3: RECEIVER - Received \$8.34, returned \$3.50, kept \$4.84

Round 4: SENDER - Sent \$3.25, received back \$7.34, kept \$9.09

Total earnings: \$26.28

Looking at the pattern:

- When I sent \$3.25 in Round 2, I got back \$4.82 (49.4% of what they received)
- When I sent \$3.25 in Round 4, I got back \$7.34 (75.3% of what they received)

The returns have been quite good, especially in Round 4. I've been sending around 65% of my endowment (\$3.25 out of \$5).



Since I've had good experiences with sending \$3.25 and building cooperation, I'll continue with this strategy. It shows trust while not risking everything.

**usage:**

```
{
 "input_tokens": 1229,
 "output_tokens": 282,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.55 to you. Your visible payoff was \$8.3.

Your total visible earnings across all rounds are \$34.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.04, so you received \$6.12. How much do you return? (0-6.12)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

```
{'return': 2.75}
```

**reasoning:**

Let me analyze this situation:

The sender sent \$2.04 out of \$5, which is about 40.8% of their endowment.

I received \$6.12 after the 3x multiplier.

The sender currently has: \$5 - \$2.04 = \$2.96

This is a more cautious send compared to what I've been seeing and sending (around \$3.25 or 65%).

Looking at my history:

- Round 1: Received \$9.78, returned \$4.00
- Round 2: Sent \$3.25, received back \$4.82
- Round 3: Received \$8.34, returned \$3.50
- Round 4: Sent \$3.25, received back \$7.34
- Round 5: Sent \$3.25, received back \$6.55
- Round 6: Received \$6.12, need to decide return

The sender showed moderate trust (40.8%). I should reciprocate proportionally.

If I return \$2.50:

- I keep: \$6.12 - \$2.50 = \$3.62
- Sender ends with: \$2.96 + \$2.50 = \$5.46

If I return \$3.00:

- I keep: \$6.12 - \$3.00 = \$3.12
- Sender ends with: \$2.96 + \$3.00 = \$5.96

If I return \$2.00:

- I keep: \$6.12 - \$2.00 = \$4.12
- Sender ends with: \$2.96 + \$2.00 = \$4.96

Given that they sent less than what I typically send, but still showed some trust, I'll return around \$2.50 to \$3.00. Let me go with \$2.75 as a middle ground.

...

**usage:**

```
{
 "input_tokens": 1374,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-pla

yer. You saw them send \$2.04, it became \$6.12, and you returned \$2.75. Your visible payoff was \$3.37.  
Your total visible earnings across all rounds are \$37.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

{'send': 3.25}

**reasoning:**

Let me analyze the previous round with this co-player:

- They sent me \$2.04 (about 41% of their endowment)
- I received \$6.12
- I returned \$2.75
- They ended with:  $\$5 - \$2.04 + \$2.75 = \$5.71$
- I ended with:  $\$6.12 - \$2.75 = \$3.37$

So they sent a moderate amount (\$2.04), and I returned \$2.75, which gave them a slight gain overall.

Now I'm the sender. Given that:

1. They sent me a moderate amount last round (\$2.04, which is about 41%)
2. I've been generally cooperative, sending around \$3.25 in other rounds
3. This is a repeated interaction with this specific co-player

I should probably send a similar or slightly higher amount than they sent to maintain cooperation. My typical send has been \$3.25, which has worked well with other players who returned good amounts.

I'll send \$3.25 to continue building trust with this co-player.

**usage:**

```
{
 "input_tokens": 1510,
 "output_tokens": 268,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$6.52 to you. Your visible payoff was \$8.27.

Your total visible earnings across all rounds are \$46.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. I'm now in Round 8 as a RECEIVER
2. My co-player sent me \$3.88, which became \$11.64
3. I need to decide how much to return

Looking at my history:

- Round 2: I was RECEIVER, got \$9.78, returned \$4.0 (kept \$5.78)
- Round 3: I was SENDER, sent \$3.25, got back \$4.82 (kept \$6.57)
- Round 3 (second part): I was RECEIVER, got \$8.34, returned \$3.5 (kept \$4.84)
- Round 4: I was SENDER, sent \$3.25, got back \$7.34 (kept \$9.09)
- Round 5: I was SENDER, sent \$3.25, got back \$6.55 (kept \$8.3)
- Round 6: I was RECEIVER, got \$6.12, returned \$2.75 (kept \$3.37)
- Round 7: I was SENDER, sent \$3.25, got back \$6.52 (kept \$8.27)
- Round 8: I'm RECEIVER, got \$11.64, need to decide

My pattern as SENDER has been consistent: I send \$3.25 each time.

AS RECEIVER:

- Round 2: received \$9.78, returned \$4.0 (return ratio =  $4.0/9.78 = 40.9\%$ )
- Round 3: received \$8.34, returned \$3.5 (return ratio =  $3.5/8.34 = 42.0\%$ )
- Round 6: received \$6.12, returned \$2.75 (return ratio =  $2.75/6.12 = 44.9\%$ )

So I've been returning around 40-45% of what I receive.

My co-player sent \$3.88, which is notably higher than the typical \$2-3 range I've observed. With \$11.64 received, applying my established return ratio suggests returning between \$4.66-\$5.24. Since they demonstrated stronger trust with a larger initial send, I'll return approximately \$5.00 to acknowledge that while staying consistent with my historical pattern of around 43%.

Looking back at my previous rounds, I notice I've consistently ensured my co-player ends up with roughly \$5.70-\$5.74 total—slightly above their initial \$5 investment. When I received \$9.78 and returned \$4.00, they netted \$5.74; at \$8.34 with a \$3.50 return, they got \$5.72; and with \$6.12 returned at \$2.75, they ended at \$5.71. This suggests I should aim for a return that keeps them in that same range. If I return \$5.63 here, they'd receive \$6.75 total, which is higher than my usual pattern. I need to consider what happens when I send larger amounts myself to calibrate appropriately....

usage:

```
{
 "input_tokens": 1437,
 "output_tokens": 1113,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.88, it became \$11.64, and you returned \$5.0. Your visible payoff was \$6.640000000000001.

Your total visible earnings across all rounds are \$52.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

```
{'send': 3.25}
```

reasoning:

Let me analyze the situation:

1. In Round 8, my co-player sent me \$3.88 (which became \$11.64), and I returned \$5.00.
2. Now in Round 9, I'm the sender against the same co-player.

Let me look at the pattern:

- When I've been sender, I've consistently sent \$3.25
- When I receive amounts, I try to return a fair portion

In Round 8, my co-player sent me \$3.88, which is higher than my usual \$3.25. They showed trust.

Looking at my returns as receiver:

- Round 3: Received \$8.34 (from \$2.78 sent), returned \$3.50
- Round 6: Received \$6.12 (from \$2.04 sent), returned \$2.75
- Round 8: Received \$11.64 (from \$3.88 sent), returned \$5.00

My return pattern seems to aim for roughly giving back a bit more than what was originally sent but not the full multiplied amount.

Now as sender against the same co-player who sent me \$3.88 in Round 8:

- They were generous with \$3.88
- I should probably match or reciprocate their level of trust
- I've been sending \$3.25 consistently, but maybe I should send more to match their \$3.88

Actually, let me stick with my consistent strategy of \$3.25. This has been working well overall and my total earnings are good at \$52.86.

usage:

```
{
 "input_tokens": 1436,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**  
**content:**

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.05 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$58.66.

This round, you are the RECEIVER against your current co-player. They sent you \$2.53, so you received \$7.59. How much do you return? (0-7.59)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

```
{'return': 3.25}
```

**Reasoning:**

Let me analyze the pattern of play:

Round 4: I was RECEIVER, saw \$2.78 sent, became \$8.34, I returned \$3.5, kept \$4.84  
Round 5: I was SENDER, sent \$3.25, became \$9.75, they returned \$7.34, I got \$9.09  
Round 6: I was SENDER, sent \$3.25, became \$9.75, they returned \$6.55, I got \$8.30  
Round 7: I was RECEIVER, saw \$2.04 sent, became \$6.12, I returned \$2.75, kept \$3.37  
Round 8: I was SENDER, sent \$3.25, became \$9.75, they returned \$6.52, I got \$8.27  
Round 9: I was RECEIVER, saw \$3.88 sent, became \$11.64, I returned \$5.00, kept \$6.64  
Round 10: I was SENDER, sent \$3.25, became \$9.75, they returned \$4.05, I got \$5.80

Now I'm RECEIVER in round 10 (wait, the prompt says round 10, but I already got round 10 info. ..). Let me re-read.

Actually, looking more carefully:

- Round 10: I was SENDER, sent \$3.25
- Now in Round 10, I'm RECEIVER from current co-player who sent \$2.53

Wait, that's confusing. Let me re-read the prompt.

"Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$4.05 to you. Your visible payoff was \$5.8. Your total visible earnings across all rounds are \$58.66.

This round, you are the RECEIVER against your current co-player. They sent you \$2.53, so you received \$7.59. How much do you return? (0-7.59)"

So the most recent previous game round was me as SENDER. Now in THIS round (round 10), I'm the RECEIVER. Got it.

Looking at my behavior as a receiver:

- Round 4: Received \$8.34 (from \$2.78 sent), returned \$3.5 (kept \$4.84, about 42% return rate)
- Round 7: Received \$6.12 (from \$2.04 sent), returned \$2.75 (kept \$3.37, about 45% return rate)
- Round 9: Received \$11.64 (from \$3.88 sent), returned \$5.00 (kept \$6.64, about 43% return rate)

I'm consistently returning around 42-45% of what I receive. This round they sent \$2.53, I got \$7.59. If I follow my pattern, I should return roughly \$3.20-\$3.40.

Returning about \$3.50 would leave them with \$5.97 total and me with \$4.09, which feels reasonably balanced....

Usage: {

```
"input_tokens": 1436,
"output_tokens": 997,
"reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:43:18  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {  
 "agent\_type": "standard",  
 "dyad\_id": "dyad\_4",  
 "move\_index": 6,  
 "opponent\_id": "Agent\_2",  
 "pairing": {  
 "agents": [  
 "Agent\_7",  
 "Agent\_2"  
 ],  
 "investor": "Agent\_7",  
 "roles": {  
 "Agent\_2": "trustee",  
 "Agent\_7": "investor"  
 },  
 "trustee": "Agent\_2"  
 },  
 "role": "investor",  
 "role\_label": "Sender",  
 "round": 10,  
 "task": "game",  
 "task\_index": 0  
 }

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may al  
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths  
 , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized  
 , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.18, it became \$12.54, and you returned \$7.0. Your visible payoff was \$5.539999999999999.

Your total visible earnings across all rounds are \$15.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous sending experience:\*\***

- Round 1: Sent \$3, received \$5.09 back - successful cooperation

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their reciprocity patterns

**\*\*Strategy:\*\***

Since my previous moderate sending amount (\$3) yielded good reciprocity, I'll use the same approach:

- Sending \$3 creates \$9 for the receiver
- Shows willingness to cooperate
- Creates significant surplus to share
- If they return even \$4, both parties benefit

This balanced approach has proven effective in building cooperative relationships.

{"send": 3}

**reasoning:**

Let me analyze the situation:

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, received \$5.09 back (good reciprocity)
- Round 2: I was receiver, got \$2.08 (became \$6.24), returned \$3.50
- Round 3: I was receiver, got \$4.18 (became \$12.54), returned \$7.00

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER again
- I'm playing against a new co-player (pairings are randomized each round)
- I have no information about this specific co-player

**\*\*Strategic considerations:\*\***

1. My previous sending experience (Round 1) was positive - sent \$3, got back \$5.09
2. I've seen varying levels of cooperation (sends of \$2.08 and \$4.18)
3. I've been fairly generous as a receiver, returning good amounts

**\*\*Options:\*\***

- Send \$0: Safe but creates no value
- Send \$2-3: Moderate trust, tests the waters
- Send \$4-5: High trust, maximizes potential gains but risky

Since my Round 1 experience was positive (sent \$3, got \$5.09 back), and I've been building cooperative relationships, I should continue with a similar strategy. Sending \$3 allows

:

1. Creation of significant value (\$9 to receiver)
2. Room for mutual benefit (if they return even \$4, I profit)
3. Not risking everything
4. Signaling willingness to cooperate

I'll go with \$3 again, which has worked well before.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 529,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.66 to you. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$21.03.

This round, you are the RECEIVER against your current co-player. They sent you \$3.4, so you received \$10.2. How much do you return? (0-10.2)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - solid trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. Their trust level is good (68% sent)
2. Should reciprocate fairly to reward and maintain cooperation
3. Both parties should benefit from the exchange

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)
- Both parties gain significantly

This rewards their trust while creating mutual benefit.

{"return": 5.50}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*Previous round:\*\***

- I sent \$3, received \$3.66 back
- Return rate was about 41% of what they received
- This was fair but not overly generous

**\*\*Current situation:\*\***

- My co-player sent \$3.40 (68% of their endowment) - good trust
- I received \$10.20
- This created a gain of \$6.80

**\*\*Considerations:\*\***

1. They showed solid trust by sending 68% of their endowment
2. I should reciprocate fairly to encourage continued cooperation
3. Need to ensure they profit from their trust

If I return \$5.50:

- They end with: \$5 - \$3.40 + \$5.50 = \$7.10 (profit of \$2.10)
- I end with: \$10.20 - \$5.50 = \$4.70 (profit of \$4.70)
- Both profit, I get slightly more which seems reasonable given they initiated the trust

Let me think about the range:

- Minimum fair return: About \$4-5 (they at least break even or profit modestly)
- Maximum generous return: About \$6-7 (they profit well, I still gain)

\$5.50 seems like a good middle ground - rewards their trust, ensures mutual benefit, and maintains cooperation.

**usage:**

```
{
```

```
"input_tokens": 1755,
"output_tokens": 512,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.4, it became \$10.2, and you returned \$5.5. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$25.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)

Both instances were profitable, showing that \$3 is a reasonable amount that encourages reciprocity.

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No specific information about their behavior
- My consistent moderate strategy has been working

**\*\*Strategy:\*\***

Sending \$3:

- Creates \$9 for the receiver
- Demonstrates cooperative intent
- Provides room for mutually beneficial returns
- Has proven effective in previous rounds

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3, received \$5.09 back - good return
- Round 4: Sent \$3, received \$3.66 back - fair return

Both times I sent \$3, I received positive returns that made the exchange worthwhile.

**\*\*Current situation:\*\***

- New co-player this round
- No specific information about their reciprocity
- My moderate sending strategy has been working well

**\*\*Strategy:\*\***

- Sending \$3 has proven to be a good balanced approach
- Creates \$9 for receiver, creating room for mutual benefit
- Shows cooperation without risking everything
- Consistent with my established pattern

I'll continue with \$3 as it's been effective.

**usage:**

```
{
 "input_tokens": 2072,
 "output_tokens": 350,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7



In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.32 to you. Your visible payoff was \$4.32.

Your total visible earnings across all rounds are \$30.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.7, so you received \$11.100000000000001. How much do you return? (0-11.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 6: I sent \$3, they returned \$2.32 (about 26% of what they received)

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They sent \$3.70 (74% of their endowment) - high trust!
- I received \$11.10
- This created \$7.40 in surplus

**\*\*Considerations:\*\***

1. They're showing high trust now by sending 74%
2. Despite lower return last round, they're signaling cooperation
3. Should reciprocate fairly to maintain the relationship

If I return \$6:

- They end with:  $\$5 - \$3.70 + \$6 = \$7.30$  (profit: \$2.30)
- I end with:  $\$11.10 - \$6 = \$5.10$  (profit: \$5.10)
- Both parties benefit significantly

This rewards their trust and encourages continued cooperation.

{"return": 6}

**reasoning:**

**\*\*Analysis of the situation:\*\***

Previous round (Round 6):

- I sent \$3 to this co-player
- They received \$9
- They returned \$2.32 to me
- This was a somewhat low return (about 26% of what they received)

Current round (Round 7):

- Same co-player (continuing interaction)
- They sent me \$3.70 (74% of their endowment)
- I received \$11.10
- This is a strong trust signal

**\*\*Considerations:\*\***

1. They showed high trust by sending \$3.70 (74%)
2. In the previous round, they returned less than I might have hoped (\$2.32 from \$9)
3. This is a continued interaction, so my response matters for future cooperation
4. I should be fair and reciprocate their trust appropriately

**\*\*Strategic thinking:\*\***

Even though they returned less in Round 6, they're now showing high trust by sending \$3.70

- I should:
- Reciprocate fairly to maintain cooperation
- Not punish them excessively for the previous lower return
- Create mutual benefit

If I return \$6:

- They end with:  $\$5 - \$3.70 + \$6 = \$7.30$  (profit of \$2.30)
- I end with:  $\$11.10 - \$6 = \$5.10$  (profit of \$5.10)
- Both profit significantly

This rewards their current trust while maintaining fair reciprocity.

**usage:**

```
{
 "input_tokens": 2393,
 "output_tokens": 586,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.7, it became \$11.100000000000001, and you returned \$6.0. Your visible payoff was \$5.100000000000001.

Your total visible earnings across all rounds are \$35.150000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

**\*\*My sending history:\*\***

- Round 1: Sent \$3 ? received \$5.09 (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 (profit: \$0.66)
- Round 6: Sent \$3 ? received \$2.32 (loss: -\$0.68)

**\*\*Current situation:\*\***

- New co-player this round (randomized pairings)
- No information about their reciprocity patterns
- My moderate strategy has mostly been effective

**\*\*Strategy:\*\***

Continue with \$3:

- Creates \$9 for the receiver
- Shows cooperative intent
- Balanced risk approach
- Has generally yielded positive outcomes (2 out of 3 profitable)

{"send": 3}

**reasoning:**

**\*\*Analysis:\*\***

**\*\*My sending track record:\*\***

- Round 1: Sent \$3 ? received \$5.09 back (profit: \$2.09)
- Round 4: Sent \$3 ? received \$3.66 back (profit: \$0.66)
- Round 6: Sent \$3 ? received \$2.32 back (profit: -\$0.68, actually a loss)

**\*\*Current situation:\*\***

- This is Round 8, so a new co-player (randomized pairings)
- No specific information about their behavior
- My \$3 strategy has been generally effective, though with varying results

**\*\*Strategy considerations:\*\***

The \$3 sending amount has been my consistent approach:

- Creates \$9 for receiver
- Shows cooperative intent but not excessive risk
- Has yielded mixed but generally positive results

I'll continue with \$3 as it's a balanced approach that:

1. Demonstrates willingness to cooperate
2. Creates significant surplus
3. Doesn't overcommit resources
4. Has worked reasonably well overall

{"send": 3}

**usage:**

```
{
 "input_tokens": 2571,
 "output_tokens": 442,
 "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.14 to you. Your visible payoff was \$7.14.

Your total visible earnings across all rounds are \$42.290000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$4.27 (85.4% of their endowment) - very high trust!
- I received \$12.81
- This created a surplus of \$8.54

**\*\*Considerations:\*\***

1. They showed exceptional trust by sending over 85% of their endowment
2. Should reciprocate generously to reward this high trust
3. Both parties should benefit substantially from the exchange

If I return \$7.50:

- They end with: \$5 - \$4.27 + \$7.50 = \$8.23 (profit: \$3.23)
- I end with: \$12.81 - \$7.50 = \$5.31 (profit: \$5.31)
- Both parties gain significantly

This rewards their exceptional trust while maintaining mutual benefit.

{"return": 7.50}

reasoning:

**\*\*Analysis:\*\***

**\*\*Current situation:\*\***

- My co-player sent \$4.27 (85.4% of their endowment) - very high trust!
- I received \$12.81
- This created a gain of \$8.54

**\*\*My receiving history:\*\***

- Round 2: Received \$6.24, returned \$3.50 (kept \$2.74)
- Round 3: Received \$12.54, returned \$7.00 (kept \$5.54)
- Round 5: Received \$10.20, returned \$5.50 (kept \$4.70)
- Round 7: Received \$11.10, returned \$6.00 (kept \$5.10)

Looking at my pattern:

- When I receive ~\$6.24: returned ~56%
- When I receive ~\$12.54: returned ~56%
- When I receive ~\$10.20: returned ~54%
- When I receive ~\$11.10: returned ~54%

I generally return about 54-56% of what I receive, which seems fair and encourages cooperation.

**\*\*For this round:\*\***

- They sent \$4.27 (very high trust - 85.4%)
- I received \$12.81
- Returning about 55% would be \$7.05

If I return \$7:

- They end with: \$5 - \$4.27 + \$7 = \$7.73 (profit: \$2.73)
- I end with: \$12.81 - \$7 = \$5.81 (profit: \$5.81)

Both parties profit substantially. This rewards their very high trust level.

Let me calculate if I should be a bit more generous given the very high trust (85.4%):

If I return \$7.50:  
- They end with:  $\$5 - \$4.27 + \$7.50 = \$8.23$  (profit: \$3.23)  
...

**usage:**

```
{
 "input_tokens": 2551,
 "output_tokens": 752,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.27, it became \$12.809999999999999, and you returned \$7.5. Your visible payoff was \$5.309999999999999.  
Your total visible earnings across all rounds are \$47.600000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 9: They sent \$4.27 (85.4% of endowment) - very high trust
- I returned \$7.50, giving them a profit of \$3.23

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They demonstrated exceptional trust
- This is an opportunity to reciprocate and build strong cooperation

**\*\*Strategy:\*\***

Since they showed very high trust (sending 85.4%), I should reciprocate with a similarly high level of trust:

If I send \$4:

- Creates \$12 for them
- Matches their trust signal
- Shows I value our cooperative relationship
- If they return fairly, both parties benefit substantially

{"send": 4}

**Reasoning:**

**\*\*Analysis:\*\***

**\*\*Previous interaction with this co-player:\*\***

- Round 9: They sent \$4.27 (85.4%), I returned \$7.50
- So they made:  $\$5 - \$4.27 + \$7.50 = \$8.23$  (profit of \$3.23)
- I made: \$5.31

**\*\*Current situation:\*\***

- Same co-player (continuing relationship)
- They showed very high trust in Round 9
- I reciprocated fairly with a generous return

**\*\*Strategy:\*\***

Since they showed exceptional trust (85.4% sent), I should reciprocate with high trust as well. This could build a strong cooperative relationship.

If I send \$4:

- Creates \$12 for them
- Shows I reciprocate their high trust
- Allows them to return generously if they choose

This is a good opportunity to match their trust level and potentially establish a highly cooperative dynamic.

Alternatively, I could send \$3.5 or even \$4.5, but \$4 seems like a strong signal of reciprocal trust without being overly risky.

Let me go with \$4 to match their high trust level.

```
Usage: {
 "input_tokens": 2556,
 "output_tokens": 448,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:43:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_2"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_2": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$4.0. Your visible payoff was \$3.5600000000000005.  
Your total visible earnings across all rounds are \$16.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my experience:

- When I sent \$3.50 in Round 2, I received \$6.73 back - a good return
- Cooperative strategies have been working well across rounds

I'll continue with a high-trust strategy.

{'send': 3.50}

**reasoning:**

Looking at my history:

- Round 1: RECEIVER - received \$10.02, returned \$5.00, profit \$5.02
- Round 2: SENDER - sent \$3.50, received back \$6.73, profit \$8.23
- Round 3: RECEIVER - received \$7.56, returned \$4.00, profit \$3.56
- Total earnings: \$16.81

My experience shows:

- Round 2: I sent \$3.50 (70%), got back \$6.73 from \$10.50 (64% return rate)
- Round 3: When I received a moderate amount (\$2.52, 50.4%), I returned \$4.00 from \$7.56 (53% return rate)

Both times I've been the sender, I sent a substantial amount and was rewarded well. The strategy of showing trust seems to be working. I should continue with a cooperative strategy.

I'll send around \$3.50 again, which has worked well for me.

**usage:**

```
{
 "input_tokens": 1400,
 "output_tokens": 305,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.95 to you. Your visible payoff was \$8.45.

Your total visible earnings across all rounds are \$25.26.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)

The cooperative high-trust strategy has been consistently successful. I'll continue with it.

{'send': 3.50}

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.50, received \$6.73 back, payoff = \$8.23
- Round 4: I sent \$3.50, received \$6.95 back, payoff = \$8.45

Both times I sent \$3.50, I received good returns. My cooperative strategy is working well. I'm building up earnings and the high-trust approach continues to pay off.

I'll stick with this successful strategy.

**usage:**

```
{
 "input_tokens": 1587,
 "output_tokens": 211,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.5 to you. Your visible payoff was \$8.0.

Your total visible earnings across all rounds are \$33.260000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed exceptional trust by sending \$4.40 (88% of their endowment)
- I received \$13.20 after the 3x multiplication
- The sender currently has only \$0.60 left

This is the highest trust I've seen so far, exceeding even Round 1 (66.8%). I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.60 + \$7.00 = \$7.60 (profit of \$2.60)
- I end with: \$13.20 - \$7.00 = \$6.20 (profit of \$6.20)

This rewards their exceptional trust while ensuring we both profit well.

{'return': 7.00}

**reasoning:**

Let me analyze this situation:

- The sender sent \$4.40 (88% of their endowment) - this is very high trust
- I received \$13.20 after the 3x multiplication
- The sender currently has \$0.60 left (\$5 - \$4.40)

This sender showed exceptional trust by sending 88% of their endowment. I should reciprocate generously.

Looking at my past behavior as receiver:

- Round 1: Received \$10.02, returned \$5.00 (about 50%)
- Round 3: Received \$7.56, returned \$4.00 (about 53%)

For this higher trust situation, I should return more generously.

If I return \$7.00:

- Sender ends with:  $\$0.60 + \$7.00 = \$7.60$  (profit of \$2.60)
- I end with:  $\$13.20 - \$7.00 = \$6.20$  (profit of \$6.20)

This rewards their exceptional trust significantly while still maintaining good profit for both.

Alternatively, if I return \$6.50:

- Sender ends with:  $\$0.60 + \$6.50 = \$7.10$  (profit of \$2.10)
- I end with:  $\$13.20 - \$6.50 = \$6.70$  (profit of \$6.70)

I think \$7.00 is appropriate given the very high trust shown (88% vs the 70% I've been sending).

**usage:**

```
{
 "input_tokens": 1827,
 "output_tokens": 548,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.4, it became \$13.200000000000001, and you returned \$7.0. Your visible payoff was \$6.200000000000001.

Your total visible earnings across all rounds are \$39.46000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.91, so you received \$11.73. How much do you return? (0-11.73)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left

This is high trust, similar to previous high-trust rounds. Comparing to my history:

- Round 1: 66.8% sent ? returned \$5.00
- Round 6: 88% sent ? returned \$7.00

At 78.2%, this falls between those two. I'll return proportionally.

If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

{'return': 6.50}

**reasoning:**

Looking at this situation:

- The sender sent \$3.91 (78.2% of their endowment), showing high trust
- I received \$11.73 after the 3x multiplication
- The sender currently has \$1.09 left (\$5 - \$3.91)

This is high trust, though not quite as high as the 88% I saw in Round 6. Let me compare to my previous responses:

Round 1: Sender sent 66.8%, I received \$10.02, I returned \$5.00 (about 50%)

Round 3: Sender sent 50.4%, I received \$7.56, I returned \$4.00 (about 53%)

Round 6: Sender sent 88%, I received \$13.20, I returned \$7.00 (about 53%)

This sender sent 78.2%, which is between Round 1 (66.8%) and Round 6 (88%).

If I return \$6.00:

- Sender ends with:  $\$1.09 + \$6.00 = \$7.09$  (profit of \$2.09)
- I end with:  $\$11.73 - \$6.00 = \$5.73$  (profit of \$5.73)



If I return \$6.50:

- Sender ends with:  $\$1.09 + \$6.50 = \$7.59$  (profit of \$2.59)
- I end with:  $\$11.73 - \$6.50 = \$5.23$  (profit of \$5.23)

Given the high trust level (78.2%), I think \$6.50 is appropriate, falling between my Round 1 return (\$5.00 for 66.8%) and Round 6 return (\$7.00 for 88%).

usage:

```
{
 "input_tokens": 2156,
 "output_tokens": 639,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.91, it became \$11.73, and you returned \$6.5. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$44.69000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

content:

Looking at my consistent success as sender:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)

The \$3.50 strategy has proven reliable and profitable. I'll continue with it.

```
{'send': 3.50}
```

reasoning:

Let me review my history as sender:

- Round 2: Sent \$3.50, received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50, received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50, received \$6.50 back (payoff \$8.00)

My sending strategy of \$3.50 has been consistently successful, with returns ranging from \$6.50 to \$6.95.

As receiver, I've been fair and reciprocal:

- Round 1: Received \$10.02, returned \$5.00
- Round 3: Received \$7.56, returned \$4.00
- Round 6: Received \$13.20, returned \$7.00
- Round 7: Received \$11.73, returned \$6.50

The cooperative strategy has worked well. I should continue with \$3.50.

usage:

```
{
 "input_tokens": 2234,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$5.32 to you. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$51.51000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my sender history:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)
- Round 8: Sent \$3.50 ? received \$5.32 back (payoff \$6.82)

Even with some variation in returns, the high-trust strategy of sending \$3.50 has been consistently profitable. I'll maintain this approach.

```
{'send': 3.50}
```

**reasoning:**

Looking at my track record as sender:

- Round 2: Sent \$3.50 ? received \$6.73 back (payoff \$8.23)
- Round 4: Sent \$3.50 ? received \$6.95 back (payoff \$8.45)
- Round 5: Sent \$3.50 ? received \$6.50 back (payoff \$8.00)
- Round 8: Sent \$3.50 ? received \$5.32 back (payoff \$6.82)

The last round had a lower return (\$5.32) but still profitable. The overall strategy of sending \$3.50 has been working well on average. I've been maintaining trust and cooperation, and while returns vary, I'm still profiting overall.

I should continue with the \$3.50 strategy as it has proven reliable over multiple rounds.

**usage:**

```
{
 "input_tokens": 2286,
 "output_tokens": 372,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.5, it became \$10.5, and you saw them return \$6.69 to you. Your visible payoff was \$8.190000000000001.

Your total visible earnings across all rounds are \$59.70000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.86, so you received \$11.58. How much do you return? (0-11.58)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$3.86 (77.2% of their endowment), showing high trust
- I received \$11.58 after the 3x multiplication
- The sender currently has \$1.14 left
- Last round, this same co-player returned \$6.69 to me when I sent \$3.50 - they were generous

This trust level is very similar to Round 7 (78.2%) where I returned \$6.50.

If I return \$6.50:

- Sender ends with: \$1.14 + \$6.50 = \$7.64 (profit of \$2.64)
- I end with: \$11.58 - \$6.50 = \$5.08 (profit of \$5.08)

This reciprocates their previous generosity and maintains fairness.

```
{'return': 6.50}
```

**Reasoning:**

Looking at this situation:

- The sender sent \$3.86 (77.2% of their endowment), showing high trust
- I received \$11.58 after the 3x multiplication
- The sender currently has \$1.14 left

In the previous round (Round 9), I was the sender against this same co-player:

- I sent \$3.50
- They returned \$6.69
- That was a generous return

Now they're showing high trust by sending \$3.86. I should reciprocate fairly.

Looking at my pattern:

- Round 1: 66.8% sent ? returned \$5.00
- Round 3: 50.4% sent ? returned \$4.00
- Round 6: 88% sent ? returned \$7.00
- Round 7: 78.2% sent ? returned \$6.50

At 77.2%, this is very similar to Round 7 (78.2%) where I returned \$6.50.

If I return \$6.50:

- Sender ends with:  $\$1.14 + \$6.50 = \$7.64$  (profit of \$2.64)
- I end with:  $\$11.58 - \$6.50 = \$5.08$  (profit of \$5.08)

This seems fair and maintains the cooperative relationship, especially since they were generous to me last round.

Actually, let me think about this more carefully. They returned \$6.69 to me when I sent \$3.50 (creating \$10.50). That's about 63.7% of what they received.

If I return around 63.7% of \$11.58 = \$7.38, that would be reciprocating at a similar rate.

But looking at my pattern, I've been returning amounts that give good profits to both parties:

- \$6.50 seems appropriate given the 77.2% trust level

...

```
Usage: {
 "input_tokens": 2256,
 "output_tokens": 680,
 "reasoning_tokens": 0
}
```